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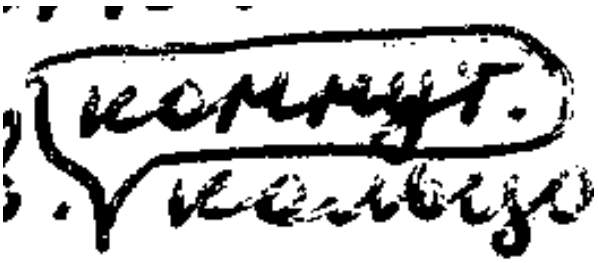
Chapter 6

GELFAND-P000459

V. Buchstaber.

Formal groups.

The def. of a formal group was given.



Original source details: GELFAND-P000459

Diagram and typed description. The abbreviation “commut.” is added in a bubble above “ring.”

commut.

A — arbitrary ring with 1;

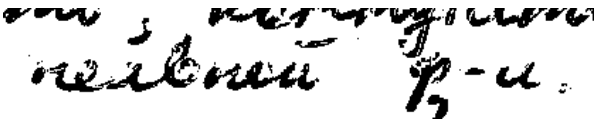
$$\sum_{i=0}^{\infty} a_i x^i, \qquad a_i \in A$$

Form. group:

$$F(x,y) = x + y + \sum_{i+k\geq 2} a_{ik} x^i y^k$$

— form. power series.

Conditions: associativity, commutativity (it follows from assoc.). Exist. of an inv. elem. from the implicit func. thm.



Original source details: GELFAND-P000459

Uncertain reading. The last words read «неявной ф-и» (“implicit func.”).

Isomorphism:

$$z = \varphi(x) = x + a_1 x^2 + \ldots; \qquad \varphi^{-1}(x) = z - a_1 z^2 + \ldots$$



Original source details: GELFAND-P000459

Uncertain reading. The sign in the first formula reads φ.



Original source details: GELFAND-P000459

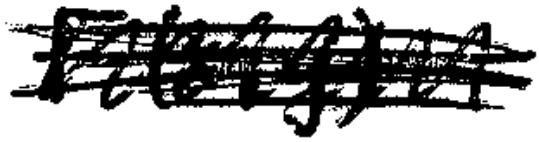
Uncertain reading. The same sign φ appears before the exponent -1.



Original source details: GELFAND-P000459

Uncertain reading. The parentheses contain x. The right-hand side uses z.

... and one is obt. from the other by a change of variables.



Original source details: GELFAND-P000459

Undeciphered passage. The entry before “and one” is crossed out. A full reading is uncertain.

Form. groups will be classified (but not up to isom. — that is a separate problem)

The series $g(x) = x + \dots$ is called a logarithm of the grp. F, if

$$F(x, y) = g^{-1}(g(x) + g(y))$$

Lazard's lemma. If A is a $\mathbb{Q}$-module, then any form. group $F(x, y)$ over A has a logarithm.



Original source details: GELFAND-P000459

Uncertain reading. The name provisionally reads «Лазара» (Lazard). The vowel after «з» may be «у».

Proof. (see Mat., $13 : 6$)

GELFAND-P000460

Consider in place of A the ring $A[[t]]$; t — time.

Then def. the shift $x \mapsto F(x, t)$ (shift of the form. var. by time t).

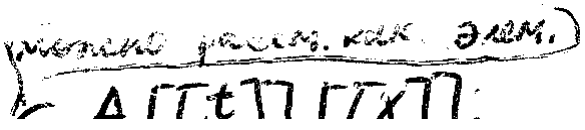
Differential:

$$\omega(x) = \psi(x)\, dx$$



Original source details: GELFAND-P000460

Uncertain reading. The first coefficient sign is set as ψ. The same letter recurs in the formulas below.



Original source details: GELFAND-P000460

Diagram and typed description. A thin line marks off the addition above the formula.

can be consid. as an elem.

$\psi(x) \in A[[t]][[x]]; \quad A \subset A[[t]]$

Inv. form (differential):

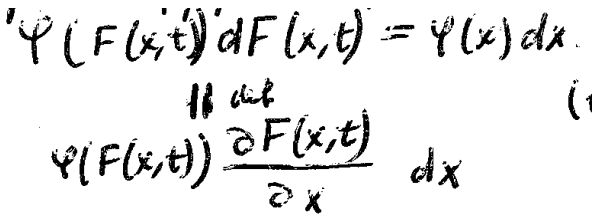


Diagram and typed description. Two vertical strokes and “or” stand between the formulas.
$\psi(F(x,t))\,dF(x,t) = \psi(x)\,dx.$

$\|$ or

$$\psi(F(x,t))\,\frac{\partial F(x,t)}{\partial x}\,dx$$

Original source details: GELFAND-P000460

$$\psi(F(x,t))\,\frac{\partial F(x,t)}{\partial x} = \psi(x)$$

(t is a parameter; the differential is taken only in x)

Set $x = 0$:

$$\psi(t)(1 + \ldots)\big|_{x=0} = \psi(0),$$

$$\psi(t) = \psi(0)\,\frac{1}{\left.\dfrac{\partial F(x,t)}{\partial x}\right|_{x=0}}.$$

These are the series that are coeffs. in the inv. differential.

Canonical form:

$$\omega(x) = \frac{1}{\left.\dfrac{\partial F(x,y)}{\partial y}\right|_{y=0}}\,dx$$

is an ~~...~~ invar. differential (simple exercise).

Uncertain reading. A heavy vertical sign lies between the words. Its reading is uncertain.



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This gives a characterization.

Consider the form. series

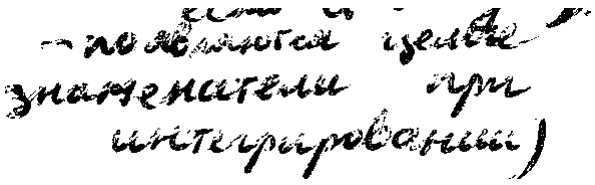
$$g(x) = \int_0^x \omega(x) = x + \ldots$$

(it is important that A is a $\mathbb{Q}$-module! — integer denominators arise during integration)

Uncertain reading. An exclamation mark follows “module.”



Original source details: GELFAND-P000460



Uncertain reading. The first word after the dash reads «появляются» (“arise”).

Original source details: GELFAND-P000460

Integrate the relation:

$$\psi(F(x,t))\,dF(x,t) = \psi(x)\,dx$$
$$g(F(x,y)) = g(x) + C;$$

Here integration is in x (t — param.), therefore $C = C(y)$.

Set $x = 0$. We obtain: $C = g(y)$ ...



Undeciphered passage. The entry after $C = g(y)$ is crossed out and illegible.

Original source details: GELFAND-P000460

$$g(F(x,y)) = g(x) + g(y)$$
$$F(x,y) = g^{-1}(g(x) + g(y)). \quad !! \qquad \blacksquare$$

Corollary Over $\mathbb{Q}$-mods. all form. laws are strictly isom.

Univ. law: consid. $g(x) = x + \sum g_i x^{i+1}$ — a series, its ring is $\mathbb{Q}[g_i]$.

$$F(x,y) = g^{-1}(g(x) + g(y));$$

Over $\overset{\downarrow}{\mathbb{Q}}[g_i]$ one obtains a univ. law. (for all $\mathbb{Q}$-modules)



Uncertain reading. The letter $\mathbb{Q}$ below the arrow is overwritten. The earlier strokes cannot be separated.

Original source details: GELFAND-P000460

We shall classify over arbitrary rings A (commut. and with 1)



Uncertain reading. The first word reads «Будем» (“we shall”).

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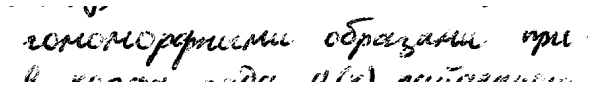


Diagram and typed description. A curved arrow leads from the universal law to the bracket before “Universality.”

[Universality:

Original source details: GELFAND-P000460

any series $F(x,y)$ over the $\mathbb{Q}$-mod. A is obt. from such a series by replacing its coefficients by their homomorphic images under a ring homom. $\mathbb{Q}[g_i]$ into A, sending g_i to the coeffs. of the series $g(x)$, found above]



Uncertain reading. The two words for "homomorphic images" have plural endings.

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GELFAND-P000461

Now consid. the general case: A is not necess. a $\mathbb{Q}$-module.

Consid. the series $F(x,y) = x + y + \sum a_{i,j} x^i y^j$

$a_{i,j}$ — alg. indep. variables.

Consid. $\mathbb{Z}[a_{i,j}]$, $\quad \deg a_{i,j} = i + j - 1$

A form. group is obt. from it

$$F(F(x,y),z) = x + y + z + \sum a_{i,j,k} x^i y^j z^k, \quad a_{i,j,k} \in \mathbb{Z}[a_{i,j}].$$
$$F(x,F(y,z)) = x + y + z + \sum a'_{i,j,k} x^i y^j z^k \qquad a'_{i,j,k} \in \mathbb{Z}[a_{i,j}]$$

Consid. in $\overset{\mathbb{Z}[a_{i,j}]}{\overline{\ldots}}$ the ideal J, gen. by the homog. elems. $\{a_{i,j,k} - a'_{i,j,k}\}$ and by the elems. $\{a_{i,j} - a_{\overline{j,i}}\}$.



Undeciphered passage. $\mathbb{Z}[a_{i,j}]$ is inserted above the cancellation. The earlier entry is illegible.

Original source details: GELFAND-P000461



Uncertain reading. There are crossed-out strokes above the second coefficient $a_{j,i}$.

Original source details: GELFAND-P000461

Now replace $a_{i,j}$ in $F(x,y)$ with $a_{i,j}$ mod J — $\overline{\ldots}$ we obtain the series $F \in \Lambda[[x,y]]$.



Undeciphered passage. The entry before "we obtain" is crossed out and illegible.

Original source details: GELFAND-P000461

$$\Lambda = \mathbb{Z}[a_{i,j}]/J$$

This is the univ. group. (clear).

Lazard's thm., Λ — a polynomial ring on generators, one in each dimension:

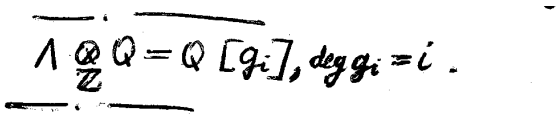
$$\Lambda = \mathbb{Z}[a_i], \quad i \geq 1, \quad \deg a_i = i.$$



Original source details: GELFAND-P000461

Uncertain reading. The penultimate letter of «одной» (“one”) is heavily overwritten.

By Lazard's lemma:



Original source details: GELFAND-P000461

Diagram and typed description. Two pairs of strokes with dots lie above and below the equality.

$$\Lambda \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}[g_i], \ \deg g_i = i.$$

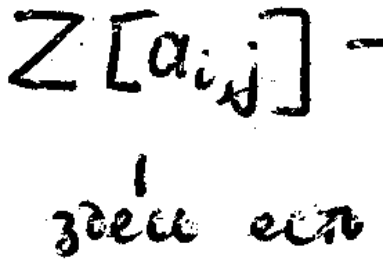
1. How many mult. generators has Λ?

Consid. $\widetilde{\Lambda} = \Lambda/(\text{decomposable elements})$.



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Uncertain reading. The word in the denominator provisionally reads «разложимым» (“decomposable”).



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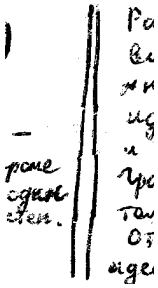
Diagram and typed description. A short vertical stroke links the first ring to the explanation.

$$\mathbb{Z}[a_{i,j}] \longrightarrow \Lambda \longrightarrow 0.$$

here there is an ideal gen. by decomp. elems. (prod. of 2 of nonzero degree)

We have: $\widetilde{\Lambda} = \sum_{i \geq 0} \widetilde{\Lambda}_i$;
multiplication is zero except when one of $\dots$ is an elem. of degree 0.

Decomp. elems. are elems. of the form $a \cdot b$, where a and b *
* are noninvertible. Take the ideal they generate, and $\widetilde{\Lambda}$ — the quotient ring.
Grading $\Rightarrow$ only elems. of degree 0 are invertible.
Hence this ideal is the image of the ideal of decomp. elems. from $\mathbb{Z}[a_{i,j}]$.



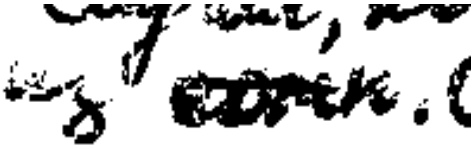
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Diagram and typed description. Two vertical strokes separate the marginal note at right.



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Uncertain reading. The left-hand sign probably reads $\widetilde{\Lambda}$ with a joined tilde. Another possible reading is χ.



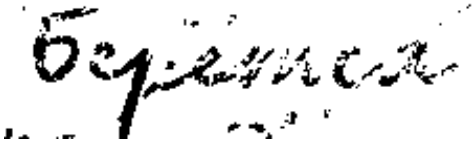
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Undeciphered passage. The short entry after “of” is crossed out and illegible.



Original source details: GELFAND-P000461

Uncertain reading. The marginal note reads «необратимы» (“are noninvertible”).



Original source details: GELFAND-P000461

Uncertain reading. «Берется» (“take”) follows «необратимы».

$\widetilde{\Lambda}_n$ — an abelian grp. under add. with gens. — gens. of the mult. grp. Λ in dimension n.

$$F(F(x,y),z) = F(x,y) + z + \sum a_{i,j} \dots (x+y)^i z^j$$



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Undeciphered passage. The entry before $(x+y)^i$ is crossed out and illegible.

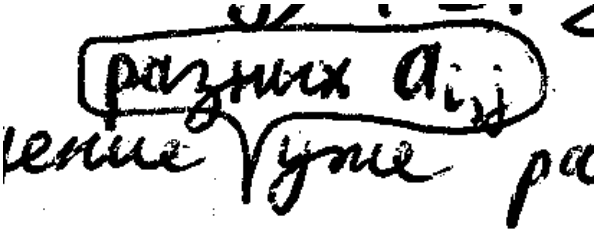


Diagram and typed description. The bubble above the line contains "of different $a_{i,j}$."

of different $a_{i,j}$

(in $\widetilde{\Lambda}$ the product already equals 0).

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$$F(x, F(y,z)) = x + F(y,z) + \sum a_{i,j} x^i (y+z)^j$$

$\widetilde{\Lambda}$ is obt. from the free ab. group gen. by $a_{i,j}$, by quotienting by the rels. ...

Let $h(x,y) = \sum h_{i,j} x^i y^j = F(x,y) - x - y$.

$$h(x,0) = 0, \quad h(0,y) = 0.$$

GELFAND-P000462

Lazard's rel.:

$$h(x,y) + h(x+y,z) = h(y,z) + h(x,y+z). \qquad (*)$$

It suffices to take homog. polynomials h.

In Lazard: methods of number theory.

Problem: find all solutions of $(*)$ [homog. polys.] over some ring K.

All solutions — a group L, Everything over the ring K

$$L \longrightarrow \mathrm{Ext}^{2,n}_{K[x]}(K,K) \longrightarrow 0.$$

Introduce on $K[x]$ a Hopf struct.: a mult. homomorphism

$$\Delta : K[x] \longrightarrow K[x] \otimes_K K[x] \qquad \text{(comultiplication)}.$$
$$x \longrightarrow x \otimes 1 + 1 \otimes x$$

A Hopf alg. has cohomology. The alg. is graded $\Rightarrow$ the cohom. is bigraded. Hence 2 indices for Ext.

Next:

$$\begin{array}{c} \text{on the left and on the right} \quad (x+y+z)^n - x^n - y^n - z^n \\ \downarrow \\ 0 \longrightarrow \{c((x+y)^n - x^n - y^n),\ a \in K\} \longrightarrow L \longrightarrow \mathrm{Ext}^{2,n}_{K[x]}(K,K) \longrightarrow 0. \end{array}$$



Diagram and typed description. The arrow from the addition points down toward the first term of the sequence.

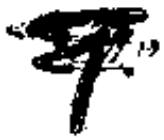
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Uncertain reading. The sign before the parentheses provisionally reads c. The separate entry reads $a \in K$. Both signs are overwritten.

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$n = 2$. Polyn. of degree 2 cxy.



Uncertain reading. The sign above "Polyn." is crossed out and illegible.

Original source details: GELFAND-P000462

$n = 3$ $h(x, y) = C_1x^2y + C_2xy^2$

$$\widetilde{\Lambda} = \mathbb{Z} + \mathbb{Z}_3.$$



Uncertain reading. The line reads $\widetilde{\Lambda} = \mathbb{Z} + \mathbb{Z}_3$.

Original source details: GELFAND-P000462

1 sol. by Lazard's lemma: $c((x + y)^n - x^n - y^n)$

$h(x, y) = x^2y + xy^2$ for all rings

$h(x, y) = x^2y$ for rings that are $\mathbb{Z}_3$-modules



Uncertain reading. The subscript on $\mathbb{Z}$ in this line is 3.

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x^2y

Consid. comodules over a Hopf alg. A — this is an ab. grp. N, for which there is a map $N \longrightarrow A \otimes N$ with nat. conds.

Choose the resolution correctly: there will be all cycles without boundaries (min. resolution). ~~by~~



Uncertain reading. The crossed-out word after the parenthesis may read «по» (“by”).

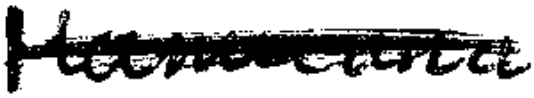
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~~H...~~



Uncertain reading. The start of the crossed-out heading shows the Cyrillic «Н».

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Undeciphered passage. The rest of the crossed-out heading is illegible.

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Calculation of the group Ext^2.

1°. K — ~~...~~ a comodule over the Hopf alg. $K[x]$.



Uncertain reading. A heavy rectangular mark is crossed out between the dash after K and “comodule.”

Original source details: GELFAND-P000462

~~...~~ $x \longrightarrow x \otimes 1 + 1 \otimes x$



Uncertain reading. A heavily crossed-out sign appears before x in the addition.

Original source details: GELFAND-P000462

$$K \longrightarrow K \otimes K[x] \quad \xleftarrow{\cdots}$$

Uncertain reading. The beginning of the crossed-out addition resembles $\leftarrow$.



Original source details: GELFAND-P000462

Undeciphered passage. The remaining strokes after $K[x]$ are illegible.



Original source details: GELFAND-P000462

$1 \longrightarrow 1 \otimes 1$ and then by linearity.

GELFAND-P000463

Consid.

$$K_*[x] = \mathrm{Hom}_K(K[x], K).$$

("diff. operators").

$$(x_*^n, x^m) = \delta_{n,m}$$

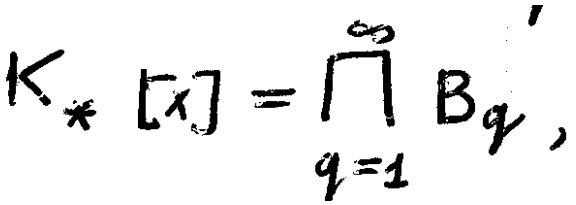
Multiplication:

$$\underline{x_*^n \cdot x_*^m = C_{n+m}^n x_*^{n+m}} \qquad \text{— } \underline{\text{table}}.$$

Here K plays no role.

It suffices to take $K = \mathbb{Z}_p$ or $\mathbb{Q}$.

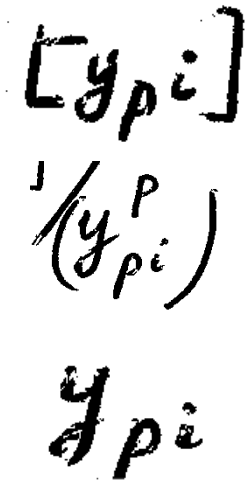
$$K_*[x] = \prod_{q=1}^{\infty} B_q, \qquad B_q = \mathbb{Z}_p[y_{p^i}]/(y_{p^i}^p)$$



Uncertain reading. The left side reads $K_*[x]$. The product has lower limit $q = 1$ and upper limit ∞.

Original source details: GELFAND-P000463

$y_1, \ldots, y_i, \ldots, y_{p^i}$ — generators.



Uncertain reading. In three places the i next to p is raised. This is set as y_{p^i}. In the denominator the separate outer exponent p on y is visible.

Original source details: GELFAND-P000463

GELFAND-P000464

30/X-72. Gelfand sem..

I. Bernstein

И.М.Г.

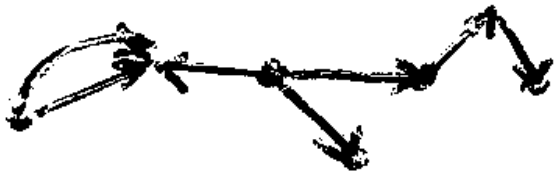
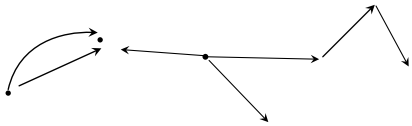


Diagram and typed description. The two arrows at left point to the same vertex. From the middle vertex arrows lead left, right and down to the right. The bent line at right has arrowheads. The small strokes at vertices have no established reading as letters.



Original source details: GELFAND-P000464

There is a directed graph. This is a category.

Ꝑ Consid. some diagrams, with at the points — vect. spaces, on the arrows maps (functoriality is not required).



Uncertain reading. A loop survives at the start of the crossed-out sign. Possibly Cyrillic «Р».

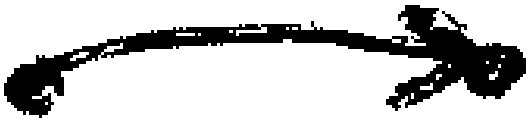
Original source details: GELFAND-P000464



Uncertain reading. The words before “diagrams” provisionally read «какие л.» (“some”).

Original source details: GELFAND-P000464

Prob. — to classify.



Original source details: GELFAND-P000464

Diagram and typed description. A slightly curved rightward arrow joins the two points.



Prob. of classif. of maps. The only invs. are dim Ker and dim Coker. Reduces to an embedding.



Original source details: GELFAND-P000464

Diagram and typed description. One loop. Its arrowhead on the left side points upward.

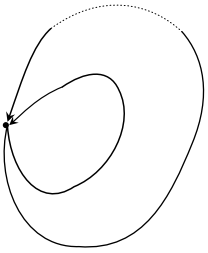


one transformation.



Original source details: GELFAND-P000464

Diagram and typed description. Two loops share the left vertex. The top of the outer loop is faint. Overlapping arrowhead strokes are visible near the vertex.



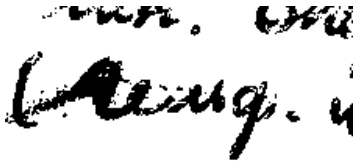
— the prob. is unsolvable (a pair of arbitrary lin. operators). ...



Original source details: GELFAND-P000464

Uncertain reading. The short entry before “it” is crossed out and illegible.

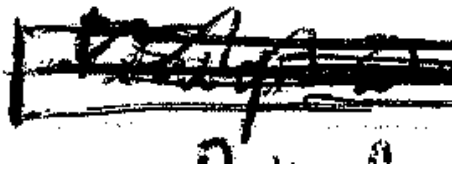
It reduces to the prob. of commuting ones (Gelf. and Ponomarev).



Original source details: GELFAND-P000464

Uncertain reading. The first letter is overwritten. The abbreviation reads «Гельф.» (Gelf.).

~~Reduc....~~



Original source details: GELFAND-P000464

Uncertain reading. The crossed-out line may begin «Свод.» (“reduc.”).

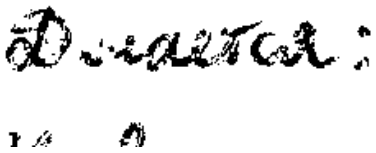


Original source details: GELFAND-P000464

Undeciphered passage. The remainder of the crossed-out line is illegible.

Space and subspace.
Space and 2 subspaces } triv.

Can be done: Space and 3 lin. subspaces.



Original source details: GELFAND-P000464

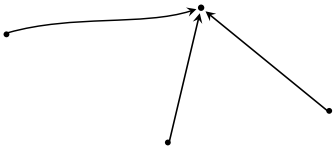
Uncertain reading. The start of the line provisionally reads «Делается» (“can be done”).

There are 9, invariants, not 8!



Original source details: GELFAND-P000464

Diagram and typed description. Three arrows point toward the upper vertex. Their arrowheads partly merge.

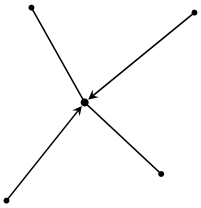


(problem).



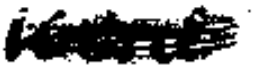
Original source details: GELFAND-P000464

Diagram and typed description. Four edges meet at the heavily inked central vertex. Overlapping strokes obscure the direction of two edges.



Space and 4 subspaces classified (Gelf. and Ponom.) — ... already an inf. number of indecomp.

Undeciphered passage. The word after the dash is crossed out and illegible.



Original source details: GELFAND-P000464

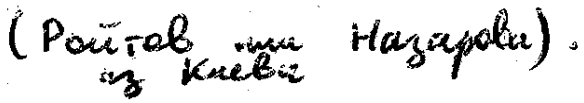
Gabriel: in which graphs are there finitely many indecomp. systems?

Answer: draw a Dynkin diagram! — these and only these are such graphs.

Exceptions: diagrams with double (generally multiple) arrows.

GELFAND-P000465

Theorem (Roytov or Nazarovy).
from Kyiv

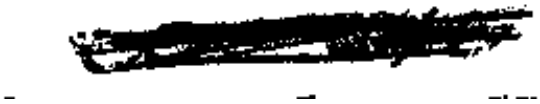


Uncertain reading. The names provisionally read «Ройтов» and «Назаровы». The addition below reads “from Kyiv.”

Original source details: GELFAND-P000465

A class of diagrams is descr. with a classific. as in Gelf. and Ponomarev.

Connection ... with singularities! — see Artin, Tyurina, Brieskorn.



Undeciphered passage. The word between “Connection” and “with singularities” is crossed out and illegible.

Original source details: GELFAND-P000465

И.Б. Category of vect. spaces over points and maps over arrows.

(The graph is fixed). Def. direct sum. This is Cat(Γ) (Γ — graph)

Morphisms — maps over each point with commut. diagrs.

- category of linear spaces.

Important: no need to ident. isomorphic ones.



Uncertain reading. The abbreviation before “isomorphic” provisionally reads «отожд.» (“ident.”).

Original source details: GELFAND-P000465

It is easy to see that $\forall$ object $= \sum$ indecomp. ones. (uniquely up to isom.).

1) Classif. Γ, with finitely many indecomp. objects.

Restrict to connected ones:



Diagram and typed description. A small angular stroke rises above the third vertex from the left.

A_n (n points)

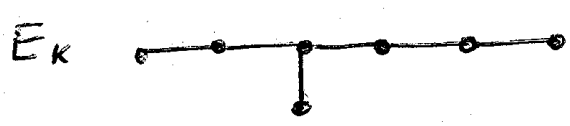
Original source details: GELFAND-P000465



Original source details: GELFAND-P000465

Diagram and typed description. There is a small leftward arrowhead on the segment between the third and fourth vertices.

D_n $n \geq 4$



Original source details: GELFAND-P000465

Diagram and typed description. There are six vertices on the horizontal chain. A branch descends from the third vertex from the left.

E_k

$k = 6, 7, 8$ (necessarily 2 segments on the left).

Place the arrows arbitrarily!!

D_4 — space and 3 subspaces.

Idea: no cycles (one obt. eigenvalues), no



Original source details: GELFAND-P000465

Uncertain reading. The abbreviation in parentheses may read «получ.» (“obt.”).



Original source details: GELFAND-P000465

Diagram and typed description. Four segments meet at the black dot. No separate vertices are visible at the ends.

— there will be an inf. number.



Original source details: GELFAND-P000465

Diagram and typed description. The chain has a fork at each end. Five vertices are marked on the chain itself.

— also imposs., since ident.



Original source details: GELFAND-P000465

Uncertain reading. The last abbreviation provisionally reads «отожд.» (“ident.”).

2) The no. of indecomp. objs. does not depend on the dir. of the arrows.
It is characterized by the set of dimensions!!
one should consid. the root syst.
γ — pos. root, α_i — simp. roots.
To each root — an indecomp. object

$$\gamma = \sum n_i \alpha_i, \qquad n_i \text{ — dimensions!!}$$





Uncertain reading. Both signs have a lower loop and two upper strokes. They are provisionally set as γ.

Original source details: GELFAND-P000465

No. = no. of simple roots.



Uncertain reading. The line reads «простых корней» (“simple roots”).

Original source details: GELFAND-P000465

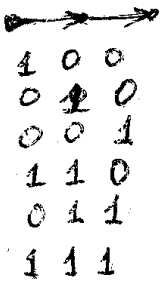


Diagram and typed description. Six rows of dimensions stand below the chain with two rightward arrows.

1	0	0
0	1	0
0	0	1
1	1	0
0	1	1
1	1	1

Original source details: GELFAND-P000465



Uncertain reading. The middle 1 in row two is overwritten. A remnant of a loop survives to its right.

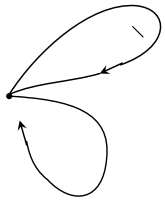
Original source details: GELFAND-P000465

3) Consid. the Γ, for which $\mathrm{Cat}(\Gamma)$ is solvable.



Original source details: GELFAND-P000466

Diagram and typed description. The two loops share a vertex. Inside the upper loop there is a short slanted stroke separate from its arrowhead.



C

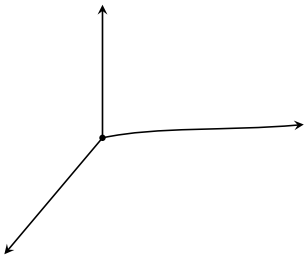
Unsolvable categ. — if there is an embedding $C \longrightarrow \mathrm{Cat}(\Gamma)$, taking nonisom. objects to nonisom. ones.

They are classif. and correspond to extended Dynkin diagrams!



Original source details: GELFAND-P000466

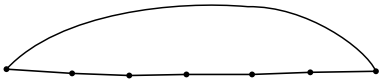
Diagram and typed description. Three arrows leave one vertex, upward, rightward and down to the left.



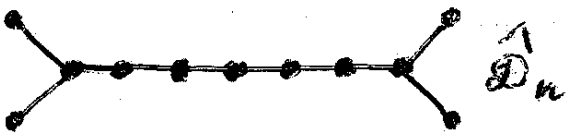


Original source details: GELFAND-P000466

Diagram and typed description. Seven points lie on a straight chain. An arc joins the endpoints.



$\widehat{A}_n$

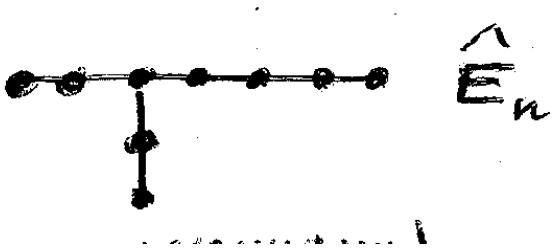


Original source details: GELFAND-P000466

Diagram and typed description. Seven points lie along the chain, with two branches at each end.

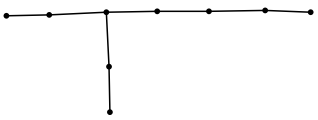


$\widehat{D}_n$



Original source details: GELFAND-P000466

Diagram and typed description. Seven points lie along the horizontal chain. Two segments with points descend from the third point from the left.



$\widehat{E}_n$

(appear in the theory of ~~grp.~~ groups generated by reflections.)



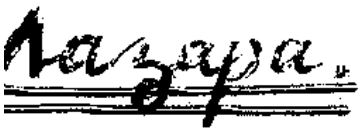
Uncertain reading. The crossed-out insertion above the line may read «гр» (“grp.”).

Original source details: GELFAND-P000466

These are all the graphs for which the prob. is solvable.

V. Buchstaber.

Lazur's thm.



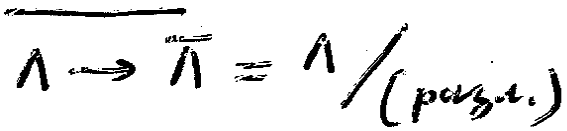
Uncertain reading. The name provisionally reads «Лазура» here. The middle vowel may be «а».

Original source details: GELFAND-P000466

$$\Lambda = \mathbb{Z}[a_1, \ldots, a_n, \ldots], \qquad \deg a_n = n \geq 1.$$

Proof.

$$\Lambda \longrightarrow \bar{\Lambda} = \Lambda/(\text{decomp.})$$



Uncertain reading. A horizontal stroke stands above the second Λ. The denominator reads «разл.» (“decomp.”).

Original source details: GELFAND-P000466

$\bar{\Lambda}$ — univ. ring for such polynomials $h(x, y)$,

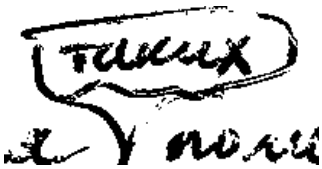


Diagram and typed description. “Such” is inserted in a bubble before “polynomials.”

Original source details: GELFAND-P000466

that $h(x, y) = h(y, x), \quad h(x, 0) = 0,$

$$h(y, z) + h(x, y + z) = h(x, z) + h(y, x + z).$$

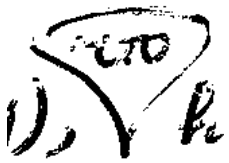
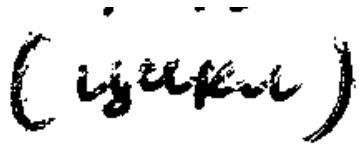


Diagram and typed description. “That” is inserted in a separate bubble before the conditions.

Original source details: GELFAND-P000466

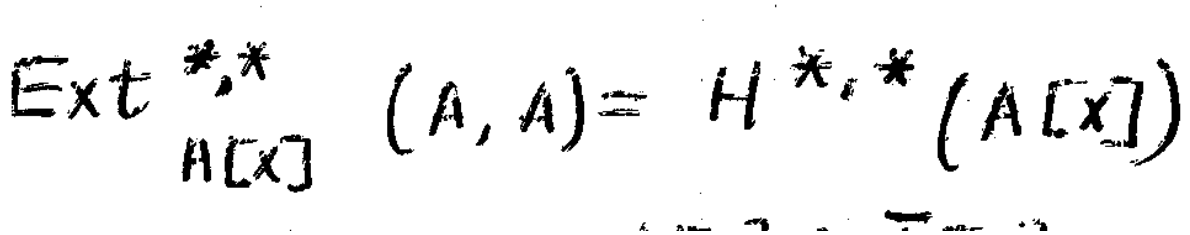
This is cohomology: $\underset{\text{(cycles)}}{dh} = 0$ (in the bar construction for polynomials)



Uncertain reading. Below dh is the word "cycles."

Original source details: GELFAND-P000466

$$\mathrm{Ext}^{*,*}_{A[x]}(A, A) = H^{*,*}(A[x])$$



Original source details: GELFAND-P000466

Uncertain reading. The subscript on Ext reads $A[x]$. There are two stars above the symbol on each side.

$$A[x] \longrightarrow A[x] \otimes \bar{A}[y]$$
$$x \longrightarrow x + y$$



Uncertain reading. Below the first arrow the manuscript reads $x \longrightarrow x + y$.

Original source details: GELFAND-P000466

Cohom. is calc. using resolutions.

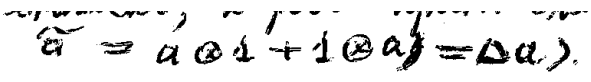
The bar constr. is one of the resolutions.

$$\mathcal{A}: \quad A[x] \xrightarrow{\Delta} A[x] \otimes \bar{A}[y] \longrightarrow A[x] \otimes \bar{A}[y] \otimes \bar{A}[z] \longrightarrow \dots$$

$\bar{A}[y]$ — polynomials with no const. term.

We obt. an acyclic complex which is used to compute cohomology

$$(\underset{h(x,y)}{\text{put everywhere}} — \tilde{a} \Rightarrow a \otimes 1 + 1 \otimes a) = \Delta a).$$



Uncertain reading. A wavy stroke stands above a. The following sign provisionally reads $\Rightarrow$. Both closing parentheses are visible.

Original source details: GELFAND-P000466

One must classif. cycles in a particular resolution.

$$0 \longrightarrow B \longrightarrow Z \longrightarrow H^2(A[x]) \longrightarrow 0$$



Original source details: GELFAND-P000466

Uncertain reading. The term after B is Z, and H has superscript 2.

$x^n \longrightarrow (x+y)^n - x^n - y^n$ (these are boundaries)

GELFAND-P000467

Passage to the obt. resol. by means of the functor $\mathrm{Hom}(\cdot, A)$



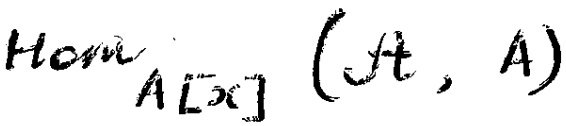
Uncertain reading. The abbreviation before "resol." reads «полученн.» ("obt.").

Original source details: GELFAND-P000467

In the acycl. resol.

$$x \longrightarrow x + y$$
$$x^n \longrightarrow (x+y)^n.$$

We must consid. $\mathrm{Hom}_{A[x]}(\mathcal{A}, A)$ (from the 1st factor est. the struct. of an $A[x]$-mod.).



Uncertain reading. The subscript on Hom is $A[x]$. The arguments are $\mathcal{A}, A$.

Original source details: GELFAND-P000467



Uncertain reading. The abbreviation after "factor" reads «уст.» ("est.").

Original source details: GELFAND-P000467

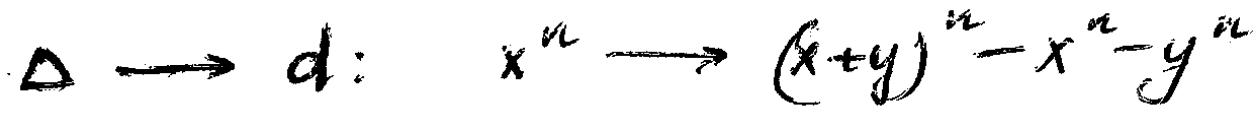
$$A \xrightarrow{d} A[y] \xrightarrow{d} A[y] \otimes A[z] \xrightarrow{d} \dots$$



Original source details: GELFAND-P000467

Uncertain reading. On this line $A[y]$ and $A[z]$ have no overbars.

$$\Delta \longrightarrow d: \qquad x^n \longrightarrow (x+y)^n - x^n - y^n$$



Original source details: GELFAND-P000467

Uncertain reading. Both subtracted terms, x^n and y^n, are visible on the next line.

$$\Delta: \quad h(x,y) \longrightarrow h\big(\underset{\|}{\Delta x}, y\big) - h\big(x, \underset{\|}{\Delta y}\big)$$
$$\qquad\qquad\quad x+z \qquad\qquad\quad y+z$$



Uncertain reading. There is a heavily inked horizontal sign between the terms. It is provisionally set as minus.

Original source details: GELFAND-P000467

$\mathcal{A}$ — a seq. of $A[x]$-modules,



Uncertain reading. A short low stroke at the end of the line resembles a comma.

Original source details: GELFAND-P000467

$A[x] \otimes y^k$ — submodules

Because of $\overline{A[y]}$:

$$x^n \longrightarrow (x+y)^n - x^n \qquad \text{(since there is no const. term)}$$

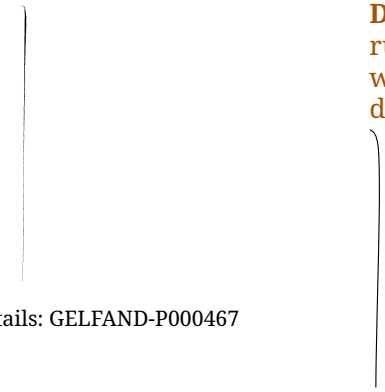


Diagram and typed description. A long line runs along the right edge of the frame. It begins with a small bend and ends just below the divider.

Original source details: GELFAND-P000467

GELFAND-P000468

13/XI-72. Gelfand sem..

L. Vaserstein.

A — assoc. ring (not necessarily with 1).

iA — ring with a formally adjoined unit.

$iA/A = \mathbb{Z}$, A — two-sided ideal in iA.

GA — mult. group of the ring.

$GA \subset A$

$(a, b) \longrightarrow a + b + ab$ This is a binary assoc. oper.

$$\| \qquad (e + a)(e + b) - e = a + b + ab$$

~~iA~~ GA — set of invertible elements with respect to this operation.



Uncertain reading. The short crossed-out entry before GA may read iA.

Original source details: GELFAND-P000468

$$a \longrightarrow 1 + a, \qquad A \longrightarrow 1 + A$$

homomorphism. into the ordinary invert. elems.

$$f : A \longrightarrow B$$
$$f(GA) \subset GB$$



Uncertain reading. The sign before GB is overwritten. It is provisionally set as $\subset$.

Original source details: GELFAND-P000468

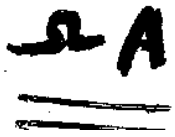
i, G — functors to the cat. of rings with 1 and the cat. of groups.

$M_k A$ — matrices $k \times k$ with ordinary multiplication.

$\widetilde{\Omega} A = \{a = (a_{ij})\}$, in $\forall$ row and column a finite no. of elems. $\neq 0$, (this is a ring)

MA — subring of finite matrices (a finite number of elems. $\neq 0$). (two-sided ideal in $\widetilde{\Omega} A$).

$\Omega A = \widetilde{\Omega} A / MA$



Uncertain reading. The letter before A on the left is overwritten. It reads Ω.

Original source details: GELFAND-P000468

$M, \widetilde{\Omega}, \Omega$ — functors.

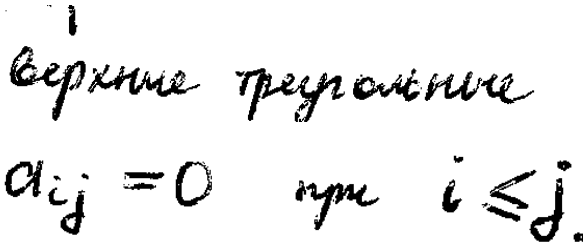
$$\underline{\underline{\Theta A}} \subset MA$$
$$|$$

upper triangular $a = (a_{ij})$: $a_{ij} = 0$ when $i \leq j$ ...



Diagram and typed description. A short vertical stroke is drawn below the double underline under ΘA.

Original source details: GELFAND-P000468



Uncertain reading. The manuscript has “upper triangular” and $a_{ij} = 0$ when $i \leq j$.

Original source details: GELFAND-P000468



Undeciphered passage. Several signs one above another are crossed out after the inequality.

Original source details: GELFAND-P000468

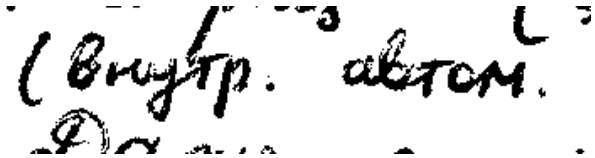
Remark. If $N \xrightarrow{\sim} N$ is a bijection, there is a map of the matrix ring (~~th.~~ autom.) — simultaneous perm. of rows and columns.



Uncertain reading. The crossed-out abbreviation before “autom.” may read «эт».

Original source details: GELFAND-P000468

(inner autom. perm. of rows and columns)



Uncertain reading. The parenthesis reads «внутр. автом.» (“inner autom.”).

Original source details: GELFAND-P000468

Even if $N \longrightarrow N$ is any map, it is assigned an endomorphism.



Uncertain reading. The first word reads «Даже» (“even”).

Original source details: GELFAND-P000468



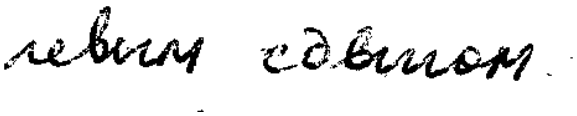
Uncertain reading. The last word is underlined and reads “endomorphism.”

Original source details: GELFAND-P000468

GMA — mult. group, MA, $GMA \subset MA$.

Simplicial structure in MA.

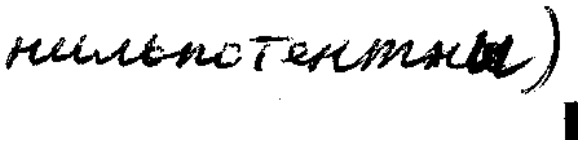
$X \subset GMA$ is called a simplex, if by a left shift and conjugation by a permutation matrix it can be taken into ΘA



Uncertain reading. The manuscript reads «левым сдвигом» (“by a left shift”).

Original source details: GELFAND-P000468

Note that $G\Theta A = \Theta A$ as a set (they are nilpotent)



Uncertain reading. The last letters are overwritten. The word reads “nilpotent.”

Original source details: GELFAND-P000468

GELFAND-P000469



Undeciphered passage. A crossed-out mark before the first line with a short stroke to its right. The reading is unresolved.

Original source details: GELFAND-P000469

In fact

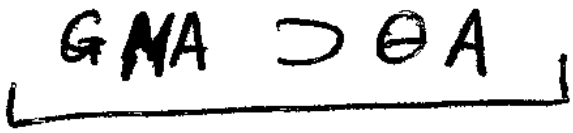


Diagram and typed description. The line under the inclusion turns up at both ends.

$\llcorner GMA \supset \Theta A \lrcorner$

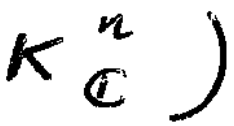
Original source details: GELFAND-P000469

Mark the vertex 0 — the identity of the group GMA.

SA — the corresponding simplicial scheme.

(Historical remark: Volodin used SA to define the functors $K^n_{\mathbb{C}}$)

in essence



Uncertain reading. The subscript reads $\mathbb{C}$, with an extra stroke inside the letter.

Original source details: GELFAND-P000469

Suppose the ring A has a unit. Then put.

$$K_n A = \pi_{n-1} SA, \qquad n \geqslant 1.$$

For $n \geqslant 2$ there is a group structure, commutative for $n \geqslant 3$.

Another def. of the group structure (commutative).

Whitehead's lemma

Consider the spheroid $s\colon Z^n \longrightarrow SA$. The group of permutations acts on SA $f\colon N \longrightarrow N$. We obtain a new spheroid s^f:

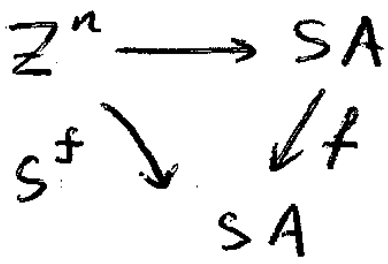
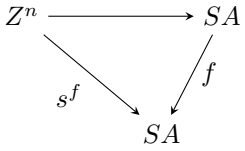


Diagram and typed description. The top arrow is unlabelled.



Original source details: GELFAND-P000469

They are always homotopic.

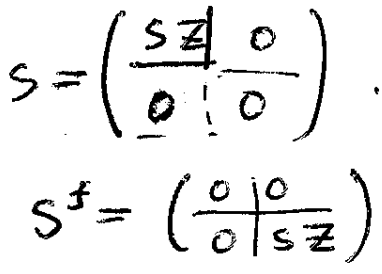
Idea of the proof:

Im $s \subset M_k A$.

$$s = \left(\begin{array}{c|c} sz & 0 \\ \hline 0 & 0 \end{array}\right).$$

Everything reduces to the case

$$s^f = \left(\begin{array}{c|c} 0 & 0 \\ \hline 0 & sz \end{array}\right).$$

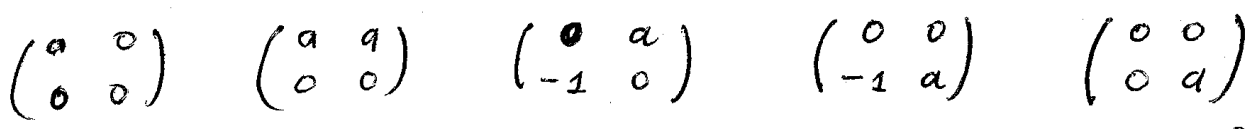


Uncertain reading. The final sign in sz has a cross-stroke and is provisionally set as a lowercase letter. Both matrices have two dividing rules.

Original source details: GELFAND-P000469

(f of a special form: $fi = i + k$, we adjust it at ∞).

$$\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} a & a \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ -1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ -1 & a \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & a \end{pmatrix}$$



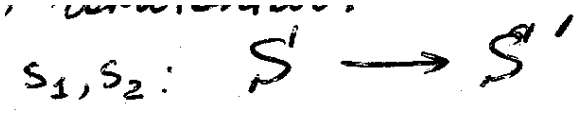
Original source details: GELFAND-P000469

Uncertain reading. The upper left zero in the third matrix is heavily retraced.

These are 5 spheroids. This is a homotopy. Each 2 neighbours are elementarily homotopic:

$$s_1, s_2 \colon S' \longrightarrow S'$$

are called elem. homotopic if $\forall$ simplices $X \subset S$, $s_1 X \cup s_2 X$ is a simplex in S'.



Uncertain reading. A short stroke is visible above the first S. The next line has $X \subset S$ without a prime.

Original source details: GELFAND-P000469

Let all a lie in one simplex X. Then $\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} a & a \\ 0 & 0 \end{pmatrix}$ lie in one simplex.

Let $a_0^{-1} \in X$. Multiply by a_0. Then they become upper triang. If all a are upper triangular, then all 5 matrices lie in one simplex.

———— · ————

The commutative group structure on $K_n A$.

$$MA \times MA \xrightarrow{\oplus} MA$$

$fi = 2i$	$f, g \colon \mathbb{N} \longrightarrow \mathbb{N}$	noncommut., nonassoc., with no
$gi = 2i + 1$	$a \oplus b = a^f + b^g$	identity elem.

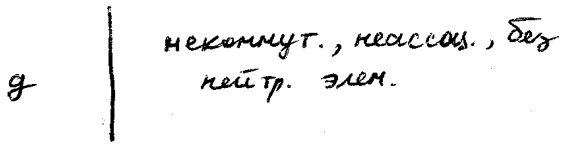


Diagram and typed description. A vertical rule separates the note about the operation.

Original source details: GELFAND-P000469

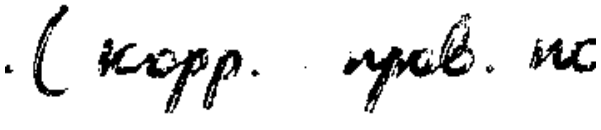


Uncertain reading. Outside the left edge of the text are a slanting stroke and a mark resembling a closing parenthesis. Their purpose is unresolved.

Original source details: GELFAND-P000469

GELFAND-P000470

it induces the map $SA \times SA \longrightarrow SA$, which gives operations on all K_i, $i \geqslant 0$. (corr. verif. by Whitehead's lemma)



Uncertain reading. The second abbreviation after «корр.» is provisionally read as «прв.».

Original source details: GELFAND-P000470

They are commutative and agree with those introduced earlier by mutual distributivity, which is verified by Whitehead's lemma.

One can define a sum — H-space. On $\widetilde{\Omega}A$ one can even add an inf. number of matrices! Hence it will have trivial K-functors.



Original source details: GELFAND-P000470

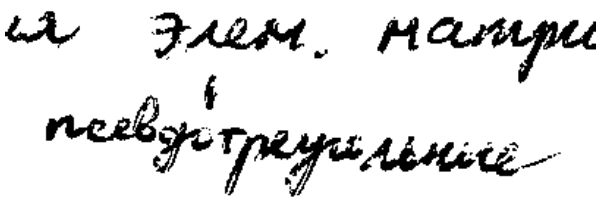
Uncertain reading. A long wavy mark is written over Ω. It is set as $\widetilde{\Omega}$.

$K_1 A = \pi_0 SA$.

This is the set of connected components in GMA.

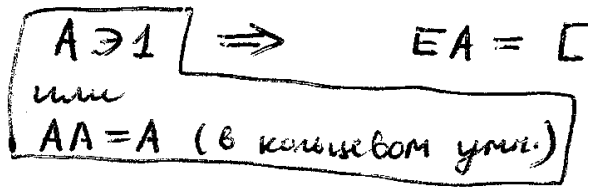
EA — the connected component of the identity. (products of elem. matrices),

pseudotriangular



Original source details: GELFAND-P000470

Diagram and typed description. "Pseudotriangular" is written under "elem." and connected to it by a stroke.



Original source details: GELFAND-P000470

Diagram and typed description. The stepped outline encloses both hypotheses. The arrow starts level with the first one.

$A \ni 1$ $\Longrightarrow$

or

$AA = A$ (in ring mult.)

$$EA = [GMA, GMA] = [EA, EA].$$

$K_2 A$ — homot. classes of loops. Loops — relations of elem. matrices.

ΘA has generators a^{ij}



Original source details: GELFAND-P000470

Undeciphered passage. The entry after a^{ij} is heavily crossed out. A short leftward arrow is visible below "Loops".

(at position i, j stands $a \in A$).

Relations: Steinberg

$$a^{ij} * b^{ij} = (a+b)^{ij},$$
$$[a^{ij}, b^{jk}] = (ab)^{ik}$$

and for some the commutator is 0.

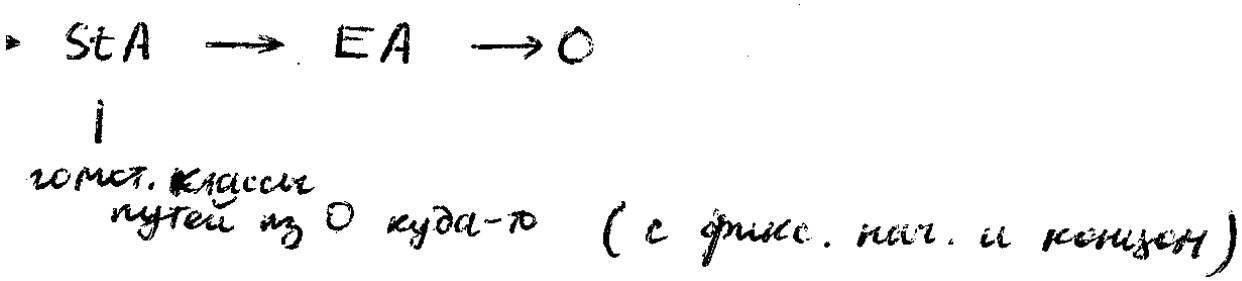
These are the defining relations.

Let StA — be the group gen. by these generators and relations.

$$0 \longrightarrow K_2 A \longrightarrow StA \longrightarrow EA \longrightarrow 0$$

$$|$$

homot. classes of paths from 0 to somewhere
(with fixed start and end)



Original source details: GELFAND-P000470

Diagram and typed description. The annotation is connected by a vertical stroke to StA.



Uncertain reading. There is a single retraced mark to the left of the formula. Its reading is unresolved.

Original source details: GELFAND-P000470

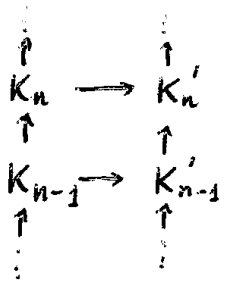
$$K_2A = \underbrace{H_2EA}_{\text{(group homology)}} \quad \sim \quad StA = \text{the universal central extension of } EA.$$

Problem: suppose there are 3 matrices in GMA. All segments are simplices. Then all 3 form a simplex.

Generalization: if any pair is a simplex, then all form a simplex. (space of tolerance)

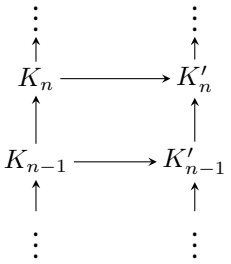
D. A. Kazhdan. Derived category.

C — an abelian category. Consider the category of complexes K_C.



Original source details: GELFAND-P000470

Diagram and typed description. The arrows point up in both columns. Two arrows point right between the columns.





Original source details: GELFAND-P000470

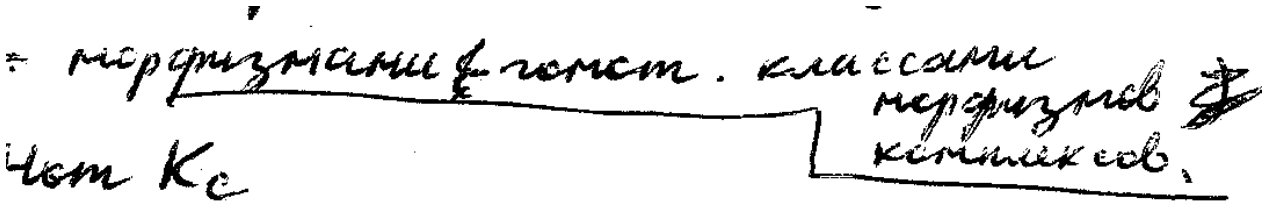
Uncertain reading. A fragment before “with morphisms” is struck out. A possible reading is «с сост. об.».

with morphisms f the homot. classes of morphisms of complexes.



Original source details: GELFAND-P000470

Uncertain reading. A single mark after “morphisms” is struck out. Its reading is unresolved.



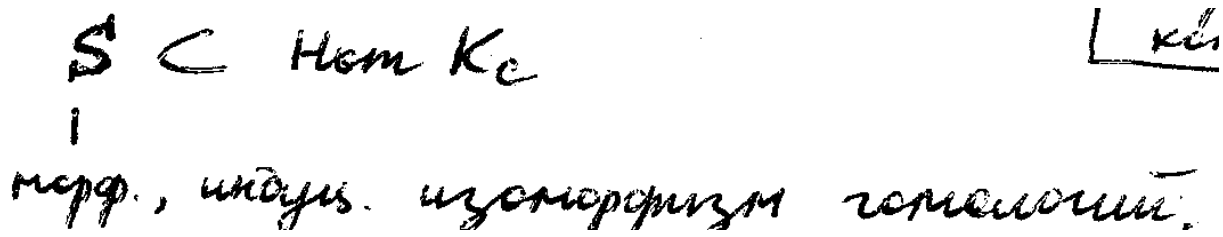
Original source details: GELFAND-P000470

Diagram and typed description. A continuous stepped underline runs beneath the two lines.

$S \subset \operatorname{Hom} K_C$

|

morph. inducing an isomorphism of homologies

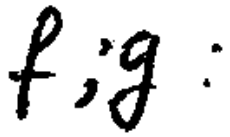


Original source details: GELFAND-P000470

Diagram and typed description. The annotation is connected by a vertical stroke to S.

GELFAND-P000471

Lemma. Let $f; g\colon K \longrightarrow K'$ be two morphisms.

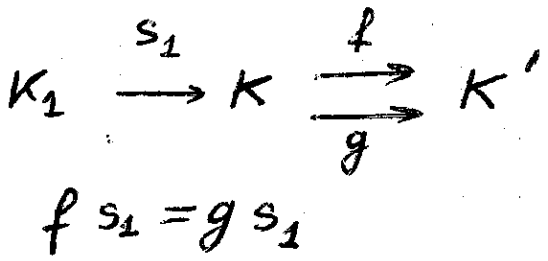


Original source details: GELFAND-P000471

Uncertain reading. Two strokes appear between f and g, here set as a semicolon.

The following statements are equivalent:

1°. $\exists K_1, s_1 \in S$:



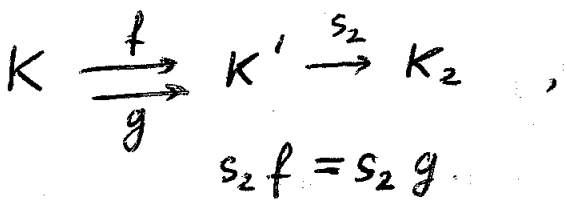
Original source details: GELFAND-P000471

Diagram and typed description. The first diagram with the equality written below it.

$$K_1 \xrightarrow{s_1} K \underset{g}{\overset{f}{\rightrightarrows}} K'$$

$$fs_1 = gs_1$$

2°. $\exists K_2, s_2 \in S$



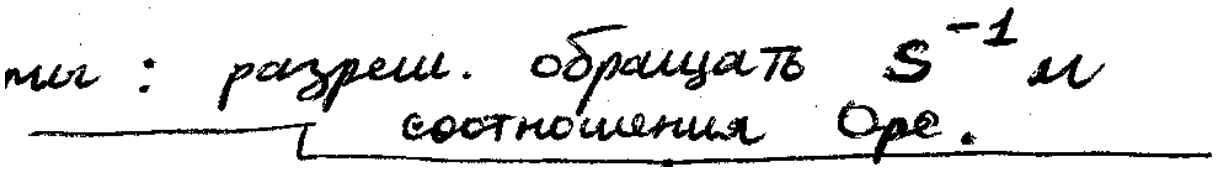
Original source details: GELFAND-P000471

Diagram and typed description. The second diagram with the equality written below it.

$$K \underset{g}{\overset{f}{\rightrightarrows}} K' \xrightarrow{s_2} K_2 \ ,$$

$$s_2 f = s_2 g.$$

L_C — the same objects, and the morphisms: allow inverting s^{-1} and Ore relations.



Original source details: GELFAND-P000471

Diagram and typed description. “Ore relations” is written below the first line, with a stepped rule.

The lemma makes it poss. to define the product correctly.



Original source details: GELFAND-P000471

Uncertain reading. The manuscript has «возможно» immediately before «корректно».

See Hartshorne. Residues and duality.



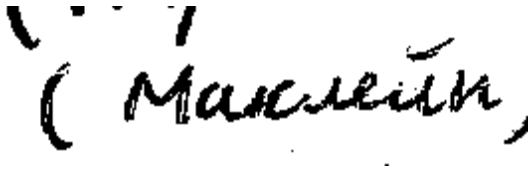
Uncertain reading. A short cross-shaped mark appears above «См.» (“See”).

Original source details: GELFAND-P000471

V. Buchstaber.

References:

	C.–E.	213–223	(Cartan and Eilenberg)
	M.	358–370	(Mac Lane)
	Adams	Mat. 3 : 4	pp. 7–19.



Uncertain reading. The parenthesis reads «Маклейн» (“Mac Lane”).

Original source details: GELFAND-P000471

$$A = \sum_{n \geqslant 0} A_n \qquad \text{— grad. ring.}$$

A_n — an A_0-module for every $n \geqslant 0$.

$$0 \longleftarrow A_0 \xleftarrow{\varepsilon} A \longleftarrow C_1 \longleftarrow C_2 \longleftarrow \cdots \longleftarrow C_s \xleftarrow{d_{s+1}} C_{s+1} \longleftarrow \cdots$$

(ε assigns A zeroth comp. to each $a \in A$).

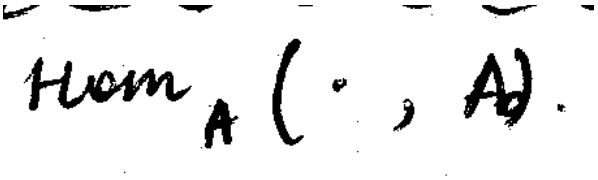


Original source details: GELFAND-P000471

Uncertain reading. A separate A is written immediately after «сопост.» (“assigns”).

Complete to a grad. free resolution.

The functor $\mathrm{Hom}_A(\cdot, A_0)$. takes d_s to $(d_s)^*$.



Uncertain reading. A small zero is visible below and to the right of the second argument A, against the parenthesis.

Original source details: GELFAND-P000471

$$H^s(A) = \mathrm{Ker}(d_{s+1})^* / \mathrm{Im}(d_s)^*$$

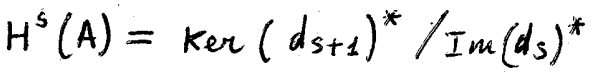


Diagram and typed description. The numerator has d_{s+1} and the denominator d_s.

Original source details: GELFAND-P000471

Theorem: 1) A free resolution exists

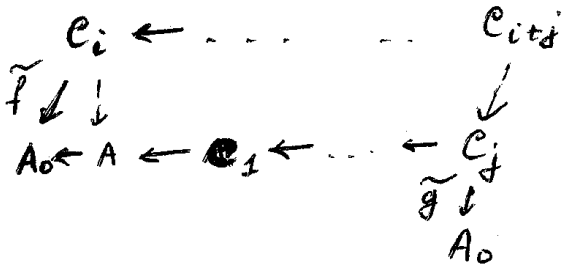
2) The groups $H^s(A)$ are independent of the resolution.

$$H(A) = \sum_{s=0}^{\infty} H^s(A) \qquad \text{— a graded ring.}$$

A def. will be given for the multiplication of Yoneda.

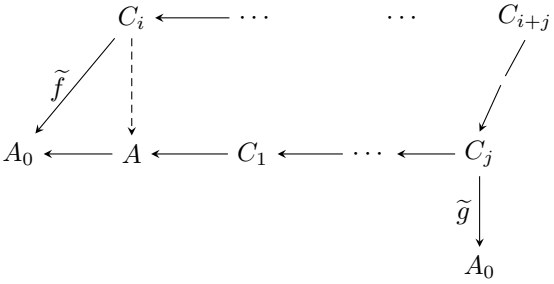
$$f \in H^i(A), \quad g \in H^j(A), \quad fg \in H^{i+j}(A),$$
$$\widetilde{f}\colon C_i \longrightarrow A_0; \qquad \widetilde{g}\colon C_j \longrightarrow A_0.$$

GELFAND-P000472



Original source details: GELFAND-P000472

Diagram and typed description. The node C_1 is heavily retraced. The arrow from C_i to A is dashed, and the arrow from C_{i+j} to C_j has a single break.



We have obtained a homomorphism $C_{i+j} \longrightarrow A_0$.



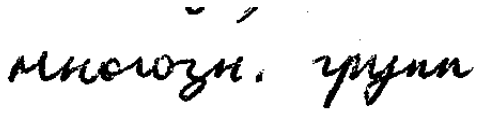
Original source details: GELFAND-P000472

Uncertain reading. The first word is read as «Получили» (“we have obtained”). Its last two letters run together.

This gives

$$A \underset{\text{assigns}}{\rightsquigarrow} H(A) \qquad \text{(a ring)}$$

$H(A)$ is generally noncommutative (!) in a genuine sense. The classification of multivalued groups rests on this.



Original source details: GELFAND-P000472

Uncertain reading. The manuscript abbreviates “multivalued” as «многозн.».

(So far in topology only commut. rings have occurred — there one always has Hopf algebras).

Standard resolution or B-resolution.

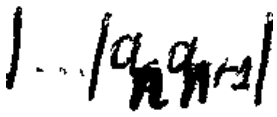
$$\overline{A} = \sum_{n>0} A_n$$

$$C_1 = A \underset{A_0}{\otimes} \overline{A} \qquad \text{Henceforth } A_0 \text{ is omitted}$$

(The starting rings are commutative!)

$$C_s = A \otimes (\overline{A})^s.$$
$$d_{s+1} \colon A \otimes (\overline{A})^{s+1} \longrightarrow A \otimes (\overline{A})^s$$

$$d_s\big(a_0[a_1|a_2|\dots|a_s]\big) = (-1)^{\varepsilon(0)} a_0 a_1 [a_2|\dots a_s] + \sum (-1)^{\varepsilon(n)} a_0 [a_1|\dots|a_n a_{n+1}|\dots a_s]$$



Uncertain reading. Two indices in the product are retraced. The reading is $a_n a_{n+1}$.

Original source details: GELFAND-P000472

$\varepsilon(n)$ — certain functions of the index, ~~For even grading $\varepsilon(n) = n$.~~

$\varepsilon(n) = n$, if all elems. are even.

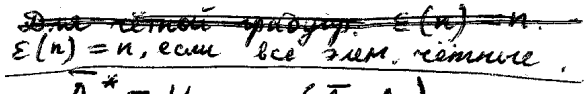


Diagram and typed description. The condition on even elements is added below the deleted line and set off by a rule.

Original source details: GELFAND-P000472

$$\mathrm{Hom}_A(C_s, A_0) \simeq (\overline{A}^*)^s, \qquad \overline{A}^* = \mathrm{Hom}_{A_0}(\overline{A}, A_0)$$
$$d_s^* \colon [\alpha_1|\alpha_2|\dots|\alpha_{s-1}] = \sum (-1)^{\varepsilon(n)} [\alpha_1|\dots|\overline{\Delta}\alpha_n|\dots|\alpha_{s-1}].$$
$$\varepsilon(n) = n, \quad \text{if } \deg \alpha_i = 2k_i$$

Def. $\overline{\Delta}$.

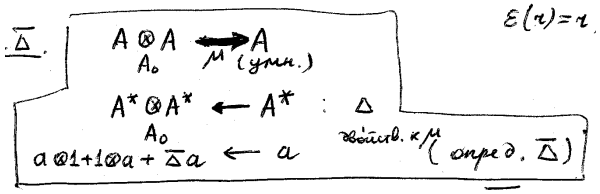
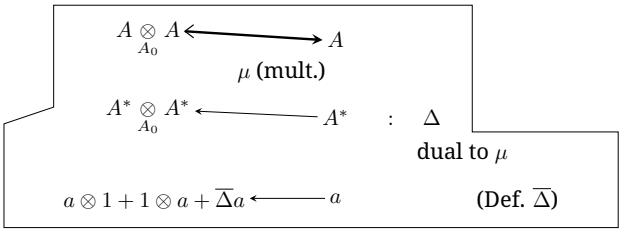


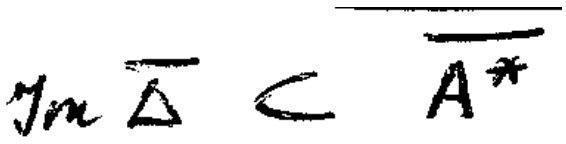
Diagram and typed description. The top arrow has a clear right head. Its retraced left end also resembles a head.



Original source details: GELFAND-P000472

In fact

$$\operatorname{Im}\overline{\Delta} \subset \overline{A^*}.$$

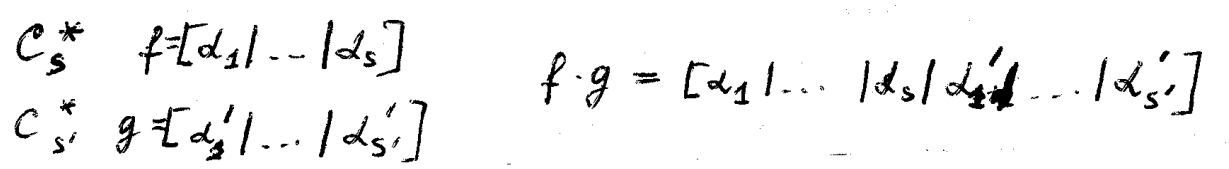


Uncertain reading. The bar on the right extends over both A and the star.

Original source details: GELFAND-P000472

This gives another definition of multiplication,

$$\begin{array}{ll} C_s^* & f = [\alpha_1|\dots|\alpha_s] \\ C_{s'}^* & g = [\alpha'_1|\dots|\alpha'_{s'}] \end{array} \qquad f \cdot g = [\alpha_1|\dots|\alpha_s|\alpha'_1|\dots|\alpha'_{s'}]$$



Original source details: GELFAND-P000472

Uncertain reading. The last index in g reads s'. The separator after the first primed term in the product is heavily retraced.

GELFAND-P000473

20/XI-72. Gelfand sem..

Kostyuchenko. One example.

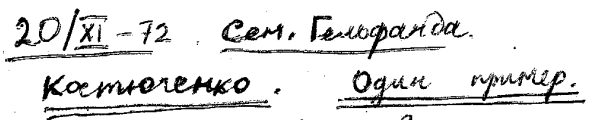


Diagram and typed description. The month is given as a Roman numeral. The name and title are double-underlined.

Original source details: GELFAND-P000473

$$\begin{aligned} v'' + 2a\lambda v' + b\lambda^2 v &= 0 \\ v(0) = v(\pi) &= 0. \end{aligned} \qquad \text{in } L_2[0,\pi].$$

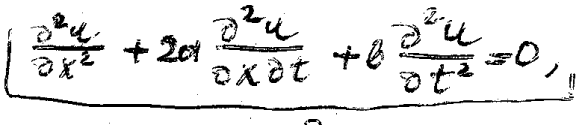


Diagram and typed description. A line under the equation turns up at both ends.

$$\frac{\partial^2 u}{\partial x^2} + 2a\frac{\partial^2 u}{\partial x \partial t} + b\frac{\partial^2 u}{\partial t^2} = 0,$$

Original source details: GELFAND-P000473

$$0 < x < \pi$$
$$0 \leqslant t < +\infty$$



Uncertain reading. The second inequality sign is retraced. The provisional reading is $0 \leqslant t < +\infty$.

Original source details: GELFAND-P000473

Suppose the eq. is elliptic.



Uncertain reading. The first word reads «Пусть» (“Suppose”).

Original source details: GELFAND-P000473

$$a^2 < b; \qquad b > 0.$$

$$v_k(x) = e^{\gamma kx} \sin kx, \qquad \gamma = \frac{-a}{\sqrt{b-a^2}}; \qquad k = \pm 1,\ \pm 2,\ \dots.$$

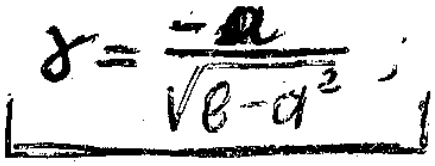
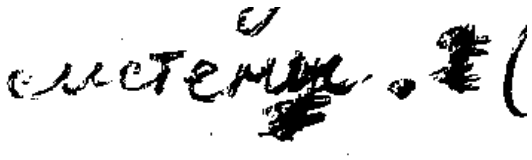


Diagram and typed description. A rule with upturned ends is drawn below the formula for γ.

Original source details: GELFAND-P000473

1. The system is complete, but it is unknown how to find the coefficients. This is the question of finding the biorthog. system.



Uncertain reading. The end of “system” is retraced. A separate mark is crossed out after the period.

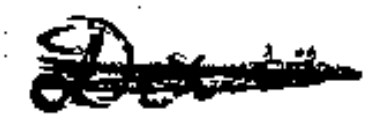
Original source details: GELFAND-P000473

(found in terms of the inverse fn. of the Christoffel–Schwarz integral)

(a biorthog. system exists)

2. This is not a basis! (not every vector expands into a conv. series in this system)

~~Even~~ Moreover: if there is an expansion in L^2, then the limit — an analytic fn.! — and satisfies inf. many boundary conditions (the converse holds too).



Uncertain reading. The crossed-out word may read «Даже» (“Even”).

Original source details: GELFAND-P000473

It cannot be summed by the Abel method of the first order.

if

$$\sum_{\lambda_k > 0} e_k e^{\lambda_k t} v_k(x)$$



Uncertain reading. The coefficient is heavily retraced. The provisional reading is e_k.

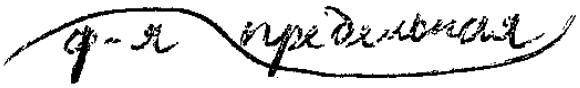
Original source details: GELFAND-P000473



Original source details: GELFAND-P000473

Undeciphered passage. Two short entries after $v_k(x)$ are heavily struck out. Their readings are unresolved.

has a limit as $t \to 0$, then the limiting fn. is analytic.



Original source details: GELFAND-P000473

Diagram and typed description. A wavy stroke runs above “fn.” and below “limiting”.

(Discrediting the Fourier method for the elliptic eq.)

With certain nonseparated conditions one can solve it.



Original source details: GELFAND-P000473

Uncertain reading. The adjective before “conditions” is abbreviated as «неразд.».



Original source details: GELFAND-P000473

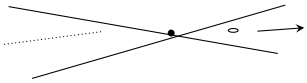
Undeciphered passage. Two marks before “Reason” are crossed out. Their readings are unresolved.

Reason for the poor behav.: the resolvent grows exp in the angles



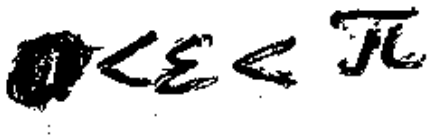
Original source details: GELFAND-P000473

Diagram and typed description. Two intersecting lines, broken strokes on the left, a dot at their crossing, and a circle and arrow on the right.



On $[0, \pi - \varepsilon]$ it is overdetermined.

On $[0, \pi + \varepsilon]$, $0 < \varepsilon < \pi$ it remains complete !!! (I. Bernstein)



Original source details: GELFAND-P000473

Uncertain reading. The lower bound is heavily retraced and is provisionally set as zero.

On $[0, 2\pi]$ many orthog. fns. arise.

A problem in the theory of entire fns..

$$\int_0^\pi f(x) e^{\gamma k x} \sin kx \, dx = 0 \quad \Longrightarrow \quad F(\lambda) = \int_0^\pi f(x) e^{\gamma \lambda x} \sin \lambda x \, dx$$

$F(\lambda)$ is an entire fn. of exp type. It is 0 at integer points $F(n) = 0$, $n > 0$.

The specific features must be taken into account.

On $[0, \pi - \varepsilon]$ the type and indicator diagram work.

Uniqueness — ?

It is unknown what happens if one replaces k by λ_k.

GELFAND-P000474

I. M. Gelfand on Arnold's work.



Original source details: GELFAND-P000474

Diagram and typed description. The name and topic are double-underlined.

$$(\mathbb{R}^n, 0) \longrightarrow (\mathbb{R}, 0)$$

$f(x) = 0$ — hypersurf.

How does the hypersurf. behave in a neighbourhood of 0 — ?

$f \in C^\infty$; $f(0) = 0$, $df(0) = 0$. (crit. point).

(In fact consider $\mathbb{C}^n \longrightarrow \mathbb{C}$)

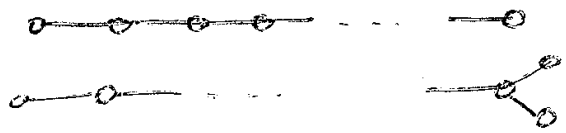
Equiv. hypersurfaces — differ by a change of variables

$$(\mathbb{R}^n, 0) \longrightarrow (\mathbb{R}^n, 0) \qquad \text{(diffeom.)}.$$

Simple hypersurf.: $f(x) = 0$

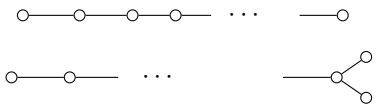
if one perturbs it, there will be finitely many orbits in jets of order N (the bound on the number of orbits does not depend on the index N).

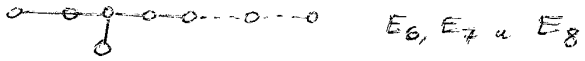
Enumeration of the simple orbits:



Original source details: GELFAND-P000474

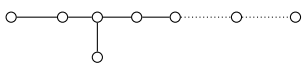
Diagram and typed description. Two chains are drawn at left. The lower chain forks at its end.





Original source details: GELFAND-P000474

Diagram and typed description. The right-hand chain has a downward branch at the third vertex. Its last links are broken.



E_6, E_7 and E_8

Next: Most orbits are simple (as a rule, the orbit is simple).



Original source details: GELFAND-P000474

Undeciphered passage. A short fragment after the opening parenthesis is struck out. Its reading is unresolved.

Nonsimple ones of codimension 6.

$$x_1^{n+1} + x_2^2 + \cdots * = 0$$

— corr. to the first type.



Uncertain reading. The first letter of the exponent $n+1$ is retraced. A separate star-like mark follows the ellipsis. A rule with upturned ends is drawn below the formula.

Original source details: GELFAND-P000474

Versal deformation.

$$F(x,\lambda) = 0, \qquad F(x,0) = f(x),$$
$$\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$$

Every other def. is equiv. to a def. induced from this one as follows:

$$G(x,\mu) = F(X(x,\mu), \Psi(\mu)) \quad \text{and} \quad X(y,0) = y, \quad \Psi(0) = 0$$



Original source details: GELFAND-P000474

Uncertain reading. The first arguments of G and X are heavily retraced and provisionally read as x. The right side clearly has $X(y,0) = y$.

Rule for constructing a vers. family: Take an orbit and a set transverse to the orbit.

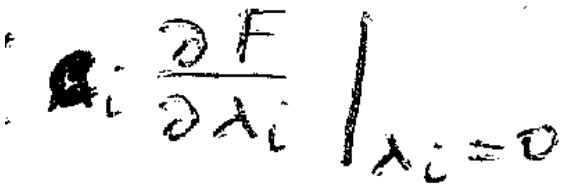


Diagram and typed description. A curved line and a short transverse segment with a point at their intersection.



Original source details: GELFAND-P000474

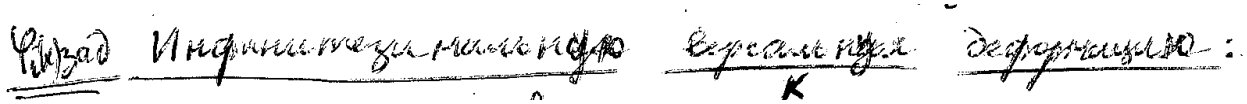
$$F(x,\lambda) = f(x) + \sum a_i \left.\frac{\partial F}{\partial \lambda_i}\right|_{\lambda_i=0} = f(x) + \sum c_i \varphi_i(x) + O(\lambda^2).$$



Uncertain reading. The multiplier in the first sum is provisionally read as a_i. The evaluation subscript is $\lambda_i = 0$.

Original source details: GELFAND-P000474

φ_i give an infinitesimal versal deformation: if $\forall$ fn. $a(x)$ is repr. in the form



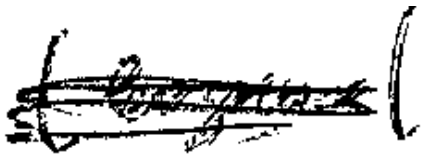
Original source details: GELFAND-P000474

Uncertain reading. The first fragment is heavily retraced and double-underlined. The provisional reading is φ_i «зад» (“give”).

$$a(x) = \sum \frac{\partial f}{\partial x_i} \cdot h_i(x) + \sum_{i=1}^{k} c_i \varphi_i(x)$$

$a(x)$ defines an infinitesim. vers. deform.

(φ_i gen. the local ring of smooth maps)



Undeciphered passage. A separate crossed-out entry precedes the parenthesis about Taylor expansion. Its reading is unresolved.

Original source details: GELFAND-P000474

(from the Taylor exp. of that fn.)

Important: the quotient ring by the Jacobian ideal $\left(\left\{\frac{\partial f}{\partial x_i}\right\}\right)$.

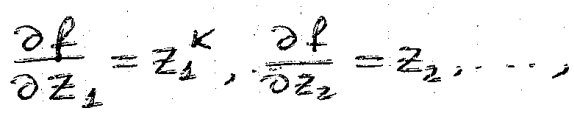
The quotient — a finite-dimensional space of dimension k.

GELFAND-P000475

$$f = z_1^{k+1} + z_2^2 + \cdots + z_n^2.$$

Then

$$\frac{\partial f}{\partial z_1} = z_1^k, \qquad \frac{\partial f}{\partial z_2} = z_2, \ \ldots,$$

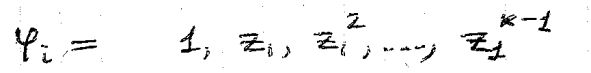


Uncertain reading. The right-hand sides are written as monomials without numerical factors.

Original source details: GELFAND-P000475

hence the quotient ring is a space of dimension k.

$$\varphi_i = 1, z_i, z_i^2, \ldots, z_1^{k-1}$$

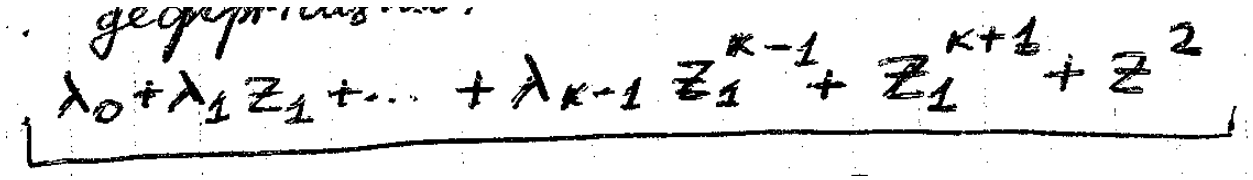


Uncertain reading. The first two powers after 1 have subscripts resembling i. The last has 1.

Original source details: GELFAND-P000475

Vers. deformation:

$$\underline{\lambda_0 + \lambda_1 z_1 + \cdots + \lambda_{k-1} z_1^{k-1} + z_1^{k+1} + z^2} \quad \text{— vers. deformation}$$



Original source details: GELFAND-P000475

Diagram and typed description. A rule with upturned ends is drawn below the polynomial. The final term is written z^2.

$\underline{F(z,\lambda) = 0.}$ The bifurc. diagram — the set of those λ, for which sing. points remain (in the vers. deformation).

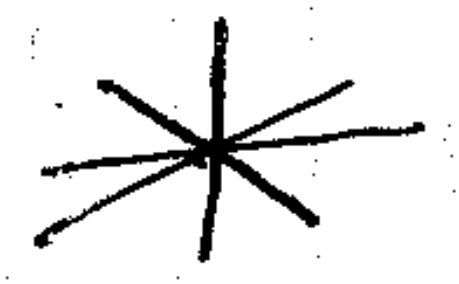
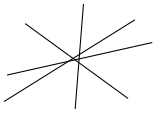


Diagram and typed description. Four segments meet at a common point to the left of the theorem.



Original source details: GELFAND-P000475

Thm.: on an open set (bifurc. diag.) the sing. points will already be simple (Morse).

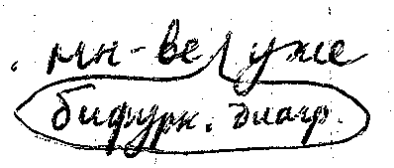


Diagram and typed description. "Bifurc. diag." is written in a bubble below "set".

Original source details: GELFAND-P000475

$z_1^{k+1} + z_2^2 = 0 \quad \| \quad \longrightarrow$ the bifurc. diag. of this point — Swallowtail.

$$z_1^{k+1} + z_2^2 + \lambda_{k-1} z_1^{k-1} + \cdots + \lambda_0$$

$$z_2^2 = -p_{k+1}(z_1); \qquad z_2 = \sqrt{-p_{k+1}(z_1)}.$$

sing. points $(z_2, P'(z_1)) = 0$, i.e.

$$\underline{P(z_1) = 0 \text{ and } P'(z_1) = 0}$$

(P has multiple roots)

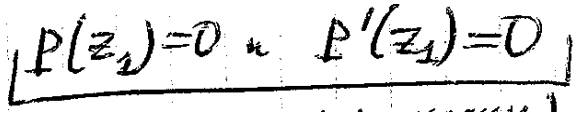


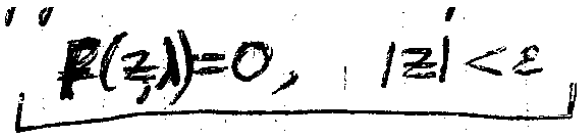
Diagram and typed description. A single rule with upturned ends surrounds the two equalities below.

Original source details: GELFAND-P000475

Consider the complement of the bifurc. diagram and its fund. group. It is the braid group of the Coxeter group.

Consider the eq.

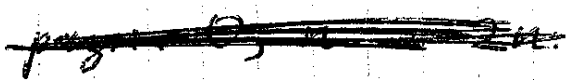
$$P(z,\lambda)=0, \quad |z|<\varepsilon \qquad \lambda \text{ such that there are no singularities}$$



Original source details: GELFAND-P000475

Uncertain reading. The first letter is heavily retraced. The provisional reading is P.

the homology of this space is $\neq 0$ only in ~~dim.~~



Original source details: GELFAND-P000475

Undeciphered passage. Traces remain after the struck-out "dim.". A full reading is unresolved.

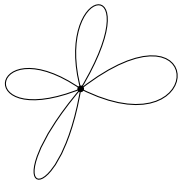
half-dimension $n-1$.

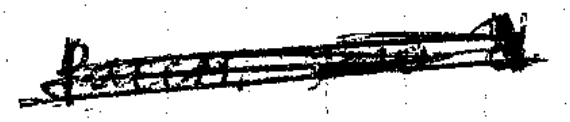
It is a bouquet of μ spheres.



Original source details: GELFAND-P000475

Diagram and typed description. Four loops with a common point depict the bouquet.





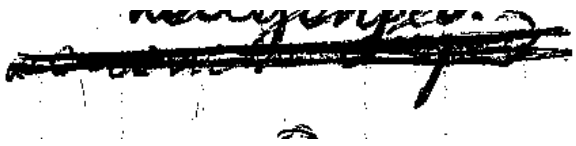
Original source details: GELFAND-P000475

Undeciphered passage. The entry to the left of H^{n-1} is heavily struck out. Its reading is unresolved.

H^{n-1} has generators $e_1\ e_2 \dots\ e_\mu$.

The intersection index defines a scalar prod. (e_i, e_k).

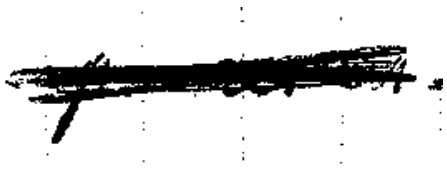
$$n-1 \equiv 2 \pmod 4 \qquad \| \qquad \begin{matrix}\text{neg. def.}\\ \text{semidef.}\\ \text{indef.}\end{matrix} \qquad e_i^2=-2 \quad \text{(always)}$$



Original source details: GELFAND-P000475

Undeciphered passage. One line between "semidef." and "indef." is crossed out. Its reading is unresolved.

monodromy: rep. of the fund. gr. in



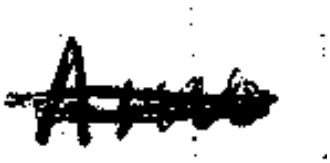
Original source details: GELFAND-P000475

Undeciphered passage. An entry after “in” is crossed out and followed by an ellipsis. Its reading is unresolved.

...

———— • ————

~~Atiyah~~



Original source details: GELFAND-P000475

Uncertain reading. The struck-out name may read «Атья» (“Atiyah”).

There are works by Atiyah, Bott,...: how to get rid of derivatives in Singer’s calculations? How can one study the topology of the world while sitting at one point? Finding explicit asymptotics for eqs. of the 2nd order (considerations of I. M. Gelfand, V. I. Arnold and others)

(will be in the next semester)

A. Shnirelman on Maslov’s work.

GELFAND-P000476

V. M. Buchstaber. (continuation).

$$A \overset{\text{addit.}}{=} \Lambda[[X]], \qquad X_n \text{ — addit. generators}$$
$$X_n \cdot X_m = C^n_{n+m} X_{n+m}$$



Original source details: GELFAND-P000476

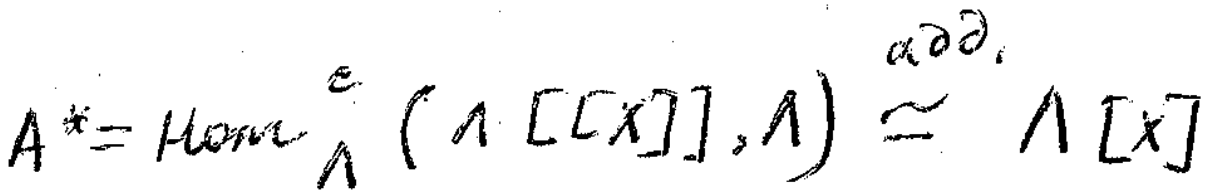
Uncertain reading. Two short parallel strokes under “addit.” read as an equals sign.

$$\begin{array}{c}\mathrm{Ext}_A^{2,*}(\Lambda,\Lambda)\\ \|\\ \mathrm{Ker}\, d_3^*/\,\mathrm{Im}\, d_2^*\end{array}$$ in terms of the B-construction

1°.

$$d_3^*\colon \overline{A}^*\otimes\overline{A}^*\longrightarrow \overline{A}^*\otimes\overline{A}^*\otimes\overline{A}^*$$

$$A^*=\mathrm{Hom}_\Lambda^c(\Lambda[[X]],\Lambda)\overset{\text{add.}}{\simeq}\Lambda[x]$$



Original source details: GELFAND-P000476

Uncertain reading. The upper index of Hom reads c and the lower one Λ.

compact (nonzero val. on a finite number).

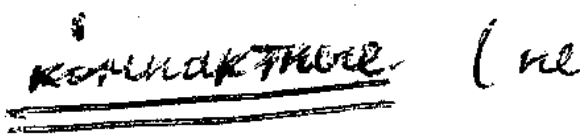
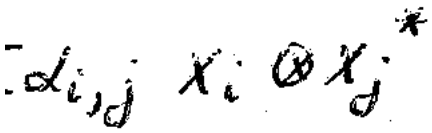


Diagram and typed description. "Compact" is written below Hom and connected to it by a stroke.

Original source details: GELFAND-P000476

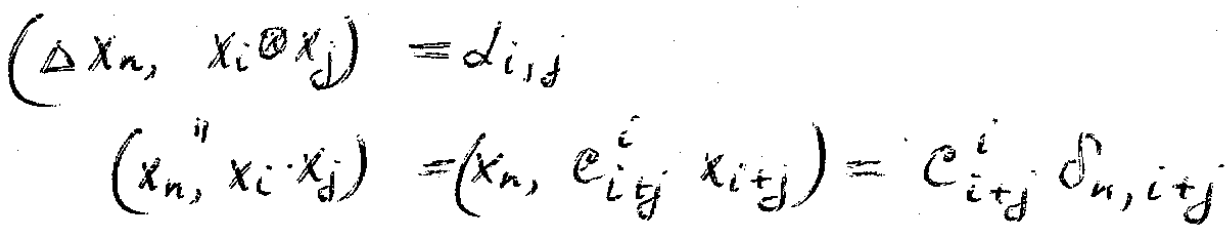
$$\Delta\colon A^*\longrightarrow A^*\otimes A^*$$
$$\Delta x_n\longrightarrow x_n\otimes 1+1\otimes x_n+\sum d_{i,j}x_i\otimes x_j^*$$



Uncertain reading. A star is visible after x_j. It is absent from the later formula for Δx_n.

Original source details: GELFAND-P000476

$$\begin{array}{c}(\Delta x_n, x_i\otimes x_j)=d_{i,j}\\ \|\\ (x_n, x_i\cdot x_j)=(x_n, C_{i+j}^i x_{i+j})=C_{i+j}^i\delta_{n,i+j}\end{array}$$



Original source details: GELFAND-P000476

Diagram and typed description. A vertical equals sign joins the two lines of the pairing.

$$\Delta x_n = x_n \otimes 1 + 1 \otimes x_n + \sum C_n^i x_i \otimes x_{n-i}$$

And put $x_n = x^n$

$$\Delta x^n = (x \otimes 1 + 1 \otimes x)^n$$
$$d_3^*\colon \overline{\Lambda[x,y]} \longrightarrow \overline{\Lambda[x,y,z]}$$

$$d_3\Bigg(\begin{array}{c} h(x,y) \\ \| \\ \sum h_{i,j}x^i y^j \end{array}\Bigg) = \sum h_{ij}(\overline{\Delta}x^i)z^j - \sum h_{i,j}x^i\overline{\Delta}y^j =$$



Uncertain reading. The calculation has d_3 without a star before $h(x,y)$. The preceding line has d_3^*.

Original source details: GELFAND-P000476

$$\overline{\Delta}x^n = \begin{array}{c} \Delta x^n \\ \| \\ (\Delta(x))^n \\ \| \\ (x+y)^n \end{array} - x^n - y^n.$$

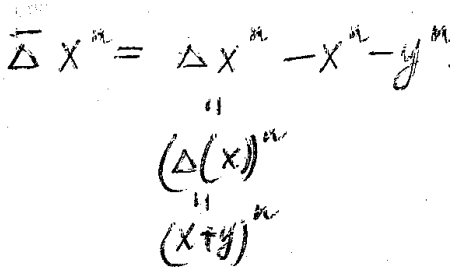
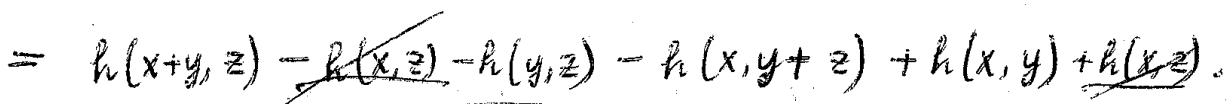


Diagram and typed description. Below Δx^n, vertical equals signs connect $(\Delta(x))^n$ and $(x+y)^n$.

Original source details: GELFAND-P000476

$$= h(x+y,z) - \cancel{h(x,z)} - h(y,z) - h(x,y+z) + h(x,y) + \cancel{h(x,z)}.$$



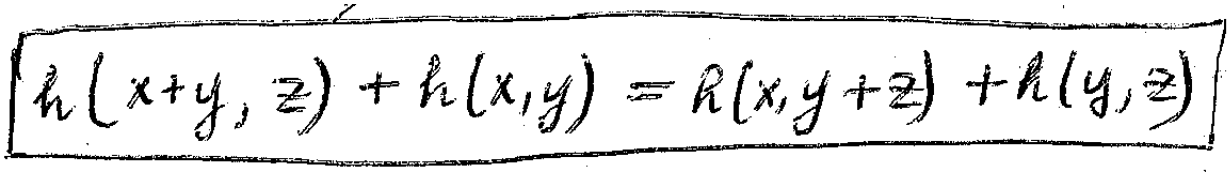
Original source details: GELFAND-P000476

Diagram and typed description. Both instances of $h(x,z)$ are crossed out. The minus and plus signs in front remain visible.

Ker d_3^*:

$$\boxed{h(x+y,z) + h(x,y) = h(x,y+z) + h(y,z)}$$

Denote these cycles by L.



Original source details: GELFAND-P000476

Diagram and typed description. The equality is enclosed in a rectangular box.

This is precisely the Lazard relation.



Original source details: GELFAND-P000476

Uncertain reading. The adjective before “relation” reads «лазаровское» (“Lazard”).

$$\underline{\operatorname{Im} d_2^*} = \Lambda\{d_2^* x^n\}$$

$$0 \longrightarrow \operatorname{Im} d_2^* \longrightarrow L \longrightarrow \operatorname{Ext}_A^{2,n}(\Lambda, \Lambda) \longrightarrow 0$$

Calculation of the group Ext.

The eq. does not involve Λ. Hence it suffices to calc. 2 cases:

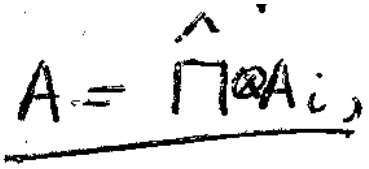
1) Λ — Z_p-module (p — prime)

2) $\Lambda = Q$

GELFAND-P000477

there remains X_{p^i}

$$\underline{A = \widehat{\prod} \otimes A_i}, \qquad A_i = Z_p[X_{p^i}]/(X_{p^i}^p) \underset{Z_p}{\otimes} \Lambda$$



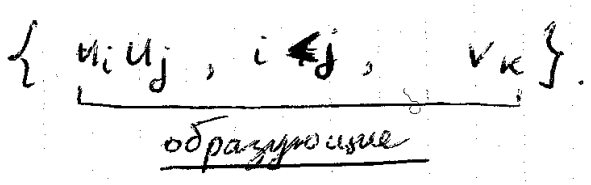
Original source details: GELFAND-P000477

Uncertain reading. The product sign has a hat. A tensor sign is written immediately to its right.

$$H^*(A_i) = \Lambda_p(u_i) \otimes Z_p[v_i], \qquad \deg u_i = p^i, \quad \deg v_i = p^{i+1}$$

$$u_i \in \operatorname{Ext}^{1,p^i}; \qquad v_i \in \operatorname{Ext}^{2,p^{i+1}}$$

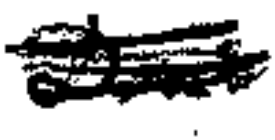
$$\operatorname{Ext}_A^{2,*} = \left\{ \underbrace{u_i u_j, \quad i < j, \quad v_k}_{\text{generators}} \right\}.$$



Original source details: GELFAND-P000477

Uncertain reading. The sign between i and j is heavily retraced. It is provisionally set as $i < j$.

~~This~~ But: symmetric ones are needed. $\underline{h(x,y) = h(y,x)}$



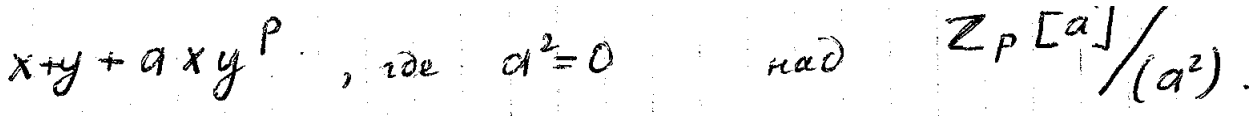
Original source details: GELFAND-P000477

Uncertain reading. The struck-out word before “But” may read «Это» (“This”).

There are noncommut. one-dimensional groups!! (if there is torsion)

(corr. to the cycles $u_i u_j$)

Example: $x + y + axy^p$, where $a^2 = 0$ over $Z_p[a]/(a^2)$.



Original source details: GELFAND-P000477

Diagram and typed description. The term axy^p and the condition $a^2 = 0$ occur on the same line.

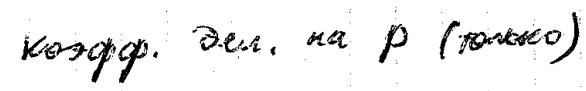
This is a noncommut. formal group!!

There are as many of them as there are cycles.

v_k corr. to commut. groups

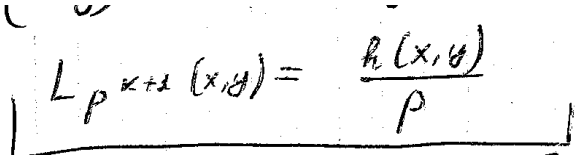
$$h(x,y) = (x+y)^{p^{k+1}} - x^{p^{k+1}} - y^{p^{k+1}}$$

coeff. divisible by p (only)



Original source details: GELFAND-P000477

Uncertain reading. The word in parentheses reads «только» (“only”).



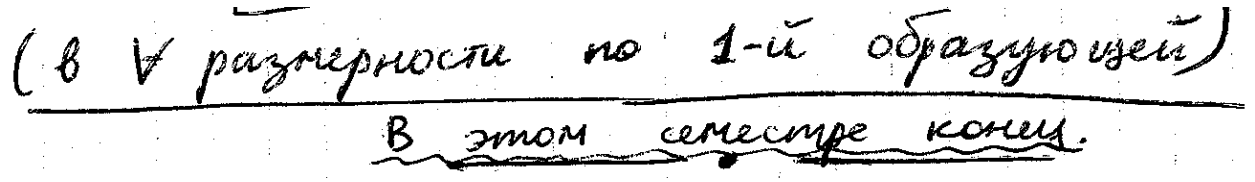
Original source details: GELFAND-P000477

Diagram and typed description. A rule with upturned ends is drawn below the formula.

$$L_{p^{k+1}}(x,y) = \frac{h(x,y)}{p}$$

(in $\forall$ dimension one generator each)

That is all for this semester.



Original source details: GELFAND-P000477

Diagram and typed description. The last line has both a wavy underline and a broken straight underline.

12/II-73. Gelfand sem.

Plan: 1. Buchstaber, Mishchenko: Topology (Sullivan's work)

2. Works by Deligne and Langlands

V. Buchstaber. On Sullivan's first paper.

M^n — manif.

A smoothing of M^n — is a sm. manif. N^n and a homot. equiv. $N^n \xrightarrow{f} M^n$

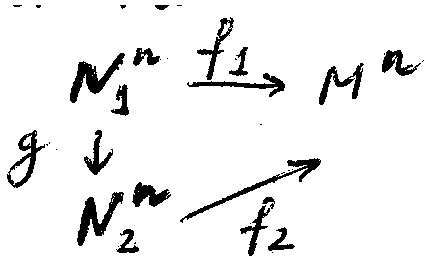


Diagram and typed description. Three arrows connect the two smooth manifolds and their common target.

$$\begin{array}{ccc} N_1^n & \xrightarrow{f_1} & M^n \\ {\scriptstyle g}\downarrow & \nearrow_{f_2} & \\ N_2^n & & \end{array}$$

Original source details: GELFAND-P000478

Equiv. $\Longleftrightarrow \exists$ a diffeom. g, such that this square commutes (up to homotopy).

$S(M^n)$ — the set of classes of equiv. smoothings.



Undeciphered passage. A short entry above "Milnor" is heavily crossed out. Its reading is unresolved.

Original source details: GELFAND-P000478

Milnor: Card $S'(S^7) = 28$.



Uncertain reading. A short stroke is visible between the first S and the parenthesis. It is set as S'.

Original source details: GELFAND-P000478

Description of $S(M^n)$ — ?

Sullivan considers in M^n a fixed piecewise-linear structure.

(M^n — a polyhedron)

and solved the prob. in this formulation:

(in place of a diffeom. — a piecewise-lin.



Undeciphered passage. A heavily crossed-out entry follows "piecewise-lin.". Its reading is unresolved.

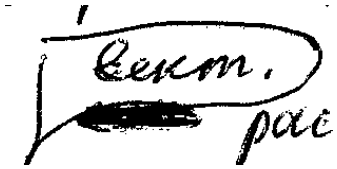
Original source details: GELFAND-P000478

homeomorphism)

Obtain $PL(M^n)$.

Here always $\underline{\underline{PL(S^n) = \{S^n\}}}$ — where the Poincaré prob. has been solved (for $n \geqslant 5$).

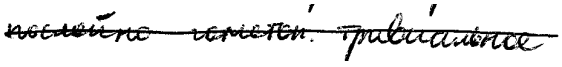
Consider on M^n (Vect.) a bundle ξ, stable (of large dim.).



Original source details: GELFAND-P000478

Undeciphered passage. The old entry before “bundle” is inked over. “Vect.” is inserted above it.

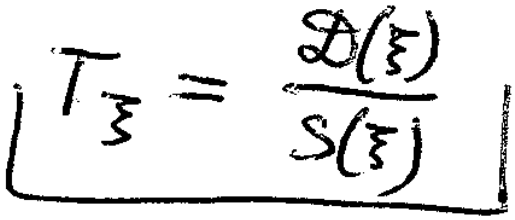
~~trivial in several degrees~~



Original source details: GELFAND-P000478

Uncertain reading. The struck-out phrase is provisionally read as «несколько степен. тривиальное».

T_ξ — the Thom complex of the bundle ξ,



Original source details: GELFAND-P000478

Diagram and typed description. The formula has a bottom rule with upturned ends.

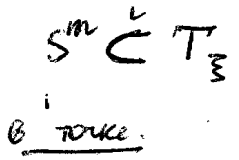
$$\underline{T_\xi = \frac{D(\xi)}{S(\xi)}}$$

D — bundle of balls,

S — bundle of spheres.

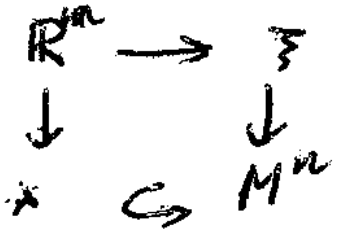
dim $\xi = m$.

$$\underset{\text{at a point.}}{S^m} \overset{i}{\subset} T_\xi$$



Original source details: GELFAND-P000478

Diagram and typed description. A stroke connects “at a point” to the left-hand sphere.

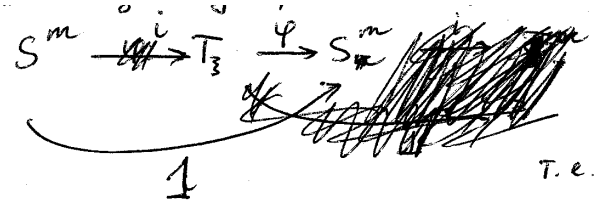


Original source details: GELFAND-P000478

Diagram and typed description. In the diagram at right, the point maps by inclusion into the base.

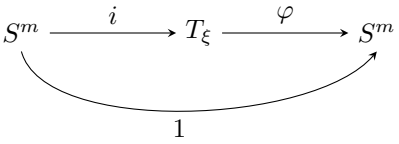
$$\begin{array}{ccc} \mathbb{R}^m & \longrightarrow & \xi \\ \downarrow & & \downarrow \\ * & \hookrightarrow & M^n \end{array}$$

The bundle ξ is called J-trivial, if $\exists$ a map φ; such that



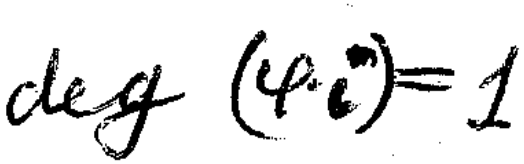
Original source details: GELFAND-P000478

Diagram and typed description. The first arrow is labelled i. The mark at the right-hand sphere and the large crossed-out entry to its right remain unresolved. The arc from the left sphere to the right sphere is labelled 1.



(homot.)

i.e. $\deg(\varphi \cdot i) = 1$



Original source details: GELFAND-P000478

Uncertain reading. A dot is visible between φ and i.

(the choice of $*$ does not matter since M^n is assumed connected).

$K_J(M^n)$ — the group of stab. bundles over M^n, each of which is J-trivial

GELFAND-P000479

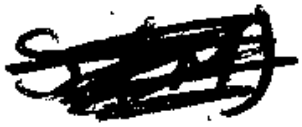
Atiyah's thm.: (Mat. $10:5$)



Original source details: GELFAND-P000479

Diagram and typed description. A short reference follows the name, «Мат. 10:5».

constr. a map



Original source details: GELFAND-P000479

Undeciphered passage. The first letter of the struck-out entry resembles S. A full reading is unresolved.

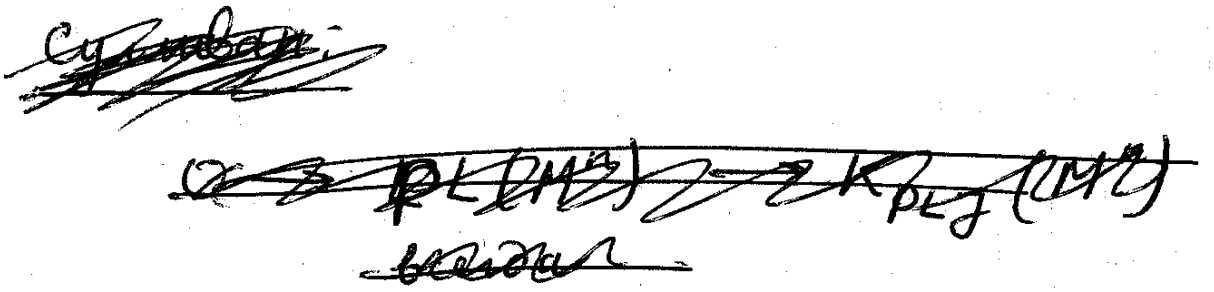
$$S(M^n) \longrightarrow K_J(M^n).$$

$N^n \longrightarrow M^n$ difference.

~~Sullivan:~~

$$~~0 \longrightarrow PL(M^n) \longrightarrow K_{PLJ}(M^n)~~$$

~~always~~



Original source details: GELFAND-P000479

Diagram and typed description. The name, sequence and word “always” are crossed out and remain legible underneath.

and

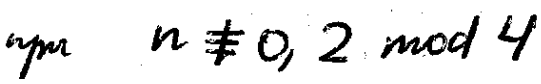
$$0 \longrightarrow PL(M^n) \longrightarrow K_{PLJ}(M^n)$$



Undeciphered passage. An entry after the second $K_{PLJ}(M^n)$ is crossed out. It may read $\to 0$.

Original source details: GELFAND-P000479

for $n \not\equiv 0, 2 \bmod 4$



Uncertain reading. The comparison sign has three horizontal strokes crossed by one stroke.

Original source details: GELFAND-P000479

$n \equiv 0 \bmod 4$	cokernel	Z
$n \equiv 2$ —//—	—//—	Z_2.

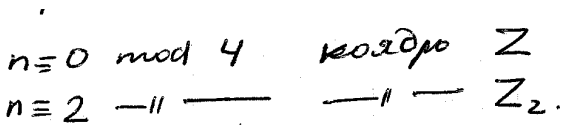


Diagram and typed description. The second row has ditto marks under “mod 4” and “cokernel”.

Original source details: GELFAND-P000479

Sullivan calc. the homot. groups of the repr. space $K_{PLJ}(M^n)$.



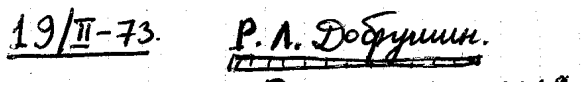
Uncertain reading. The loop on the descending stroke is retraced. The last letter of «группп» may be a second «п».

Original source details: GELFAND-P000479

For $K_J(M^n)$ it has not been done. A seq. for $S(M^n)$ exists too.

GELFAND-P000480

19/II-73. R. L. Dobrushin.



Original source details: GELFAND-P000480

Diagram and typed description. The month is written as a Roman numeral. Two lines with cross-strokes are drawn below the name.

On field theory.

Gelfand: 2 problems: — · —

1) quantize existing eqs.



Original source details: GELFAND-P000480

Uncertain reading. The end of the abbreviation is provisionally read as «существующ.» (“existing”).

2) write eqs. that actually correspond (phys.)



Original source details: GELFAND-P000480

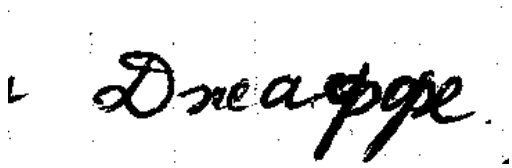
Uncertain reading. The abbreviation after “actually” reads «ответ.».

— · —

A method of constr. objects satisfying the Wightman axioms (probab. objects).

Principal author — Nelson.

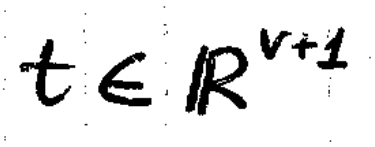
One can (almost) obtain the res. of Glimm and Jaffe.



Original source details: GELFAND-P000480

Uncertain reading. The manuscript writes Jaffe as «Джеаффе», with «е» before «а».

Random variable: ξ_t, $t \in \mathbb{R}^{\nu+1}$



Original source details: GELFAND-P000480

Uncertain reading. The first sign in the exponent resembles Latin v or Greek ν. It is provisionally set as ν.

1) it is Eucl. invari.

2) Markov prop.

GELFAND-P000481

26/II-73 V. M. Buchstaber. $hs(M^n)$ — (homotopic) smoothings.

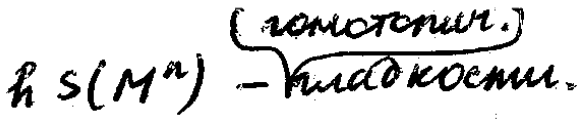


Diagram and typed description. “(homotopic)” is inserted above “smoothings”, with a pointer to the word.

Original source details: GELFAND-P000481

$ht(M^n)$ — homot. triangulations.
is described.

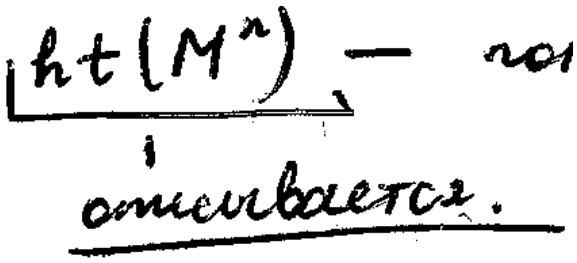


Diagram and typed description. “Is described.” is connected by a stroke to the bracket below $ht(M^n)$.

Original source details: GELFAND-P000481

Hauptvermutung — a homeomorphism (of manifolds) can be replaced by a piecewise-lin. homeom. (if M is simply conn. and there is in H^3 no 2-torsion)

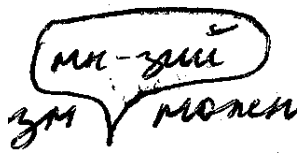
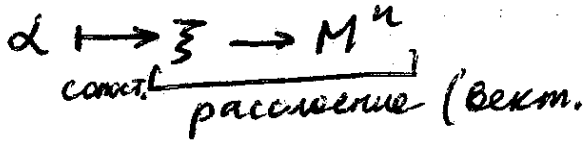


Diagram and typed description. “Of manifolds” is inserted in a callout after “a homeomorphism”.

Original source details: GELFAND-P000481

Let $\alpha \in hs(M^n)$ (hom. smoothing).

$$\alpha \underset{\text{assoc.}}{\mapsto} \underbrace{\xi \longrightarrow M^n}_{\text{bundle (vect.)}} \qquad \text{smooth.}$$



Uncertain reading. The abbreviation under the first arrow is provisionally read as «сопст.». Another possible reading is «сопост.».

Original source details: GELFAND-P000481

$$f\colon N^n \longrightarrow M^n;$$
$$\exists\, g\colon M^n \longrightarrow N^n; \qquad f \circ g \sim 1, \quad g \circ f \sim 1.$$
$$\tau_{M^n} + g^* \nu_{N^n} \qquad (\nu_{N^n} = -\tau_{N^n} \text{ mod triv.})$$

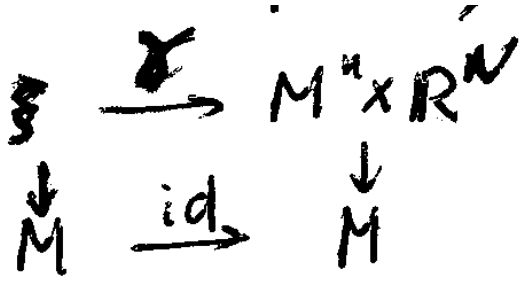


Original source details: GELFAND-P000481

Uncertain reading. The sign for the normal bundle resembles v. It is provisionally set as ν.

Ex. Prove that 2 equiv. smoothings induce isom. bundles ξ.

The resulting smooth bundle ξ is J-trivial, i.e.



Original source details: GELFAND-P000481

Diagram and typed description. The square has an arrow labelled γ with a heavily retraced sign.

$$\begin{array}{ccc} \xi & \xrightarrow{\gamma} & M^n \times R^N \\ \downarrow & & \downarrow \\ M & \xrightarrow{\mathrm{id}} & M \end{array}$$

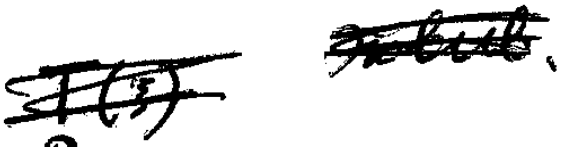
and γ is a proper map and a homot. equiv.



Original source details: GELFAND-P000481

Uncertain reading. The sign before “is” is heavily retraced. It is provisionally set as γ.

~~$T(\xi)$ equiv.,~~

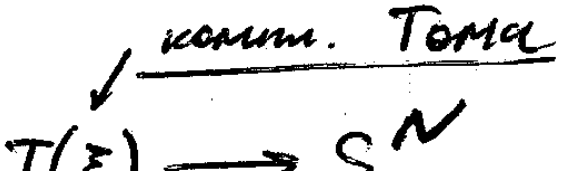


Original source details: GELFAND-P000481

Uncertain reading. After the crossed-out $T(\xi)$, «эквив.» (“equiv.”) is a provisional reading. «трив.» (“triv.”) remains possible.

Thom complex

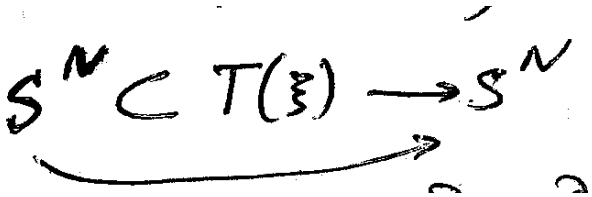
2nd def. — $\exists$ a map $T(\xi) \longrightarrow S^N$, such that



Original source details: GELFAND-P000481

Diagram and typed description. “Thom complex” is underlined with an arrow pointing to $T(\xi)$.

the composite



Original source details: GELFAND-P000481

Diagram and typed description. The lower arrow denotes the composite.

$$S^N \subset T(\xi) \longrightarrow S^N$$

is a homot. equiv.

Equivalence — thm. of Dold (technical, trivial in idea)

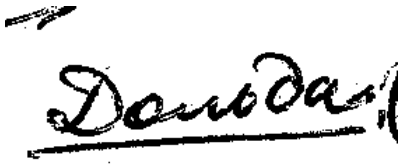


Diagram and typed description. Dold's name is underlined.

Original source details: GELFAND-P000481

GELFAND-P000482

5/III-73 Gelfand's Sem..

Theorem. ~~F. A. Berezin~~ Lomonosov



Uncertain reading. The heavily crossed-out name is provisionally read as "F. A. Berezin".

Original source details: GELFAND-P000482

R — a family of pairwise commut. (in Banach space B) op., containing a completely cont. op. $A \neq 0$.

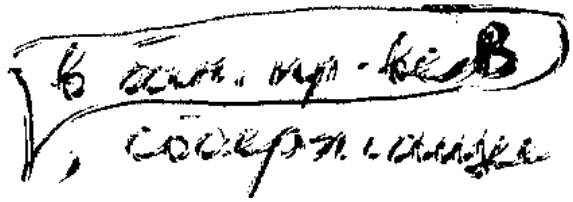
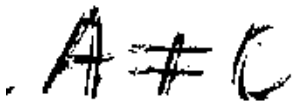


Diagram and typed description. "In Banach space B" is written in a callout after "op.".

Original source details: GELFAND-P000482



Uncertain reading. The last sign is open on the right and resembles C. It is provisionally set as 0.

Original source details: GELFAND-P000482

Then R has a common (nontriv.) invariant subsp..

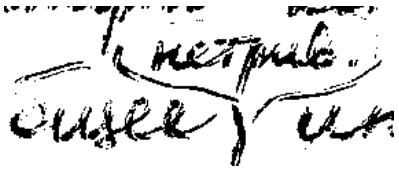


Diagram and typed description. "Nontriv." is inserted between "common" and "invariant".

Original source details: GELFAND-P000482

(generalization of the Aronszajn–Smith theorem)

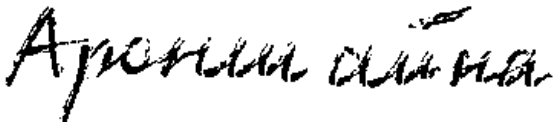


Diagram and typed description. The first surname reads «Ароншайна» in Russian.

Original source details: GELFAND-P000482

The proof works in the case of an LCS, short and simple.



Original source details: GELFAND-P000482

Uncertain reading. A short sign before "short" is heavily crossed out. Its reading is unresolved.

Proof. S' — a closed ball, $r = 1$ centered at x_0, such that
(in B)



Original source details: GELFAND-P000482

Diagram and typed description. "In B" is inserted below "ball" in a callout.



Original source details: GELFAND-P000482

Diagram and typed description. The first occurrence of the ball symbol S' is primed. The manuscript subsequently writes S.

$$\inf_{x \in S} \|Ax\| > 1/2$$

$$\begin{matrix} Ax_0' \neq 0 \\ \|x_0'\| = 1 \end{matrix} \quad \text{and we take} \quad \|A\| \cdot x_0' = x_0.$$

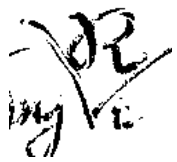


Original source details: GELFAND-P000482

Uncertain reading. The letter between the norm bars is heavily retraced. A provisional reading is $\|A\|$.

$\overline{AS}$ — a compact set that does not contain zero.

Consid. the algebra $\mathcal{R}$ with 1, generated by the family R (without closure).



Original source details: GELFAND-P000482

Diagram and typed description. The symbol inserted above "algebra" is provisionally set as $\mathcal{R}$.

$$\forall y \in B, \qquad \mathcal{R}y \quad \text{— lin. set.}$$

It is easy to see that $\mathcal{R}y$ is invar. under $\mathcal{R}$.

$\overline{\mathcal{R}y}$ — an inv. subsp..

We must prove that it can be nontriv..

Let $y \in \overline{AS}$. We must check that $\exists y$; $\mathcal{R}y$ is not dense.

Suppose otherwise, then



Original source details: GELFAND-P000482

Uncertain reading. The crossed-out word after "then" is undeciphered. «Оно» ("it") is a possible reading.

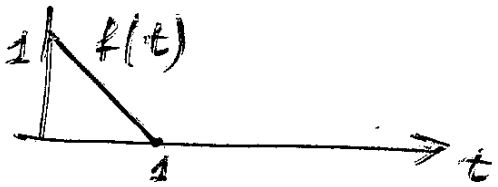
$$\forall y \in \overline{AS} \quad \exists T \in \mathcal{R}, \qquad \|Ty - x_0\| < 1.$$

From the compactness of $\overline{AS}$ it follows that $\exists T_1, \ldots, T_n$

$$\forall y \ \exists i \qquad \|T_i y - x_0\| < 1,$$

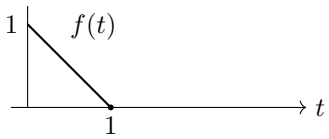
$$\forall y \in AS \quad \{T_1 y, \ldots, T_n y\} \cap S \neq \varnothing.$$

Let $f(t) \in C(0, +\infty)$



Original source details: GELFAND-P000482

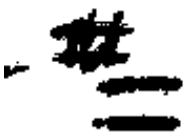
Diagram and typed description. The graph of $f(t)$, the marks 1 on both axes and the point on the t-axis.



GELFAND-P000483

Consid.

$$\Phi y \overset{\#}{=} \frac{\sum_1^n f(\|T_i y - x_0\|) T_i y}{\sum_j f(\|T_j y - x_0\|)} \qquad \text{(convex comb.)}$$



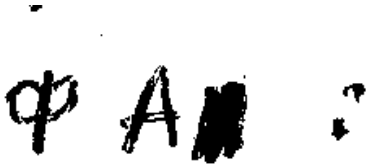
Original source details: GELFAND-P000483

Uncertain reading. A sign resembling # is written above and to the left of the equals sign.

$$\Phi\colon \overline{AS} \longrightarrow S$$

This is a cont. function on $\overline{AS}$.

Consid. $\Phi A\colon S \longrightarrow S$,



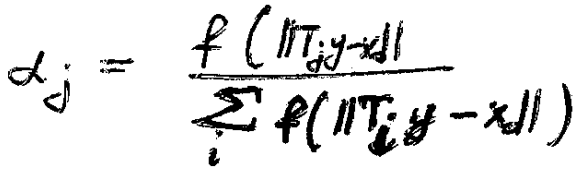
Original source details: GELFAND-P000483

Uncertain reading. A sign after ΦA is overwritten. It may be x.

where the image of ΦA is compact and it is cont..

By Schauder's thm. $\exists$ a fixed point, i.e.,

$$\sum_1^n \alpha_i T_i A x = x, \qquad \alpha_j = \frac{f(\|T_j y - x_0\|}{\sum_i f(\|T_i y - x_0\|)}$$



Original source details: GELFAND-P000483

Diagram and typed description. Both arguments show x_0. The closing parenthesis after the norm is absent in the numerator.

$T = \sum \alpha_i T_i A$ is completely continuous.

$\exists x\colon Tx = x$. Hence an eigenvect. with eigenval. 1 — a nontriv. invar. subsp. (even finite-dimensional). Hence $\mathcal{R}$ has an invar. subsp..



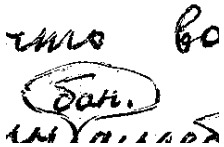
Original source details: GELFAND-P000483

Diagram and typed description. End mark with an oval above two crossing strokes and a dot to the right.



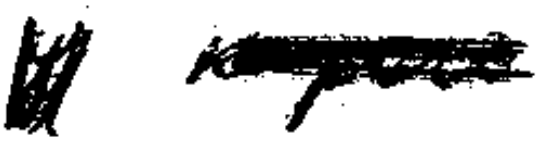
(This does not mean there is a finite-dimensional one in general).

Corollary. If a comm. (Banach) algebra has an elem.,



Original source details: GELFAND-P000483

Diagram and typed description. "Banach" is inserted after "comm.".



Uncertain reading. A sign resembling «и» (“and”) and an undeciphered word are crossed out after “elem.”.

Original source details: GELFAND-P000483

(by which)
mult. is completely cont., then ∃ a nontriv. ideal.

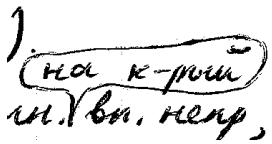


Diagram and typed description. “By which” is inserted after “mult.”.

Original source details: GELFAND-P000483

Henkin. (announcement)

Let there be in C^n a convex dom. $D \subset C^n$, given by the eq. $\rho(z) < 0$.

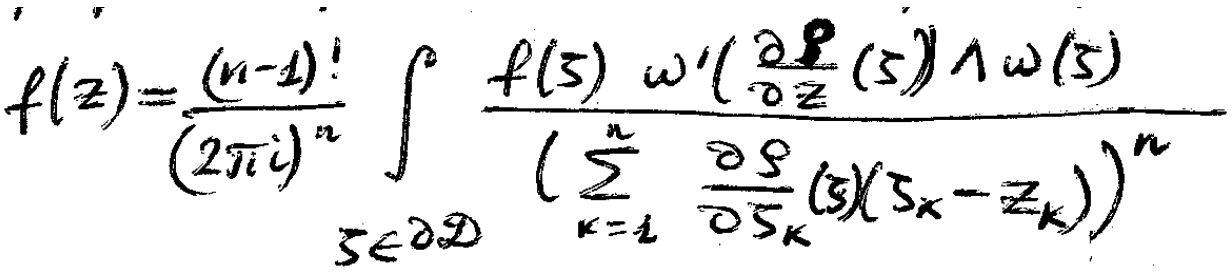


Uncertain reading. The letter D before the inclusion sign is heavily retraced.

Original source details: GELFAND-P000483

Leray integral formula for bounded holom. functions:

$$f(z) = \frac{(n-1)!}{(2\pi i)^n} \int_{\zeta \in \partial D} \frac{f(\zeta)\, \omega'\left(\frac{\partial \rho}{\partial z}(\zeta)\right) \wedge \omega(\zeta)}{\left(\sum_{k=1}^{n} \frac{\partial \rho}{\partial \zeta_k}(\zeta)(\zeta_k - z_k)\right)^n}$$



Original source details: GELFAND-P000483

Uncertain reading. The sign ρ in the numerator of the derivative in ω' is heavily retraced.

1) Leray-type integral:

$$f_g(z) = \cdot\ \cdot \int^{g(\zeta)\cdots}_{\cdots} \qquad g(\zeta) \text{ — a function on the boundary.}$$

$$f \longrightarrow [f(\zeta)g(\zeta) - f_g(\zeta)] \quad \text{— a completely cont. operator}$$

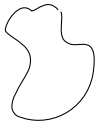
and a Hölder condition with exp. $1/2$

2) $\rho(z)$ — strictly plurisubharm..



Original source details: GELFAND-P000483

Diagram and typed description. Contour to the left of “Then”.



Then D — ~~dom.~~ is a closed subset of a strictly convex dom., approaching the boundary transversely.



Original source details: GELFAND-P000483

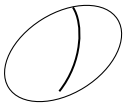
Uncertain reading. The crossed-out word is provisionally read as «обл.» (“dom.”), possibly «область» (“domain”).

(follows from Oka’s thm.)



Original source details: GELFAND-P000483

Diagram and typed description. A tilted oval with an interior arc. The arc reaches the upper edge and ends near the lower edge.



3) in item 2) write out the extension from the subset with estimates! (in the class of bounded functions)

The converse is also true: such a subset is ~~a domain~~ a bounded domain of holomorphy.



Original source details: GELFAND-P000483

Uncertain reading. The crossed-out word is provisionally read as «область» (“domain”), possibly «обл.» (“dom.”).

4) Corollary: for functions anal. inside and cont. in the closure, the space of maximal ideals $=$ the closure of the domain.

GELFAND-P000484

F. A. Berezin.

Quantization problem.

M — sympl. manifold (ω — a form [illegible] with matrix ω_{ik}; ω^{ik} — inverse matrix)



Uncertain reading. A heavily retraced sign follows “form”. Possible readings are 2 or ξ.

Original source details: GELFAND-P000484

$$[f, g] = \sum \omega^{ik} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^k}$$

Quantization: a family of assoc. rings A_h with elements $f(x)$ (functions on M)

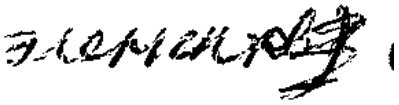
$$f_1 * f_2 = g(x \mid h)$$

$$\left.\begin{array}{ll} 1) & g(x \mid 0) = f_1(x) \cdot f_2(x) \\ 2) & \dfrac{\partial}{\partial h}(f_1 * f_2 - f_2 * f_1)\Big|_{h=0} = [f_1, f_2] \end{array}\right\} \quad \text{(correspondence principle)}.$$

Several further reqs. (less obligatory)

3) all A_h — algebras with 1, corresp. $f_0 \equiv 1$.

4) A_h — an algebra with trace (def. on a distinguished set of elements in A_h).



Uncertain reading. The ending of the word “elements” was rewritten. The earlier reading is unresolved.

Original source details: GELFAND-P000484

$$\mathrm{Sp}(f_1 * f_2) = \mathrm{Sp}(f_2 * f_1)$$

and

$$\mathrm{Sp}\, f = c \int f \, d\mu,$$

where $d\mu$ — the nat. measure on the manifold (a power of ω).



Diagram and typed description. The parenthetical reads «степень ω» (“a power of ω”).

Original source details: GELFAND-P000484

Req. of math. naturalness — functoriality with respect to canonical Diffeomorphisms (with respect to some family of maps — some fixed category).

A solution is known when $M \in$ a certain class (2 classes of Kähler manifolds).

1) dom. in C^n
2) compact symm. manifolds,

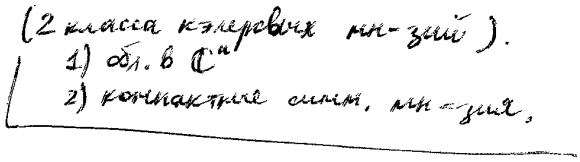


Diagram and typed description. A common stroke runs to the left of both items and below them.

Original source details: GELFAND-P000484

Fock space.



Uncertain reading. A sign above the start of the heading is heavily blacked out. Its reading is unresolved.

Original source details: GELFAND-P000484

$$M = C^n \qquad \omega = \sum dz_i \wedge d\bar{z}_i.$$

Functions — entire functions $f(z, \bar{z})$ (expan. in x, y in a series converg. everywhere)

$$(f_1 * f_2)(z, \bar{z}) = \frac{1}{h^n} \int f_1(z, \bar{v}) f_2(v, \bar{z}) e^{-\frac{1}{h}(z-v)(\bar{z}-\bar{v})}\, dv\, d\bar{v}$$

Important: the structure of lin. rep. of this ring.

Let $\varphi(z)$ — ordinary entire functions satisfying

$$(\varphi, \varphi) = \frac{1}{h^n} \int |\varphi(z)|^2 e^{-\frac{z\bar{z}}{h}}\, dz\, d\bar{z} < +\infty \qquad \text{(space } F_2\text{)}$$

There acts a lin. rep. of the ring A_h:

$$(\hat{f}\varphi)(z) = \frac{1}{h^n} \int f(z, v) \overline{\varphi(v)}\, e^{-\frac{1}{h}(v-z)\bar{v}}\, dv\, d\bar{v},$$

f — Wick symbol.



Diagram and typed description. The bar is over $\varphi(v)$. The preceding factor $f(z, v)$ has no bar over v.

Original source details: GELFAND-P000484

One more ring of operators:

$$(\hat{f}\varphi)(z) = \frac{1}{h^n}\int \mathring{f}(v,\bar{v})\varphi(v)e^{-\frac{1}{h}(v-z)\bar{v}}\,dv\,d\bar{v},$$

$\mathring{f}$ — anti-Wick symbol.

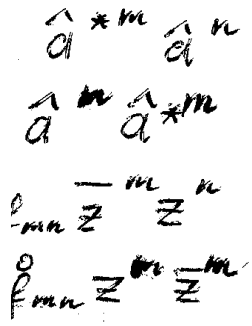
$$(\hat{a}_k\varphi)(z) = \frac{\partial\varphi}{\partial z_k}, \qquad (\hat{a}_k^*\varphi)(z) = z_k\varphi.$$

$$\hat{f} = \sum f_{mn}\hat{a}^{*m}\hat{a}^n \quad \longleftrightarrow \quad f(z,\bar{z}) = \sum f_{mn}\bar{z}^m z^n$$
$$\hat{f} = \sum \mathring{f}_{mn}\hat{a}^n\hat{a}^{*m} \quad \longleftrightarrow \quad \mathring{f}(z,\bar{z}) = \sum \mathring{f}_{mn}z^n\bar{z}^m.$$



Original source details: GELFAND-P000484

Uncertain reading. The sign at the upper right of the first $\hat{f}$ is overwritten. A possible reading is $*$.



Original source details: GELFAND-P000484

Uncertain reading. The exponents on the first $\hat{a}$ and the first z of the lower row are heavily retraced. Both are provisionally read as n.

GELFAND-P000485

Consid. $e_{\bar{v}}(z) = e^{\frac{1}{h}\bar{v}z}$ (coherent states or Poisson vectors).

~~$f\ldots$~~

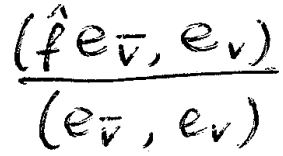


Original source details: GELFAND-P000485

Uncertain reading. The letter f is visible in the crossed-out entry. The rest is undeciphered.

Consid. now

$$\hat{f} \longleftrightarrow f(v,\bar{v}) = \frac{(\hat{f}e_{\bar{v}}, e_v)}{(e_{\bar{v}}, e_v)} \quad \text{(extension)} =$$
$$= \mathrm{Sp}(\hat{f}P_v).$$



Original source details: GELFAND-P000485

Diagram and typed description. In both rows of the fraction the first e has subscript $\bar{v}$ and the second e has subscript v.

(P_v — orthog. proj. onto e_v)

$$\hat{f} = \int \mathring{f}(v, \bar{v}) P_v \, dv \, d\bar{v} \qquad \text{(ext. of anti-Wick symbols).}$$

Duality:

1.

Wick ↑ op. ⟶ symbol
anti-Wick symb. ⟶ operator



Diagram and typed description. A heavily retraced upward arrow before "op." points toward the wavy underline. Its connection with the two rows is unresolved.

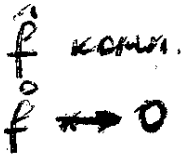
Original source details: GELFAND-P000485

2.

$\hat{f}$ — bd. $\Longrightarrow$ f bd.
$\mathring{f}$ — bd. $\Longrightarrow$ $\hat{f}$ bd.

3.

$\hat{f}$ comp. $\Longrightarrow$ $f \to 0$
$\mathring{f} \to 0 \Longrightarrow$ $\hat{f}$ comp.



Uncertain reading. The arrow toward 0 in the second row is heavily retraced. It is provisionally set as $\to$.

Original source details: GELFAND-P000485

4. Nuclearity (analog.)

$$\operatorname{Sp} \hat{f} = \frac{1}{h^n} \int f(z, \bar{z}) \, dz \, d\bar{z} = \frac{1}{h^n} \int \mathring{f}(z, \bar{z}) \, dz \, d\bar{z}.$$

Conversion formula:

$$f(z, \bar{z}) = \frac{1}{h^n} \int \mathring{f}(v, \bar{v}) e^{-\frac{1}{h}(z-v)(\bar{z}-\bar{v})} \, dv \, d\bar{v}.$$

At $h/2$ one obtains the Weyl symbol. (Gelfand's remark: (these are symbols) of Poincaré .)

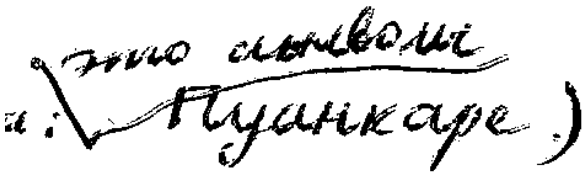


Diagram and typed description. "These are symbols" is inserted above, before "of Poincaré".

Original source details: GELFAND-P000485

General construction.

H — Hilbert space, M — set with a measure, $\alpha \in M$, measure $d\alpha$.

e_α — overcompl. system, if

$$(f, f) = \int |(f, e_\alpha)|^2 \, d\alpha.$$

Consid. then the correspondence:

$$\hat{A} \longrightarrow \frac{(\hat{A}e_\alpha, e_\alpha)}{(e_\alpha, e_\alpha)} = \underset{\mathrm{Sp}(\hat{A}\hat{P}_\alpha)}{A(\alpha)} \qquad \text{(covar. symbol)}.$$

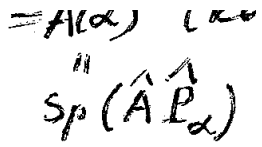


Diagram and typed description. Two strokes and the trace expression appear below $A(\alpha)$.

Original source details: GELFAND-P000485

contravar. symbol.

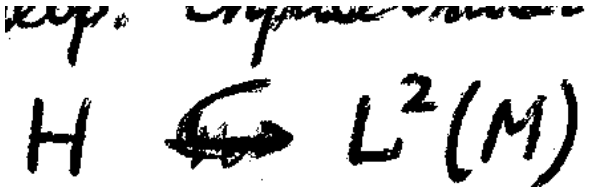
$$\hat{A} = \int \mathring{A}(\alpha)\hat{P}_\alpha \, d\alpha.$$

Earlier: $e_\alpha = e^{\bar{v}z}$, $\alpha = (v, \bar{v})$.

All thms. hold.

Another method:

$H \hookrightarrow L^2(M)$, $\hat{T}$ — orthog. projector from $L^2(M)$ into H.



Uncertain reading. The heavily retraced sign between H and $L^2(M)$ is provisionally read as a rightward inclusion arrow.

Original source details: GELFAND-P000485

$$(\hat{B}f)(\alpha) = \mathring{A}(\alpha)f(\alpha) \quad \text{— operator in } L^2(M).$$

Then $\hat{A} = \hat{T}\hat{B}\hat{T}$.

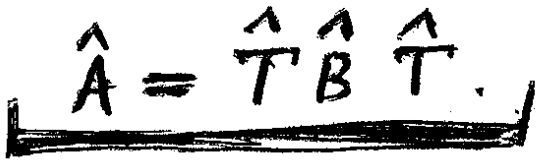


Diagram and typed description. A stroke runs below and to either side of the formula.

Original source details: GELFAND-P000485

GELFAND-P000486

Example: $e^{i(p,x)}$ in bounded domains.

Let $\varphi(x)$ be an arbit. convex (downward) function of a real var. x.



Original source details: GELFAND-P000486

Diagram and typed description. “Downward” is inserted above, after “convex”. An arc is drawn beneath the word.

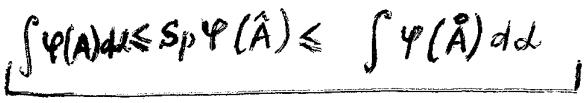
Then

$$\int \varphi(A)\, d\mu \leqslant \operatorname{Sp} \varphi(\hat{A}) \leqslant \int \varphi(\mathring{A})\, d\alpha$$



Original source details: GELFAND-P000486

Uncertain reading. The left differential is provisionally read as $d\mu$. Another possible reading is $d\alpha$.



Original source details: GELFAND-P000486

Diagram and typed description. A stroke runs below and to either side of the inequality.

Left side — a consequence of Jensen’s ineq..

Right side — $\rightarrow$

$$\boxed{\operatorname{Sp}(P\varphi(A)P) \geqslant \operatorname{Sp} \varphi(PAP)}$$



Original source details: GELFAND-P000486

Uncertain reading. The sign before the box is provisionally set as a right arrow. It may be crossed out.

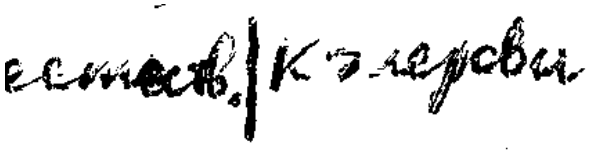


Original source details: GELFAND-P000486

Uncertain reading. On the right side the parenthesis merges with adjacent strokes. A provisional reading is $\operatorname{Sp} \varphi(PAP)$.

Consid. the domain $G \subset \mathbb{C}^n$,

in it there exists a nat. | Kähler metric.



Original source details: GELFAND-P000486

Diagram and typed description. A vertical stroke precedes “Kähler”.

We construct a ring ~~of~~ from real-anal. functions $f(z, \bar{z})$



Uncertain reading. A short sign after “ring” is crossed out. The provisional reading is «от» (“of”).

Original source details: GELFAND-P000486

$$f_1 * f_2 = \int f_1(z, \bar{v}) f_2(v, \bar{z}) G_h(v, \bar{v}, z, \bar{z})\, dv\, d\bar{v}$$

$$G_h = c(h) \left[\frac{K(z, \bar{z}) K(v, \bar{v})}{K(z, \bar{v}) K(\bar{z}, v)} \right]^{1/h},$$

(one can prove that $K(z, \bar{v}) \neq 0$)

where K — the Bergman kernel fn..

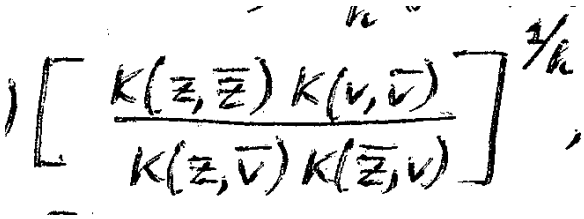


Diagram and typed description. The diagonal factors are in the numerator. The second denominator factor is $K(\bar{z}, v)$. The exponent is $1/h$.

Original source details: GELFAND-P000486

$$c(h) = h^{-\dim G} c_1(h),$$

$c_1(h)$ is cont. as $h \to 0$. (arbitrary)

$\longrightarrow$ unknown if $\exists$ a quantiz. also satisfying 3).

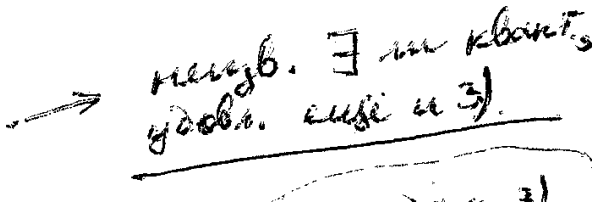


Diagram and typed description. The rightward arrow from “(arbitrary)” points to the underlined marginal note.

Original source details: GELFAND-P000486

Then all reqs. are met except 3).

on the basis of 3).
in a nat. way.

If G — a homog. dom., this too is met and $c_1(h)$ is specified

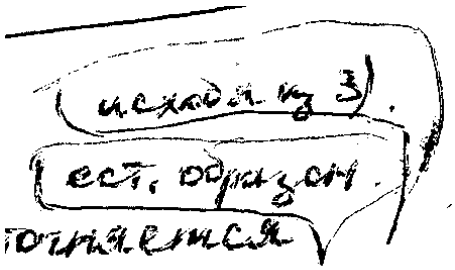


Diagram and typed description. Two callouts, one above the other, appear on the right above “is specified”.

Original source details: GELFAND-P000486

If G is even symmetric, and motions — autom. of the ring, then only this.
Overcompl. system here (for a homog. dom.)

$$M = G \subset \mathbb{C}^n = \mathbb{R}^{2n}$$

$e_v(z) = (K(z, \bar{v}))^{1/h}$ (for a homog. domain).



Uncertain reading. A sign below the exponent $1/h$ is crossed out. It may be an earlier exponent.

Original source details: GELFAND-P000486

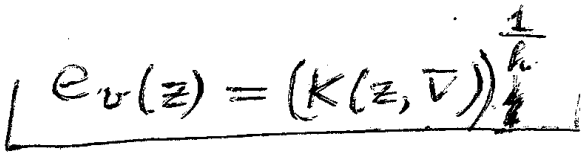


Diagram and typed description. A stroke runs below and to either side of the formula.

Original source details: GELFAND-P000486

Hilbert space:

$$H = \{f : \int |f(z)|^2 K^{1/h}(z, \bar{z})\, d\mu(z) < +\infty \quad f \text{ (holom.)}\}.$$



Uncertain reading. A heavy additional stroke below $<$ resembles a crossed-out "2". A possible reading is $\leqslant$.

Original source details: GELFAND-P000486

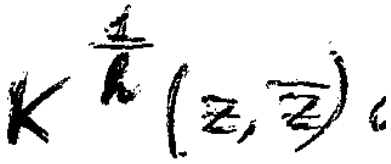


Diagram and typed description. The weight has exponent $1/h$. There is no visible minus before the exponent.

Original source details: GELFAND-P000486

$\mu(z)$ — an invar. measure generated by the Bergman metric.
A larger space — all functions.
In the gen. case it is unknown whether $K(z, \bar{v})$ can become 0.
Covar. symbols: continuations are analytic on all of $G \times \bar{G}$.

(complex)
Compact case. Comp. symmetric space.

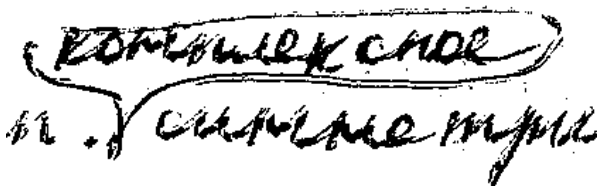


Diagram and typed description. "Complex" is inserted in a callout after "Comp.".

Original source details: GELFAND-P000486

Consid. the potential of the Kähler metric, i.e. $\Phi(z, \bar{z})$

GELFAND-P000487

Consid.

$$e^{\frac{1}{h}\Phi(z,\bar{z})} = K_h(z,\bar{z}) \qquad \left(\Phi \text{ is def. by the cond. } \omega^{ij} = \frac{\partial^2 \Phi}{\partial z_i \partial \bar{z}_j}\right).$$

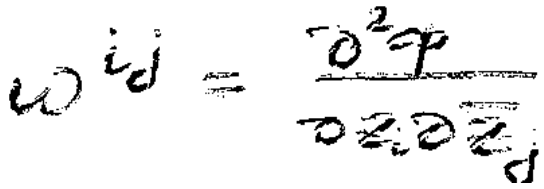


Diagram and typed description. The two indices i, j on ω are written above.

Original source details: GELFAND-P000487

At $h = \underline{\underline{\frac{1}{n}}}\,\Phi(z,\bar{z})$ extends and there exists a single-valued fn.

$$K_h(z,\bar{v})$$

and the whole preceding theory.

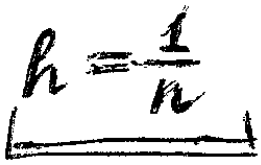


Diagram and typed description. The formula $h = 1/n$ has a double stroke underneath with turned-up ends.

Original source details: GELFAND-P000487

Then the Hilbert space H_α is finite-dimensional and dim is easily computed.



Uncertain reading. The heavily retraced subscript is provisionally read as α. Another possible reading is h.

Original source details: GELFAND-P000487

Lobachevsky plane:

$$K^{1/h} = (1 - z\bar{z})^{+2/h}$$

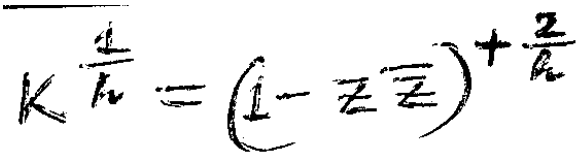


Diagram and typed description. A plus sign is written before $2/h$.

Original source details: GELFAND-P000487

Autom. of the space generate autom. of the ring.

One obtains a projective rep. of the Lobachevsky (autom. of) plane.

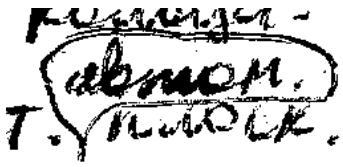


Diagram and typed description. “Autom.” is inserted in a callout after “rep.”.

Original source details: GELFAND-P000487

GELFAND-P000488

12/III-73 Gelfand's Sem..

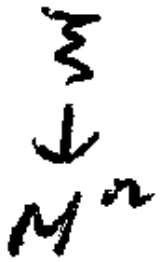
V. M. Buchstaber.

$S(M^n)$ — set of smoothings on M^n
$hS(M^n)$ — homot. smoothings
$t(M^n)$ — triangulations
$ht(M^n)$ — hom. triangul.

— · —

Consid. the set $hS(M^n)$.

On it, consid. smooth vect. bundles.



Original source details: GELFAND-P000488

Diagram and typed description. An arrow runs down from ξ to M^n.

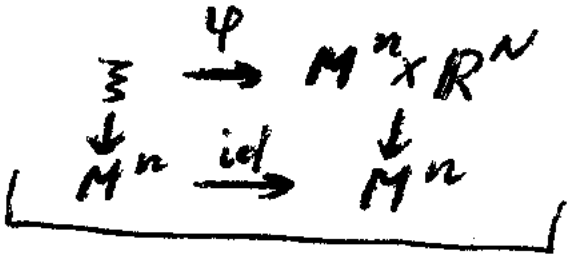
$$\begin{array}{c} \xi \\ \downarrow \\ M^n \end{array}$$

J-triv. bundles (fibrewise homot. triv.), i.e.



Original source details: GELFAND-P000488

Uncertain reading. The first abbreviation is provisionally read as «псл.», possibly «посл.», for "fibrewise".



Original source details: GELFAND-P000488

Diagram and typed description. Square with an underline whose ends turn upwards.

$$\begin{array}{ccc} \xi & \xrightarrow{\varphi} & M^n \times \mathbb{R}^N \\ \downarrow & & \downarrow \\ M^n & \xrightarrow{\mathrm{id}} & M^n \end{array}$$

$N = \dim \xi$



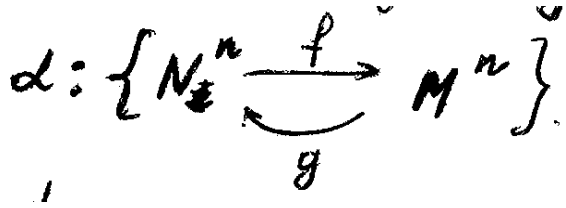
Original source details: GELFAND-P000488

Diagram and typed description. The equality has a double underline.

φ — a proper homotop. equivalence.

$\Phi(M^n)$ — set of smooth J-triv. bundles on M^n. (up to trivial ones).

Let



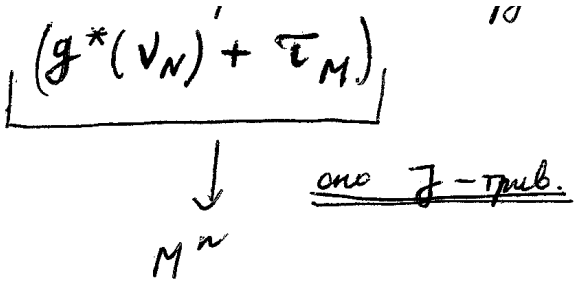
Original source details: GELFAND-P000488

Diagram and typed description. The heavily retraced subscript on N is provisionally set as ξ. Another possible reading is ζ.

$$\alpha : \{ \quad N^n_\xi \underset{g}{\overset{f}{\rightleftarrows}} M^n \quad \}$$

g — a homot. inverse.

From α we construct an elem. of the group $\Phi(M^n)$



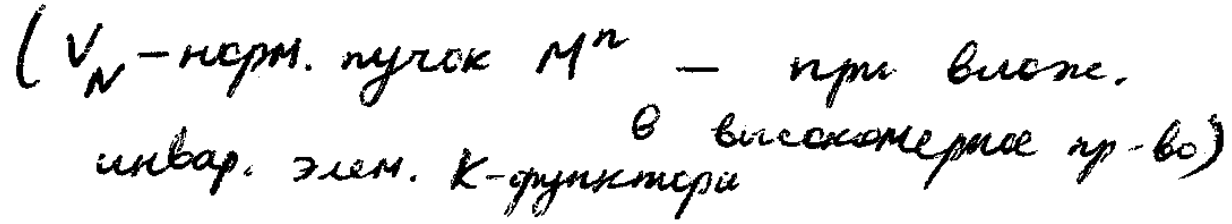
Original source details: GELFAND-P000488

Diagram and typed description. A stroke runs below the sum. An arrow points from it to the base.

$$\underbrace{(g^*(\nu_N) + \tau_M)}_{\text{it is } J\text{-triv.}} \downarrow M^n$$

(ν_N — norm. bundle of M^n — for an embedd. in a high-dimensional space)

invar. elem. of the K-functor.



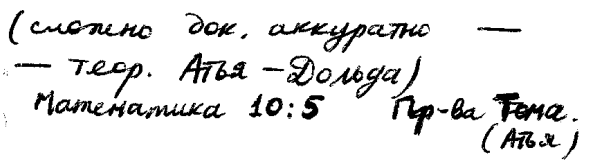
Original source details: GELFAND-P000488

Uncertain reading. The explanation for ν_N reads M^n.

$$g \text{ — ident.} \Longrightarrow \nu_N + \tau_M \underset{\text{def}}{=} \text{triv.}$$

$$g \text{ — hom. equiv.} \Longrightarrow \ldots$$

(difficult to prove carefully —
— Atiyah–Dold thm.)
Matematika 10:5 Thom spaces. (Atiyah)



Original source details: GELFAND-P000488

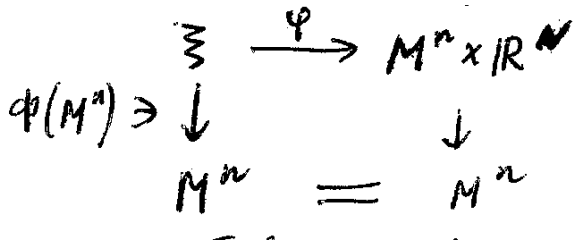
Diagram and typed description. The note and reference are to the right of the calculation.

Let $\pi_1(M^n) = 0$.

Exact seq.

$$hS(M^n) \longrightarrow \Phi(M^n) \longrightarrow P_n$$

$$P_n = \begin{cases} 0, & n \neq 4k,\ 4k+2 \\ \mathbb{Z}, & n = 4k \\ \mathbb{Z}_2, & n = 4k+2 \end{cases}$$

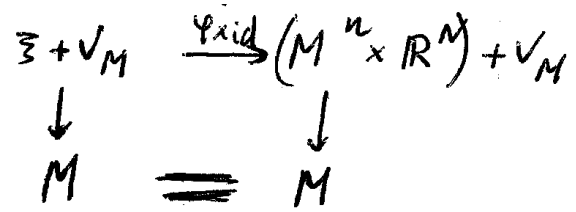


Original source details: GELFAND-P000488

Diagram and typed description. The bottom horizontal connection is written as an equals sign.

$$\begin{array}{ccc} & \xi & \xrightarrow{\varphi} & M^n \times \mathbb{R}^N \\ \Phi(M^n) \ni & \downarrow & & \downarrow \\ & M^n & = & M^n \end{array}$$

Let us add ν_M:



Original source details: GELFAND-P000488

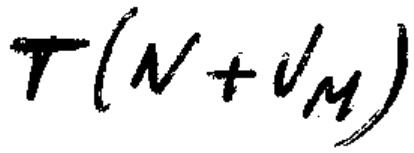
Diagram and typed description. The normal bundle is added to both upper entries.

$$\begin{array}{ccc} \xi + \nu_M & \xrightarrow{\varphi \times \mathrm{id}} & (M^n \times \mathbb{R}^N) + \nu_M \\ \downarrow & & \downarrow \\ M & = & M \end{array}$$

$\varphi \times \mathrm{id}$ — a proper homot. equiv..

Hence

$$T(\xi + \nu_M) \longrightarrow T(N + \nu_M) \qquad (*)$$



Original source details: GELFAND-P000488

Diagram and typed description. The argument on the right reads $N + \nu_M$.

GELFAND-P000489

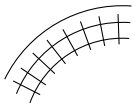
Remark on the props. of the Thom complex, norm. bundle.

$M^n \subset \mathbb{R}^L$.



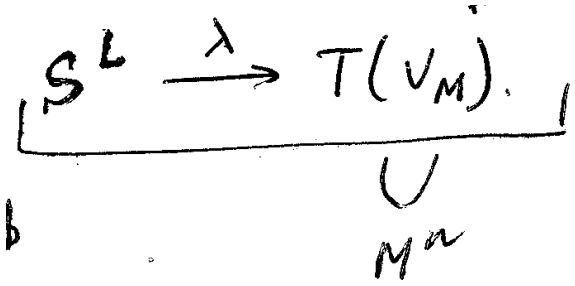
Original source details: GELFAND-P000489

Diagram and typed description. Three arcs and transverse strokes.



The space of the tube is the space of the norm. bundle.

There is a nat. map



Original source details: GELFAND-P000489

Diagram and typed description. The arrow λ and its endpoints have a stroke underneath. Below $T(\nu_M)$ are $\cup$ and M^n.

$$\underbrace{S^L \xrightarrow{\lambda} T(\nu_M).}_{\cup\ M^n}$$



Original source details: GELFAND-P000489

Uncertain reading. A sign below the left part is crossed out. Possible readings are β and φ.

The map λ is transversely regular on M^n. Its preimage is M^n. Only this will be used.

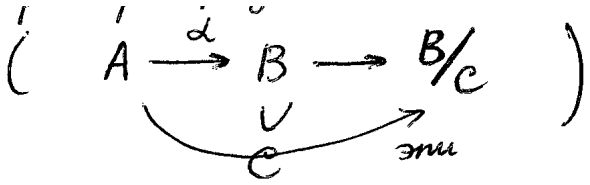
$\exists$ a map $S^L \xrightarrow{\lambda} T(\nu_M)$



Original source details: GELFAND-P000489

Uncertain reading. An entry following the second formula is crossed out. Its reading is unresolved.

and an embedding $M \subset T(\nu_M)$, which is transversely reg. on M.



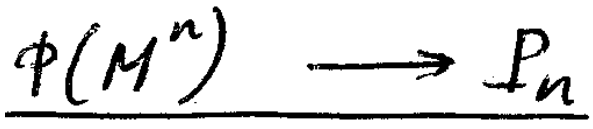
Original source details: GELFAND-P000489

Diagram and typed description. The lower arc goes from A to B/C. Its endpoint is labelled "epi".

$$\Big(A \xrightarrow{\alpha} B \longrightarrow B/C \Big).$$
$$\cup$$
$$C \qquad \text{epi}$$

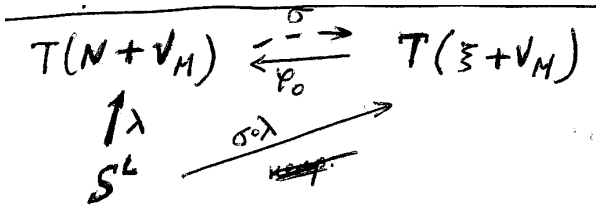
—— · ——

The map $\phi(M^n) \longrightarrow P_n$.



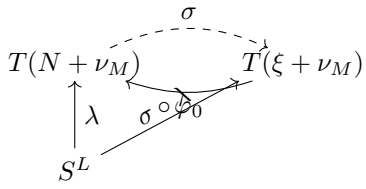
Original source details: GELFAND-P000489

Uncertain reading. The sign before (M^n) is provisionally set as ϕ.



Original source details: GELFAND-P000489

Diagram and typed description. The reverse arrow σ from $T(N+\nu_M)$ is dashed.



~~cont.~~



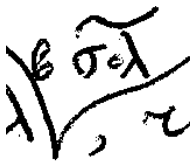
Original source details: GELFAND-P000489

Uncertain reading. The crossed-out abbreviation below the diagonal arrow is provisionally read as «непр.» (“cont.”).

φ_0 is a homot. equiv. by construction. σ — a hom. inverse.

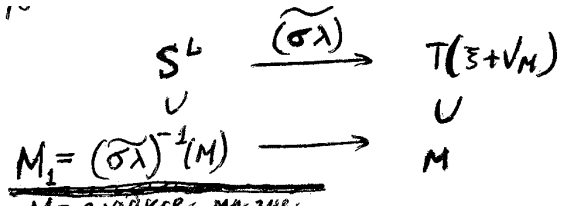
$\sigma \circ \lambda$ — a cont. map. $\quad M \subset T(\xi + \nu_M)$

We homotope $\sigma \circ \lambda$ (to $\widetilde{\sigma \circ \lambda}$), so that on M it becomes transversely reg..



Original source details: GELFAND-P000489

Diagram and typed description. The callout after $\sigma \circ \lambda$ reads “to” followed by the same composite with a tilde.



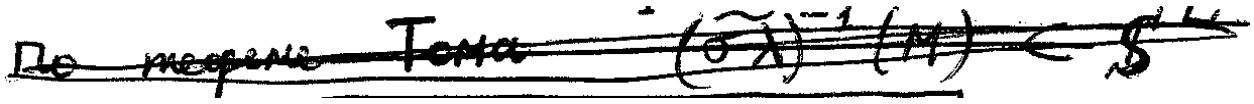
Original source details: GELFAND-P000489

Diagram and typed description. The preimage M_1 is below S^L. The two rows have $\cup$ signs between them.

$$\begin{array}{ccc} S^L & \xrightarrow{(\widetilde{\sigma\lambda})} & T(\xi+\nu_M) \\ \cup & & \cup \\ M_1 = (\widetilde{\sigma\lambda})^{-1}(M) & \longrightarrow & M \end{array}$$

M_1 — a smooth manifold. $\dim M_1 = \dim M$.

~~By Thom's theorem $(\widetilde{\sigma\lambda})^{-1}(M) \subset S^L$.~~



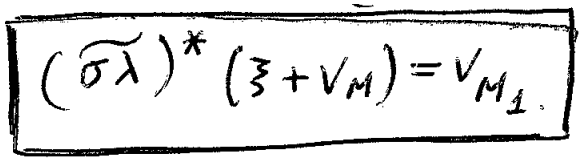
Original source details: GELFAND-P000489

Diagram and typed description. The entire line is crossed out.

Important:

$$\boxed{(\widetilde{\sigma\lambda})^*(\xi + \nu_M) = \nu_{M_1}.}$$

(reformul. of transverse regularity)



Original source details: GELFAND-P000489

Diagram and typed description. The equality is enclosed in a rectangle.

$\widetilde{\sigma\lambda} : M_1 \longrightarrow M$.

Thus from the bundle ξ we have constr. a manifold M_1 and its cont. map

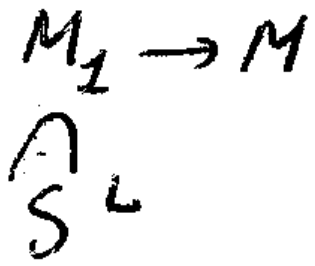


Diagram and typed description. Below M_1 are $\cap$ and S^L.

$$\begin{array}{ccc} M_1 & \longrightarrow & M \\ \cap & & \\ S^L & & \end{array}$$

Original source details: GELFAND-P000489

M_1 with norm. bundle $\nu_{M_1} = (\widetilde{\sigma\lambda})^*(\xi + \nu_M)$. (depends on ξ)

There arose a ~~problem~~ surgery situation.



Uncertain reading. The crossed-out word is provisionally read as «задача» (“problem”).

Original source details: GELFAND-P000489

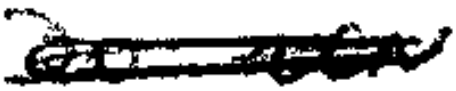
A problem in smooth topology: how uniquely is M_1 determined.

GELFAND-P000490

Milnor–Kervaire thm.. All M_1 are obtained from one another by Morse surgeries. (necess. and suff.)

Is there a manifold there homot. equiv. to M.

P_n — the obstruction group



Uncertain reading. A word after “obstructions” is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000490

for surgery to a homotop. equiv.

$$\sigma_0 : M_1 \longrightarrow M$$

The seq. is exact.

$$? \longrightarrow hS(M^n).$$

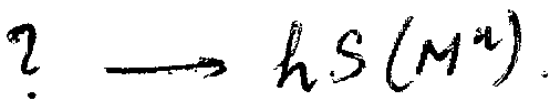
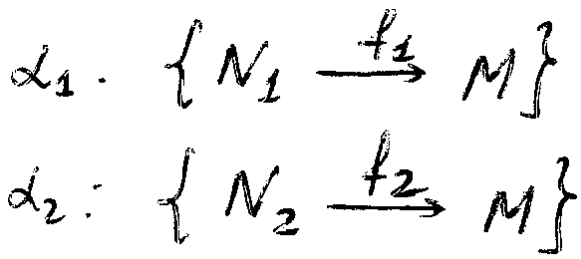


Diagram and typed description. A question mark is written in place of the first term.

Original source details: GELFAND-P000490

$$\alpha_1. \quad \{N_1 \xrightarrow{f_1} M\},$$
$$\alpha_2 : \quad \{N_2 \xrightarrow{f_2} M\}.$$



Original source details: GELFAND-P000490

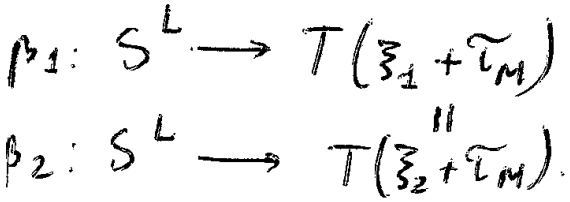
Diagram and typed description. A period follows α_1, and a colon follows α_2.

$\xi_1 \in \phi(M)$ is constr. from α_1.



Original source details: GELFAND-P000490

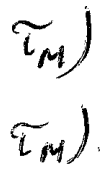
Diagram and typed description. The sign before (M) is set as ϕ.



Original source details: GELFAND-P000490

Diagram and typed description. Two arrows point to Thom complexes. Two vertical strokes stand between the targets.

$$\begin{array}{ccc} \beta_1 : S^L & \longrightarrow & T(\xi_1 + \tau_M) \\ & & \| \\ \beta_2 : S^L & \longrightarrow & T(\xi_2 + \tau_M). \end{array}$$



Original source details: GELFAND-P000490

Uncertain reading. The wavy stroke may be part of τ_M or a separate tilde over τ. It is provisionally set as τ_M.

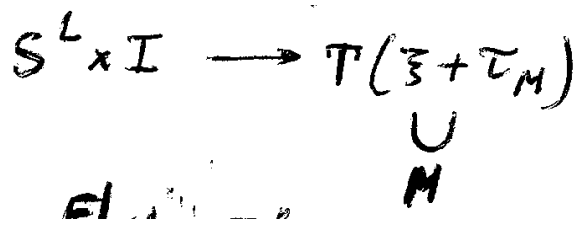
We obtain the problem:



Original source details: GELFAND-P000490

Uncertain reading. An entry between “problem” and “diss.” is crossed out. Its reading is unresolved.

Novikov’s diss.:



Original source details: GELFAND-P000490

Diagram and typed description. Below the target are $\cup$ and M.

$$\begin{array}{ccc} S^L \times I & \longrightarrow & T(\xi + \tau_M) \\ & & \cup \\ & & M \end{array}$$

Film:

$$F\big|_{S^L\times 0} = \beta_1$$
$$F\big|_{S^L\times 1} = \beta_2$$

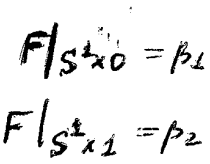


Diagram and typed description. The two restrictions of F at 0 and 1.

Original source details: GELFAND-P000490



Uncertain reading. The sign before the quantifier is overwritten. Its reading is unresolved.

Original source details: GELFAND-P000490

$\exists$ h-cobordisms among the films?

GELFAND-P000491

19/III-73 Gelfand's Sem.



Uncertain reading. A short entry before the date is crossed out. It may contain digits.

Original source details: GELFAND-P000491

Arnold. Consid. the germ of a fn. $f : (\mathbb{C}^n, 0) \longrightarrow (\mathbb{C}, 0)$. 0 — a crit. point.



Diagram and typed description. A yellow callout icon with three black strokes is visible above the line. Its text is unavailable in this frame.

Original source details: GELFAND-P000491

Consid. the space of k-jets. The group of k-jets of diffs. acts.

$j^k f$ k-suff. (the subsequent ones have no effect)

m — number of moduli. Thm.. $m = 0 \Longrightarrow$ the list

$$\begin{array}{ll} A_k & x^{k+1} \\ D_k & x^2 y + y^{k-1} \\ E_6 & x^3 + y^4 \\ E_7 & x^3 + xy^3 \\ E_8 & x^3 + y^5 \end{array}$$

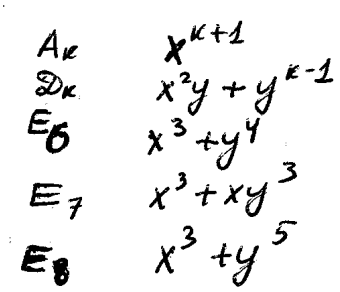


Diagram and typed description. Several strokes in the list are retraced, including those in E_6 and E_8.

Original source details: GELFAND-P000491

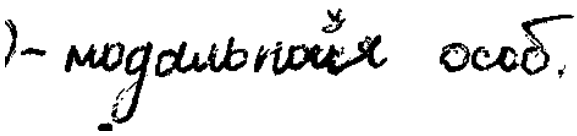
Stab. equiv. $\oplus$ with nondeg. quadr. forms.

The quadr. form is positive definite.

GELFAND-P000492

An analyt. fn., up to an analyt. change of var., uniquely det. the change of var.

0-modality sing. — after changing the coeffs. $\exists$ only a finite number of variants.



Uncertain reading. The ending was overwritten. The line is provisionally read as «модульная», with «модульных» also possible.

Original source details: GELFAND-P000492

— · —

$f = \varepsilon$ level line: Intersection with a ball — an even-dimensional manifold with boundary.

$$f^{-1}(\varepsilon) \cap B_\delta$$



Diagram and typed description. Two closed curves are joined by inward-curving sides.



Original source details: GELFAND-P000492

Intersection index



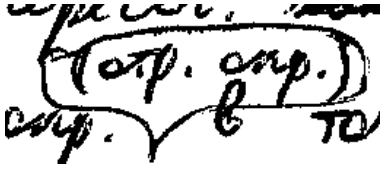
Original source details: GELFAND-P000492

Uncertain reading. An entry after "intersection" is crossed out. Its reading is unresolved.

of real homology in the middle dim. (integral)

(neg. def.)

It is definite in sign exactly in the listed cases, provided $n \equiv 3 \pmod 4$. (otherwise add extra variables) — Tyurin's thm..

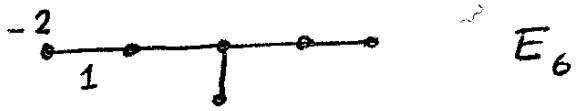


Original source details: GELFAND-P000492

Diagram and typed description. "(Neg. def.)" is inserted in a callout after "definite in sign".

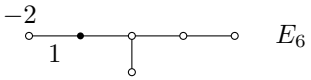
$$f^{-1}(\varepsilon) \cap B_\delta \approx \underbrace{S^{n-1} \vee \ldots \vee S^{n-1}}_{\mu}.$$

$\mu =$ dim of the local ring.



Original source details: GELFAND-P000492

Diagram and typed description. Diagram of E_6. The leftmost vertex is marked -2 and the first edge is marked 1.





Original source details: GELFAND-P000492

Uncertain reading. An entry to the right of E_6 is crossed out. Its reading is unresolved.

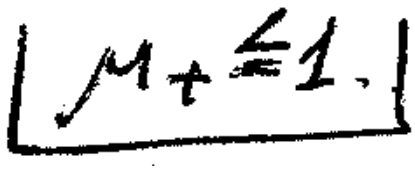
The quadr. form is given by the Dynkin diag.

$\boxed{m = 1.}$ — one also finds a list (long).

First: $\underline{x_1^3 + x_2^3 + x_3^3}$.

$$\underline{\mu = \mu_+ + \mu_0 + \mu_-}$$

$\underline{\mu_+ \leqslant 1.} \Longrightarrow$ always $m = 1$.



Original source details: GELFAND-P000492

Uncertain reading. The sign before 1 is retraced. It is provisionally set as $\leqslant$. The conclusion reads $m = 1$.

For all of them

$$\mu_+ \leqslant 1 \Longrightarrow \mu_0 = 1 \qquad \text{(always degen.)}$$

$\mu_+ = 1$. $axyz + x^k + y^\ell + z^m$ — three-index series $T_{k\ell m}$.

A. S. Mishchenko.

Sullivan: localization, periodicity and ... Galois-

$$K(X)/J\text{-equiv.} = J(X) \quad \text{— a finite group.}$$

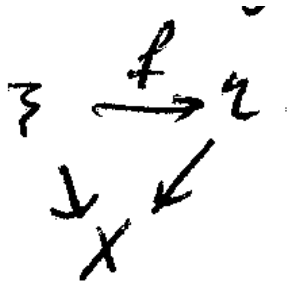
Adams: it is described in terms of the cohom. ops. of K-theory ψ^k.

$$\psi^k(\xi) = \xi^k \quad \text{and additive} \quad \Longrightarrow \quad \text{this determines them uniquely}$$

$(\dim \xi = 1) \qquad k = 2, 3, \ldots$

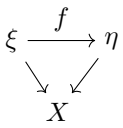
Thus $J(X)$ is determined by the group $K(X)$ as a module over these ops..

J-equiv. — fibrewise homot. equiv..



Original source details: GELFAND-P000492

Diagram and typed description. The arrow f joins ξ and η. Both letters have arrows to X.

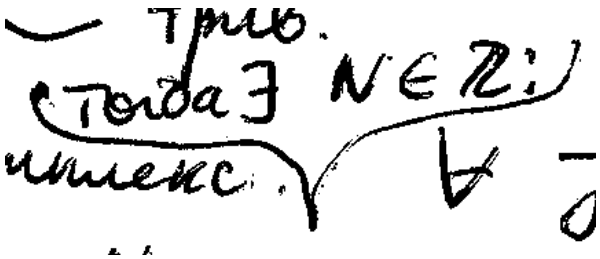


f — proper. $\deg f = 1$ on fibres.

It was proved (Adams, Atiyah):

$$\xi \overset{J}{\sim} \text{triv.}.$$

X — a finite complex. (then $\exists N \in \mathbb{Z}$:)



Original source details: GELFAND-P000492

Diagram and typed description. "Then $\exists N \in \mathbb{Z}$:" is inserted after "complex".

$\forall$ J-triv. bundles ξ $(\forall k)$ $\exists$



Original source details: GELFAND-P000492

Diagram and typed description. The quantifier $\forall k$ is inserted between ξ and $\exists$.



Original source details: GELFAND-P000492

Uncertain reading. A word before “bundle” is crossed out. Its reading is unresolved.

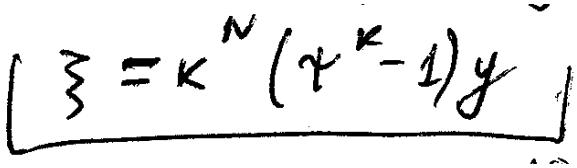
a bundle y:



Original source details: GELFAND-P000492

Uncertain reading. An entry after the colon is crossed out. Its reading is unresolved.

$$\underline{\xi = k^N(\psi^k - 1)y}$$



Original source details: GELFAND-P000492

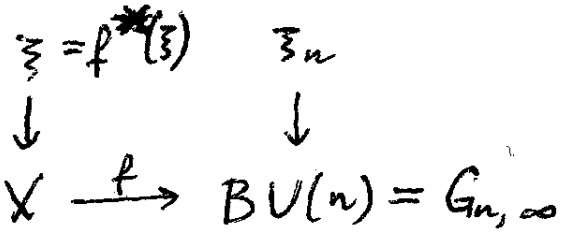
Diagram and typed description. A stroke runs below and to either side of the equality.

The converse — the Adams conjecture: $\forall$ finite compl. X $\exists N$; $\forall k$

$$k^N(\psi^k - 1)\xi \overset{J}{\sim} 0.$$

GELFAND-P000493

Idea.



Original source details: GELFAND-P000493

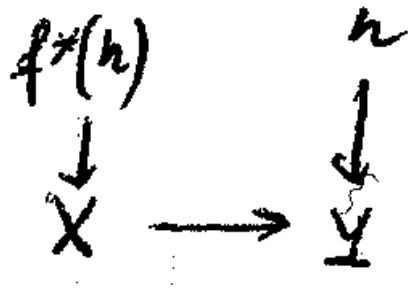
Diagram and typed description. The left side reads $\xi = f^*(\xi)$ and the right side ξ_n.

$$\begin{array}{ccc} \xi = f^*(\xi) & & \xi_n \\ \downarrow & & \downarrow \\ X & \xrightarrow{\ f\ } & BU(n) = G_{n,\infty} \end{array}$$



Original source details: GELFAND-P000493

Uncertain reading. The sign after f is heavily retraced and provisionally set as a star.



Original source details: GELFAND-P000493

Diagram and typed description. The second diagram shows $f^*(\eta)$ and η.

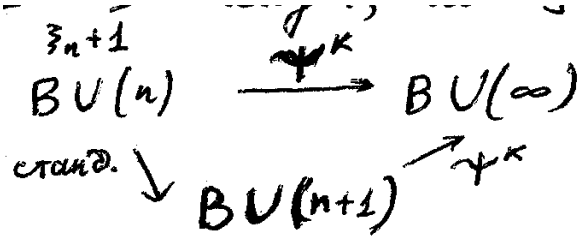
$$\begin{array}{ccc} f^*(\eta) & & \eta \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

$$\psi^k(f^*(\eta)) = f^*\psi^k(\eta).$$

Hence the op. ψ^k is det. by specifying a bundle on $\underline{\underline{BU(n)}}$. $(\psi^k(\xi_n))$.

From ψ^k one constructs a map $BU(n) \longrightarrow BU(\infty)$ (unknown dimension).

$\psi^k(1) = 1 \Longrightarrow$ it follows that $\exists$ a sequence of maps.



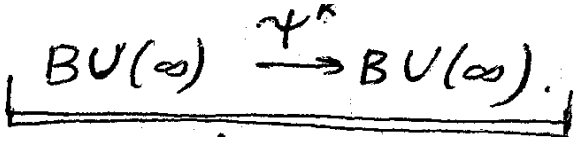
Original source details: GELFAND-P000493

Diagram and typed description. Above $BU(n)$ is $\xi_n + 1$.

$$\begin{array}{ccccc} \xi_n + 1 & & & & \\ BU(n) & & \xrightarrow{\psi^k} & & BU(\infty) \\ & \searrow^{\text{std.}} & & \nearrow_{\psi^k} & \\ & & BU(n+1) & & \end{array}$$

Thus ψ^k is given by a single map.

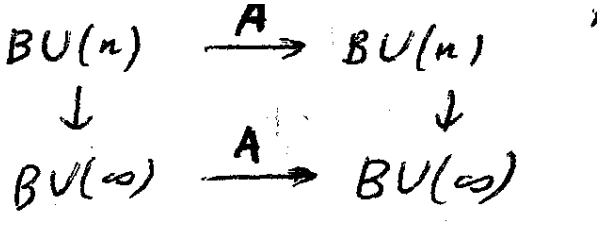
$$\underline{\underline{BU(\infty) \xrightarrow{\psi^k} BU(\infty).}}$$



Original source details: GELFAND-P000493

Diagram and typed description. A double stroke runs below the arrow, with turned-up ends.

Suppose there is a cohom. op. $BU(\infty) \xrightarrow{A} BU(\infty)$, and suppose it preserves dim.

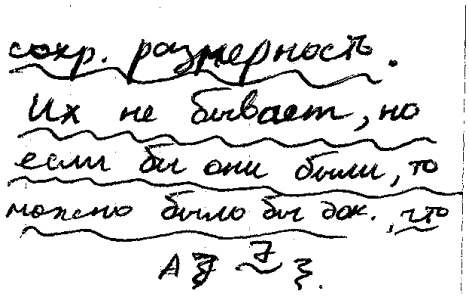


Original source details: GELFAND-P000493

Diagram and typed description. The square has A on both horizontal arrows.

$$\begin{array}{ccc} BU(n) & \xrightarrow{A} & BU(n) \\ \downarrow & & \downarrow \\ BU(\infty) & \xrightarrow{A} & BU(\infty) \end{array}$$

preserves dimension. They do not occur, but if they did, one could prove that $A\xi \overset{J}{\sim} \xi$.



Original source details: GELFAND-P000493

Diagram and typed description. The note on the right has a wavy underline.

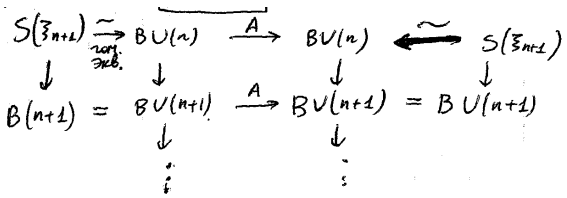
Then $A\xi \overset{J}{\sim} \xi$.

spheres



Original source details: GELFAND-P000493

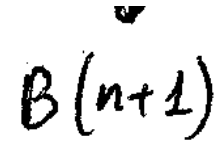
Diagram and typed description. The frame around “spheres” is open on the left.



Original source details: GELFAND-P000493

Diagram and typed description. The diagram has two levels and vertical ellipses.

$$\begin{array}{ccccccc} S(\xi_{n+1}) & \xrightarrow[\text{equiv.}]{\sim} & BU(n) & \xrightarrow{A} & BU(n) & \xleftarrow{\sim} & S(\xi_{n+1}) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ B(n+1) & = & BU(n+1) & \xrightarrow{A} & BU(n+1) & = & BU(n+1) \\ & & \downarrow & & \downarrow & & \\ & & \vdots & & \vdots & & \end{array}$$



Original source details: GELFAND-P000493

Diagram and typed description. The lower-left node reads $B(n+1)$.

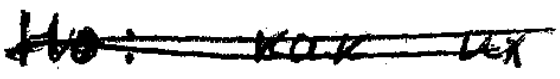


Original source details: GELFAND-P000493

Uncertain reading. The upper-right arrow is heavily retraced. The arrowhead is visible on the left.

A fibrewise homot. equiv. is obtained.

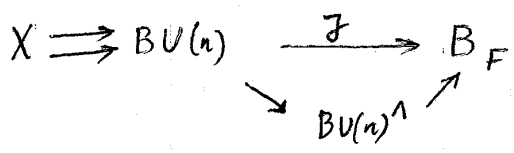
~~But: how to~~



Original source details: GELFAND-P000493

Diagram and typed description. “But: how to” is crossed out.

There are none, but how do we “find” them?

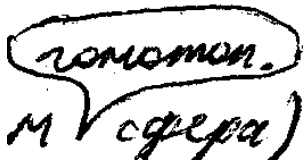


Original source details: GELFAND-P000493

Diagram and typed description. Two arrows go from X to $BU(n)$.

$$\begin{array}{ccccc} X & \rightrightarrows & BU(n) & \xrightarrow{J} & B_F \\ & & & \searrow \quad \nearrow & \\ & & & BU(n)^\wedge & \end{array}$$

(classif. space of bundles with fibre a (homotopy) sphere)

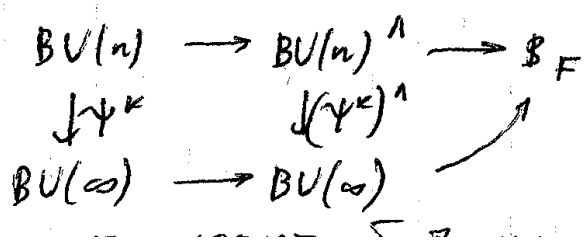


Original source details: GELFAND-P000493

Diagram and typed description. “Homotopy” is inserted before “sphere”.

J — a functor.

Suppose $\exists BU(n)^\wedge$, to which the Adams operations transfer, i.e. $\exists$ an op. $(\psi^k)^\wedge$, for which the diag. commutes.



Original source details: GELFAND-P000493

Diagram and typed description. The lower-right arrow bends up toward B_F.

$$\begin{array}{ccccc} BU(n) & \longrightarrow & BU(n)^\wedge & \longrightarrow & B_F \\ \downarrow{\scriptstyle \psi^k} & & \downarrow{\scriptstyle (\psi^k)^\wedge} & & \uparrow \\ BU(\infty) & \longrightarrow & BU(\infty) & & \end{array}$$

Perhaps, it may be that $(\psi^k)^\wedge$ preserves dimension.

(orientations — dim.-preserving maps)

These spaces were constr. by Sullivan.

GELFAND-P000494

$BU(n)^\wedge$ will be the compl. Grassmannian.

On



Original source details: GELFAND-P000494

Uncertain reading. A short entry after “On” is crossed out. Its reading is unresolved.

S^{2n-1} the group $U(n)$.

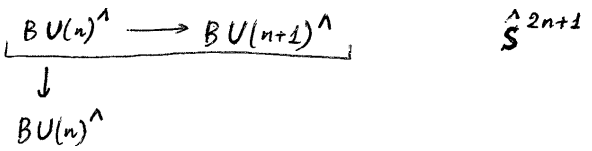


Original source details: GELFAND-P000494

Uncertain reading. The sign in the exponent is heavily retraced. It is provisionally set as $2n-1$, possibly $2n+1$.

We obtain bundles (spher.) on $BU(n)$.

Sullivan: another space $\widehat{S}^{2n-1}$ and the group G_n. Consid. the bundle and the classif. space.



Original source details: GELFAND-P000494

Diagram and typed description. A stroke runs below the horizontal arrow. The separate entry on the right is $\widehat{S}^{2n+1}$.

$$\underbrace{BU(n)^\wedge \longrightarrow BU(n+1)^\wedge}_{\downarrow} \qquad \widehat{S}^{2n+1}$$
$$BU(n)^\wedge$$

and the fibrewise homot. type will be preserved.



Original source details: GELFAND-P000494

Undeciphered passage. The line is heavily crossed out. A continuous reading is unresolved.

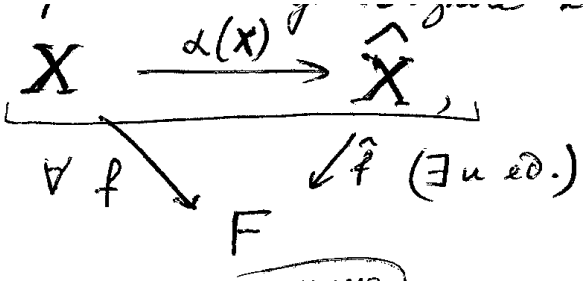
Simply connected compls. with fin. gen. homot. grps. are considered.

Claim. $\overset{\forall X}{\exists}$



Original source details: GELFAND-P000494

Diagram and typed description. $\forall X$ is inserted above $\exists$.



Original source details: GELFAND-P000494

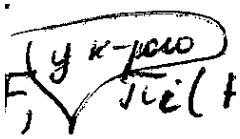
Diagram and typed description. A stroke runs under the upper arrow. Two arrows point to the bottom node.

$$\underbrace{X \xrightarrow{\alpha(X)} \widehat{X},}$$
$$\forall f \qquad \widehat{f}\ (\exists \text{ and unique})$$
$$F$$

(functor)

$\alpha(X)$ — a transformation of the ident. functor.

($\forall F$, (for which) $\pi_i(F)$ are finite).



Original source details: GELFAND-P000494

Diagram and typed description. "For which" is inserted after F.

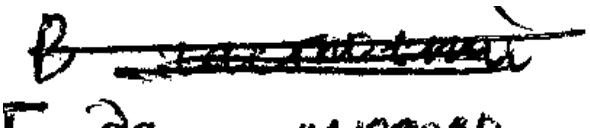


Original source details: GELFAND-P000494

Uncertain reading. An entry before "are finite" is blacked out. Its reading is unresolved.

$\pi_i(\widehat{X})$ are profinite ($\varprojlim$ fin. groups). (in partic., compact).

~~In particular~~



Original source details: GELFAND-P000494

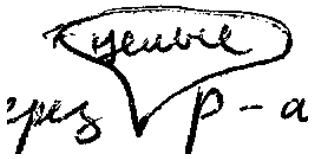
Uncertain reading. The crossed-out words are provisionally read as “In particular”.

Then one can compute:

$$\pi_i(\widehat{X}) = \widehat{\pi_i(X)}$$

(profin. completion). Neighb. — subgroups of fin. index and their limit.

$Z^\wedge$ is descr. by p-adic (integral) numbers.



Original source details: GELFAND-P000494

Diagram and typed description. “Integral” is inserted before p.

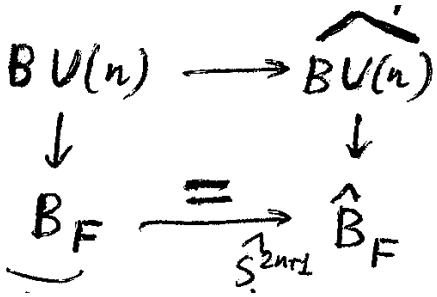
The homot. groups of $\widehat{X}$ behave like integral p-ad. numbers.

~~Action of the Galois grp..~~



Original source details: GELFAND-P000494

Diagram and typed description. The heading is crossed out.

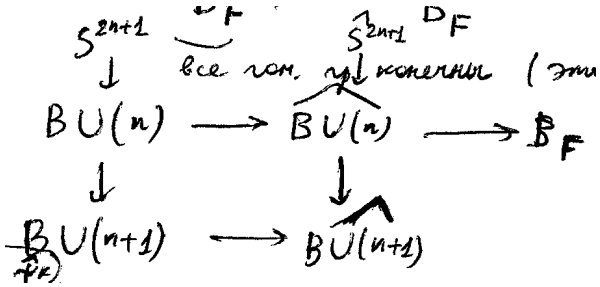


Original source details: GELFAND-P000494

Diagram and typed description. The lower arrow points right and has an equals sign above it.

$$\begin{array}{ccc} BU(n) & \longrightarrow & \widehat{BU(n)} \\ \downarrow & & \downarrow \\ B_F & \xrightarrow{=} & \widehat{B}_F \end{array}$$

all homot. grps. are finite (these are stable homot. groups of spheres)

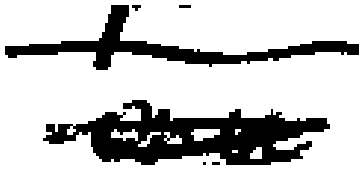


Original source details: GELFAND-P000494

Diagram and typed description. The lower horizontal arrow points right. There is an additional stroke at its left end. The arrow from $\widehat{S}^{2n+1}$ crosses the neighbouring writing.

$$\begin{array}{ccccc} S^{2n+1} & & \widehat{S}^{2n+1} & & \\ \downarrow & & \downarrow & & \\ BU(n) & \longrightarrow & \widehat{BU(n)} & \longrightarrow & B_F \\ \downarrow & & \downarrow & & \\ BU(n+1) & \longrightarrow & \widehat{BU(n+1)} & & \end{array}$$

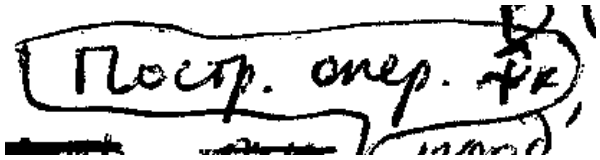
~~Suppose~~



Original source details: GELFAND-P000494

Uncertain reading. The following short word is crossed out. Its reading is unresolved.

(The constr. op. ψ_k) extends the Adams ops. and preserves dimension.

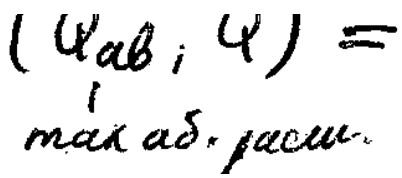


Original source details: GELFAND-P000494

Uncertain reading. The small sign to the right of ψ is provisionally set as subscript k. Another possible reading is $\widehat{\psi}^k$.

Consid. the Grassmannians: $G_{N,n}$. There are finitely many aff. alg. charts, with eqs. over Z (scheme over Z). — an alg. set over Q. $\Rightarrow$ the Galois grp. acts there $\mathrm{Gal}(\mathbb{C} : Q)$ — acts on $G_{N,n}$. It is discontinuous in the usual sense, but continuous in the Zariski top.. Consid. Étale top. ...

$$\mathrm{Gal}(\mathbb{C} : Q) \longrightarrow \mathrm{Gal}\Big(\underset{\text{max. ab. ext.}}{Q_{ab}} ; Q \Big) = \widehat{Z}^*$$



Original source details: GELFAND-P000494

Diagram and typed description. Below Q_{ab} is the label “max. ab. ext.”.

$$H_i(V_c, A) \approx H_i(V_{et}, A)$$
$$\widehat{V}_c \sim \varprojlim N(V_{et}), \qquad N \text{ — nerve.}$$

GELFAND-P000495

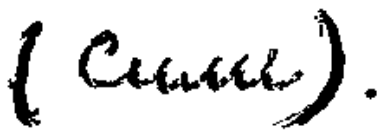
26.3.73

Statement of the res. of Atiyah, Patodi, Singer. (I. M. Gelfand).

Let A — be a self-adj. ell. op.

$$\zeta_A(s) = \mathrm{Tr}\, A^{-s} = \sum \lambda_i^{-s}$$

at $s = 0$ has no pole (Seeley).

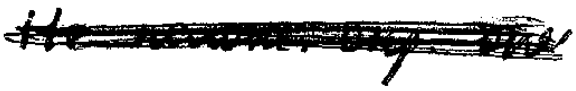


Original source details: GELFAND-P000495

Uncertain reading. The Cyrillic spelling of the name is provisionally read as «Сили», possibly «Силли».

$\zeta_A(0)$ — Atiyah and others.

Consid. In place of the ζ-fn., L-series.



Original source details: GELFAND-P000495

Undeciphered passage. The crossed-out phrase to the right may begin “Not”. The rest is illegible.

Let A — not pos. def. (self-adj. ellip.)

$$\eta(s) = \sum_{\lambda \neq 0} (\operatorname{sign} \lambda_i)\, |\lambda_i|^{-s}$$

Difficult theor.: $s = 0$ — is no pole !!

Important example — 1st-order op.

General prob.: no pos. def. Consid. $\sum$ (mult. characters of λ_i) (L-series). On compl. numbers there are more complicated mult. characters.

Example.

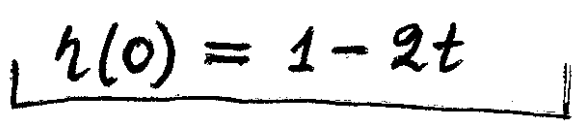


Original source details: GELFAND-P000495

Uncertain reading. The crossed-out entry before A partly overlaps A.

$$A = i\frac{d}{dx} + t \qquad \text{on } [0, 2\pi] \qquad \text{periodic}$$

$$\eta(0) = 1 - 2t$$



Original source details: GELFAND-P000495

Manuscript excerpt. An open-topped bracket runs under the formula.

Eigenvals. $n + t$, $n = -2, -1, 0, 1, \ldots$

$\eta(0)$ — the "asymmetry" of the operator:

$$\sum_{\lambda_i > 0} 1 - \sum_{\lambda_i < 0} 1.$$

S. I. Gelfand. Account of Manin's last lecture:

p-adic integration (Mazur, Swinnerton — Dyer)



Original source details: GELFAND-P000495

Uncertain reading. The middle letters of the first part of the name run together. Provisionally read as Swinnerton.

GELFAND-P000496

Z_p — integral p-adic numbers, a fn. is defined on them. It is to be integrated.
(p-adic vals.)
lit. L. G. Schnirelmann, Dwork, Grauert, Krasner, Tate
$Z_p \supset A_i = p^i Z_p$ — syst. of neighb. of 0. The measure is def. on sets $a + A_i$
$\mu(a + A_i)$ — a p-adic number (not nec. integral).

$$a \in Z_p \quad \text{or} \quad a \in Z_p/p^i$$

There must be finite additivity; reduces to the cond.:

$$\mu(a + A_i) = \sum_{t=0}^{p-1} \mu(a + tp^i + A_{i+1})$$

2 examples: 1) invar. measure: indep. of a, but dep. on i. Then

$$\mu(a + A_i) = p^{-i} \qquad \text{(only possibility).}$$

The measure of a small set is large! — far from 0

$$\left(\left\| \frac{a}{b} \right\| = p^{-n}, \quad \text{where } \frac{a}{b} = p^n \frac{a_1}{b_1}, \quad a_1, b_1 \text{ rel. prime to } p \right).$$

2) More important example. (Leopoldt — Kubota measures).

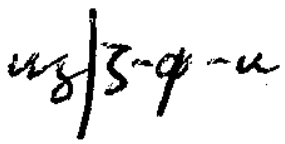
2 parameters n and a

$$(n, p) = 1, \qquad 0 \le a \le p^i - 1$$

$$\mu_n(a + A_i) = \left[\frac{na}{p^i} \right] - \frac{n-1}{2}$$

$$\mu_2(a + A_i) = \begin{cases} +1/2, & \text{if } a > \dfrac{p^i}{2} \\ -1/2, & \text{if } a < \dfrac{p^i}{2} \end{cases}$$

It is addit. !! The measure of $\forall$ set is bounded
(it is obtained by the Mellin transf. from the | ζ-fn. of Leopoldt — Iwasawa)



Uncertain reading. A vertical stroke appears between “from” and ζ. Its function is unresolved.

Original source details: GELFAND-P000496

$$Q_p \ni f(x), \quad x \in Z_p \qquad \int_{Z_p} f(x)\mu(x) \qquad \text{in Riemann's sense}$$

R_m — representatives of $Z_p/p^m Z_p$ (1 from $\forall$ class)
This is a partition. Form

$$S(R_m) = \sum_{a \in R_m} f(a)\mu(a + A_m)$$

they converge under conds. on the measure and the fn.

Moderately growing measure — if it grows more slowly than the invariant one

$$\lim_{m\to\infty} \max |\mu(a + A_m)|p^{-m} = 0$$

f satisfies the Lipschitz cond. if

$$|f(a) - f(b)| < Cp^{-m}, \quad \text{if } |b - a| < p^{-m}$$

(in partic., if given by a conv. power series)

If the measure grows moderately and the fn. satisfies the Lipschitz cond., then, $\exists$ a limit, called the integral.

GELFAND-P000497

If we consid. the ordinary $\zeta(s) = \sum 1/n^s$. Then

$\zeta(1 - 2k)$ — a rat. number $\forall k > 0$ (expr. through Bern. numbers).



Uncertain reading. The crossed-out entry in the left margin resembles a sum. An s is visible on the right. The full reading is unresolved.

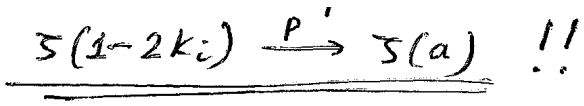
Original source details: GELFAND-P000497

Let k_i — ~~p-adic numbers~~ be integers and in the p-adic sense

$$k_i \longrightarrow a \in Z_p.$$

Iwasawa — Leopoldt:

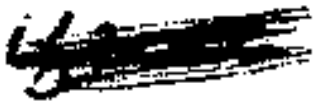
$$\underline{\zeta(1 - 2k_i) \xrightarrow{p} \zeta(a)} \quad !!$$



Manuscript excerpt. The right-hand side reads $\zeta(a)$, with p over the arrow.

Original source details: GELFAND-P000497

We obtain a cont. fn. on Z_p —



Uncertain reading. A short word is crossed out. Its reading is unresolved.

Original source details: GELFAND-P000497

p-adic var. It is the Mellin transf. of the measure of example 2.

A. Gabrielov. Dynkin diagr. of isol. sing. of fns.

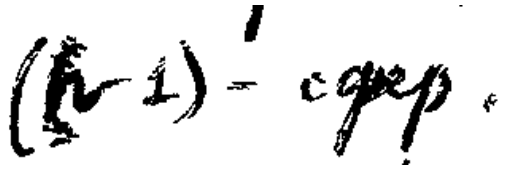
$f(x)$ — complex-analytic in a neighb. of 0, 0 — an isol. sing.

Ball of rad. r_0 (small). Consid.

$$f(x) = \delta, \quad |x| < r_0$$

Choose also δ.

For small δ this is a complex set of dim. $(n-1)$, homot. equiv. to a bouquet of $(n-1)$-spheres.



Original source details: GELFAND-P000497

Uncertain reading. The first letter in the sphere dimension is heavily retraced. It is set provisionally as $(n-1)$, possibly $(k-1)$.

and van. cycles.

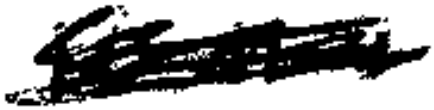


Original source details: GELFAND-P000497

Uncertain reading. The retraced mark at the start of the line is provisionally read as “and”.

GELFAND-P000498

9/IV-73 yr.



Original source details: GELFAND-P000498

Uncertain reading. A short entry to the right of the date is crossed out. Possibly «Сем.» (“Sem.”).

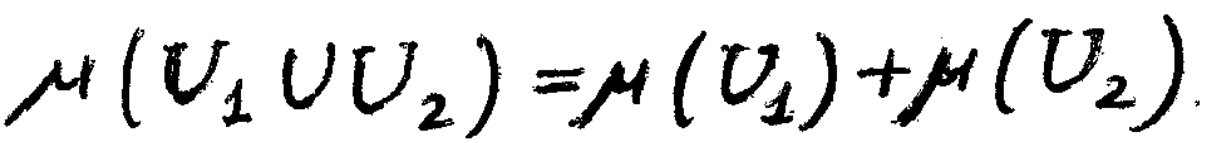
S. I. Gelfand. (account of lectures by Yu. I. Manin)

$Q_p \supset Z_p$,

{A measure μ on Z_p with vals. in the fin.-dim. space W over Q_p.}

$$\mu : U \longmapsto \mu(U) \in W, \qquad U \subset Z_p$$

$$\mu(U_1 \cup U_2) = \mu(U_1) + \mu(U_2).$$



Original source details: GELFAND-P000498

Manuscript excerpt. The additivity formula in the manuscript.

In Q_p a basis of open-and-closed sets $A_n = p^n Z_p$. $\quad U = (a + A_n)$.



Original source details: GELFAND-P000498

Uncertain reading. The letter before the subscript p is heavily retraced. Provisionally read as Q_p.

Riemann integral.

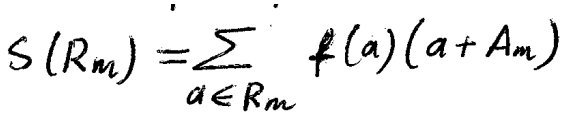
f — a fn. on Z_p with vals. in Q_p.

$m \geq 1$,

R_m — representatives of $Z_p / p^m Z_p$. (a finite set).

Riemann sum

$$S(R_m) = \sum_{a \in R_m} f(a)(a + A_m)$$



Original source details: GELFAND-P000498

Manuscript excerpt. The entry immediately after $f(a)$ is $(a + A_m)$.

Theor.: as $m \to \infty$ under certain conds. on the fn. and measure $\exists$ a limit

$$\lim_{m \to \infty} S(R_m) = \int_{Z_p} f(a)\mu(a) \in W$$

Conditions: μ is bounded, and the fn. is loc. constant

Leopoldt — Kubota measures:

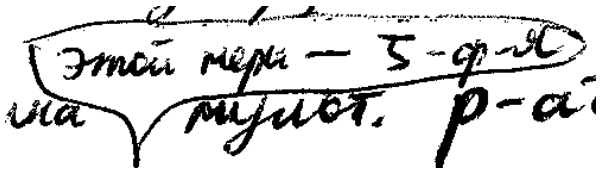
k — an integer, $(k, p) = 1$,

$$\mu_k(a + A_n) = \left[\frac{ka}{p^n}\right] - \frac{k-1}{2}$$

$0 \leq a \leq p^n - 1$

Easy to check (useful) that it is finitely addit.

(Mellin transf.



Original source details: GELFAND-P000498

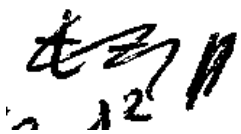
Diagram and typed description. A callout between “Mellin” and “mult.”.

Of this measure — ζ-fn.

mult. p-adic char. on Z_p)

Mult. char.: p-adic number to a p-adic power.

$$\chi_t(a) = e^{t \ln a}, \qquad a = 1 \bmod p.$$

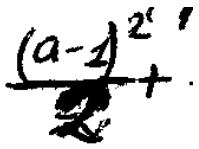


Original source details: GELFAND-P000498

Uncertain reading. The crossed-out entry begins with t. The following signs are illegible.

$t \in Z_p$ (for convergence)

$$\ln a = (a-1) - \frac{(a-1)^2}{2} + \dots$$



Uncertain reading. A 2 is visible in the denominator. The other strokes below the fraction bar are illegible.

Original source details: GELFAND-P000498

2nd cl. of measures using modular forms:

$$H = (z \mid \operatorname{Im} z > 0)$$

Acts

$$SL(2,R)/(\pm 1) \supset \Gamma = SL(2,Z)/(\pm 1)$$

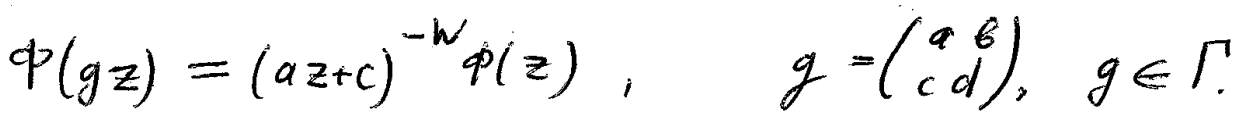
Consid.

$$\varphi(z) = \sum_{n=1}^{\infty} \lambda_n e^{2\pi i n z}$$

(var. invar. decr. fns.)

Autom. form of weight w

$$\varphi(gz) = (az+c)^{-w}\varphi(z), \qquad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad g \in \Gamma.$$



Original source details: GELFAND-P000498

Manuscript excerpt. The factor is written $(az+c)^{-w}$.

GELFAND-P000499

Further condition: φ eigenfn. with respect to all Hecke operators.

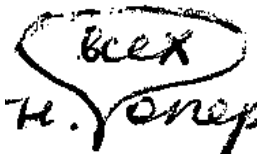
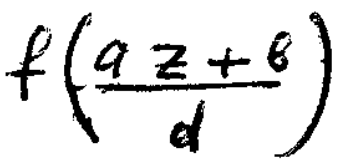


Diagram and typed description. "All" is inserted in a callout above the line.

Original source details: GELFAND-P000499

$$(T_k\varphi)(z) = \sum_{\substack{ad=k \\ 0 \le b \le d-1}} f\left(\frac{az+b}{d}\right)$$



Manuscript excerpt. The summand has f. The left-hand side has φ.

Original source details: GELFAND-P000499

(a, b, d, k — integers)

$T_k\varphi$ — also a modular form

Let $\lambda_1 = 1$. Then

$$T_k\varphi = \lambda_k\varphi \qquad (\lambda_k \text{ — coeff. of } e^{2\pi ikz}).$$

_____ . _____

$$p: \qquad \chi : (\mathbb{Z}/p^n\mathbb{Z})^* \to \mathbb{C}^*$$

(primitive character) (p^n — conductor of the char.)

(on $p^{n-1}\mathbb{Z}/p^n\mathbb{Z}$ nontriv.)



Uncertain reading. The sign between n and 1 is inked over. Provisionally set as $n - 1$.

Original source details: GELFAND-P000499

$$\varphi_\chi(z) = \sum \lambda_n \chi(n) e^{2\pi inz} \quad \text{— also modular.}$$

$$L(\varphi, \chi, 1) = \text{[factor illegible]} \int_0^{i\infty} \varphi_\chi(z)\, dz$$



Uncertain reading. The factor before the integral is heavily overwritten. Possibly C or a longer expression.

Original source details: GELFAND-P000499

Has the interpretation

$$L(\Phi, \chi, s) = C \int_0^{i\infty} \Phi_\chi(z) z^{s-1}\, dz = \sum \frac{\lambda_n \chi(n)}{n^s} \qquad (L\text{-fn.})$$

p-adic part.

χ — also a loc. const. fn. on integral p-adic numbers

(on the invertible ones as before, elsewhere 0).

Let $\chi(-1) = 1$

Aim: $L(\varphi, \chi, 1)$ repr. by $\int \chi$ on p-adic numbers.

$$\int \chi(a)\mu(a) = \textcircled{?}\, L(\varphi, \chi, 1).$$



Manuscript excerpt. The question mark is circled in the manuscript.

Original source details: GELFAND-P000499

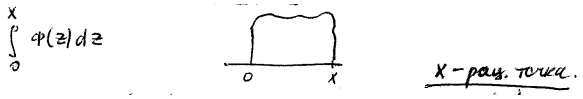
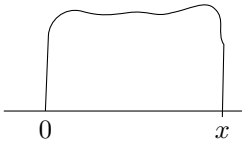


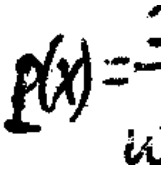
Diagram and typed description. The line from 0 to x runs above the baseline. There are no arrows.

$\int_0^x \varphi(z)\, dz$  x — rat. point.

Original source details: GELFAND-P000499

(essentially an $\int$ over a cycle in the quotient space)

$$\ell(x) = \frac{1}{\omega} \operatorname{Re} \int_0^x \Phi(z)\, dz \text{ — an alg. integer}$$



Uncertain reading. The initial letter is provisionally set as ℓ. Its form is unusual.

Original source details: GELFAND-P000499

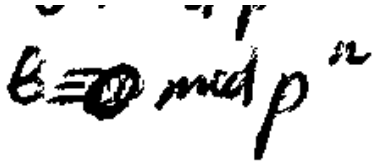
from a fixed ext. of Q.

$\exists\omega$ (transcend.)

(in many cases)

$$T^2 - \lambda_p T + p^{w-1} = 0, \qquad \pi, \bar{\pi} \text{ — roots}, \qquad \lambda_p \neq 0 \bmod p.$$

$$\mu(a + A_n) = \lim_{\substack{m\to\infty \\ m\geq n}} \frac{1}{\pi^m + \bar{\pi}^m} \sum_{\substack{b \bmod p^m \\ b\equiv 0 \bmod p^n}} \ell\left(\frac{b}{p^m}\right)$$



Original source details: GELFAND-P000499

Uncertain reading. The residue after $\equiv$ is heavily retraced. Its shape suggests 0, possibly a.



Original source details: GELFAND-P000499

Uncertain reading. The letter before the fraction is provisionally set as ℓ, as in the preceding formula.

lim exists.

Final result:

$$\pi^{k-1} \int_{\mathbb{Z}_p} \chi^{-1}(a)\mu(a) = -\frac{p^k}{\omega g(\chi)} L(\Phi, \chi, 1)$$

GELFAND-P000500



Original source details: GELFAND-P000500

Uncertain reading. The crossed-out word is provisionally read as «Числа» (“Numbers”).



Original source details: GELFAND-P000500

Diagram and typed description. Two lines with transverse hatching run under the name.

I. M. Gelfand.

$$1^k + 2^k + \ldots + n^k = P_{k+1}(n) \quad \text{— Bernoulli polynomial.}$$

$$\sum_{i=1}^{n} f(i), \qquad F'(x) = f(x).$$

$$F(n+1) - F(n) = f(n) + \frac{f'(n)}{2!} + \frac{f''(n)}{3!} + \ldots$$
$$F(n) - F(n-1) = f(n-1) + \frac{f'(n-1)}{2!} + \frac{f''(n-1)}{3!} + \ldots$$
$$F(2) - F(1) = f(1) + \frac{f'(1)}{2!} + \ldots$$



Uncertain reading. A heavily retraced entry in the right margin. Possibly “36” or “38”. A short separate stroke is visible below and to the left.

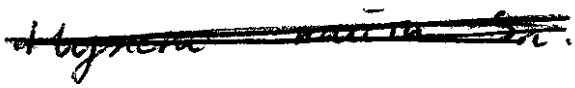
Original source details: GELFAND-P000500

Hence

$$F(n+1) - F(1) = S_n + \frac{S'_n}{2!} + \frac{S''_n}{3!} + \ldots$$

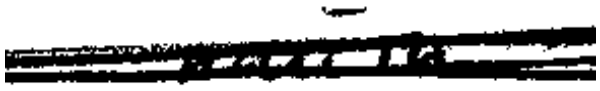
$$\left\| \begin{array}{l} S_n = \sum_{i=1}^{n} f(i) \\ S'_n = \sum_{i=1}^{n} f'(i) \\ \quad - \quad - \quad - \quad - \\ S_n^{(k)} = \sum_{i=1}^{n} f^{(k)}(i) \end{array} \right.$$

Similarly:



Uncertain reading. The crossed-out entry may begin “and take” and end with S_i or Σ_i.

Original source details: GELFAND-P000500



Undeciphered passage. Pen strokes cover the middle of the crossed-out entry. Its reading is unresolved.

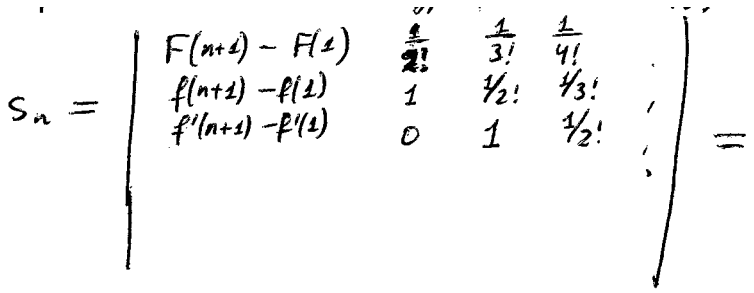
Original source details: GELFAND-P000500

$$f(n+1) - f(1) = S'_n + \frac{S''_n}{2!} + \dots$$
$$f'(n+1) - f'(1) = S''_n + \dots$$

We shall consid. this as a syst. of eqs. in $S_n, S'_n, S''_n, \dots$

Hence

$$S_n = \begin{vmatrix} F(n+1) - F(1) & \frac{1}{2!} & \frac{1}{3!} & \frac{1}{4!} & \cdot \\ f(n+1) - f(1) & 1 & 1/2! & 1/3! & \cdot \\ f'(n+1) - f'(1) & 0 & 1 & 1/2! & \cdot \\ & & & & \cdot \end{vmatrix} =$$



Original source details: GELFAND-P000500

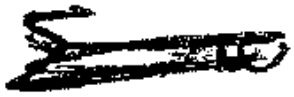
Manuscript excerpt. Three rows are written. The dots descend at the right and the bars extend below the written rows.

Expand along the 1st column:

$$= B_0\big(F(n+1) - F(1)\big) + B_1\big(f(n+1) - f(1)\big) + B_2\big(f'(n+1) - f'(1)\big) + \dots$$

But: B do not depend on the form of the fn.!

Take $f(x) = e^{tx}$. Then $f'(x) = te^{tx}$. Geom. progr.



Undeciphered passage. A crossed-out expression on the left begins with $\sum$. The rest and any limits are illegible.

Original source details: GELFAND-P000500

$$S_n = \frac{e^{t(n+1)} - e^t}{e^t - 1}$$ expand in t

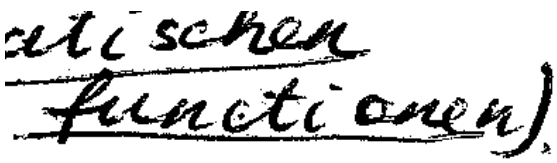
in a power series — we get the coefficients.

Bernoulli numbers in arbitrary fields.

Math. Z., 1898 Hurwitz (Analogue of the Bernoulli numbers for Gaussian numbers)

Hurwitz, Math. Ann., 1899, pp. 196–226

(Über die Entwicklungskoeffizienten der Lemniskatischen functionen).



Uncertain reading. The initial letter of the last word is provisionally read as lowercase “f”.

Original source details: GELFAND-P000500